



UNIVERSITY OF SAN CARLOS
SCHOOL OF ENGINEERING

PHYSICS ENFORCEMENT ON MACHINE
LEARNING SURROGATES OF THE ACTIVATED
SLUDGE PROCESS
DISSERTATION

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Declaration

I hereby declare that the work presented in the dissertation entitled “Physics Enforcement on Machine Learning Surrogates of the Activated Sludge Process” is original and the result of my own research. Formulations and ideas taken from other sources are cited as such.

Cebu City, Philippines, 2026

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Abstract

Activated sludge treatment links biological conversion, nutrient transformations, solids separation, and recirculation. Mechanistic models describe these relationships, but repeated simulation can be costly when many operating conditions are examined. Statistical surrogates offer faster response evaluation. Their usefulness depends on more than agreement with reference data because a prediction can have a small average error while violating conservation or producing negative component concentrations. This research develops a common framework for prediction, physical enforcement, interpretation, and operating assessment. A steady-state reactor benchmark compares several learning families through training-size studies, nested validation, and separately simulated beyond-domain conditions. A common post prediction procedure imposes stoichiometric conservation and nonnegativity while allowing paired assessment of prediction error and correction size. A structured second-order regression separates operating effects, influent effects, curvature, and component coupling. Its checked output procedure distinguishes invariant-aware fitting from final-state feasibility. A connected plant surrogate extends enforcement to the mixer, reactor train, clarifier outlets, and solids inventory. Surrogate-selected operating controls are assessed through independent mechanistic solution and comparison with direct optimization under common engineering priorities. The theoretical and methodological framework treats predictive accuracy, physical admissibility, interpretability, and decision quality as distinct requirements. Its guarantees concern the declared process representation and imposed constraints. Empirical treatment outcomes and model performance are reserved for subsequent evaluation.

Keywords. Activated sludge process, machine learning surrogate, wastewater treatment, conservation, nonnegativity, interpretable regression, operating optimization.

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Principal Symbols

Symbol	Meaning
F, P, K	Numbers of component states, reaction processes, and independent stoichiometric invariants
ν	Stoichiometric matrix with processes in rows and components in columns
ρ	Vector of mechanistic process rates
A	Full-row-rank stoichiometric invariant operator
V_0, Z	Orthonormal bases of the stoichiometric right null space and the null space of A
c_{in}, c_{out}	Influent state and mechanistic reference effluent state
$c_{raw}, \tilde{c}$	Raw component prediction and corrected component prediction
I_{comp}	Fixed component-to-reported-composite map
D_i	Hydraulic dilution rate for the stated reactor and flow basis
ω, e_{SO}	Net dissolved-oxygen transfer rate and dissolved-oxygen coordinate selector
u	Standalone-reactor operating input vector
ϕ, B, Γ, R	Second-order feature vector, driver coefficients, learned component coupling, and coupled-system matrix
$\hat{C}, \Phi$	Auxiliary fitted component states and training feature matrix
N	Number of observations in the stated training-size design
n_r, L	Numbers of biological reactor stages and clarifier layers
ϑ	Connected-plant control vector
q_P, q_C, q_E, q_U	Flow ratios for the reactor train, clarifier feed, overflow, and underflow relative to fresh flow
m, c_i, c_E, c_U	Mixer, reactor-stage, overflow, and underflow concentrations
g_E, g_U	Flow-weighted clarifier outlet component responses
χ, M_{cl}	Joint plant response and clarifier solids inventory
D_χ	Positive scaling matrix for the plant response
J	Dimensionless engineering objective

Symbols specific to a formulation are defined with their equations. A numerical

coordinate is interpreted using the unit of its corresponding component.

Chapter 1

Research Problem and Purpose

1.1 Activated sludge prediction as an engineering problem

Wastewater treatment protects receiving waters by removing or transforming substances carried by domestic and industrial discharges. The substances of concern include biodegradable organic matter, nutrients, and suspended particles. Their effects depend on the receiving environment and the conditions of exposure. Madhav et al. (2020) describe how pollutant sources connect to environmental and health effects. Lin et al. (2022) show why these effects cannot be reduced to a single measure of water quality. These concerns give treatment prediction a practical purpose. Engineers need to understand how operating choices affect several aspects of treated water at the same time.

The activated sludge process uses microbial communities to transform wastewater constituents. Organic matter can be converted into biomass and oxidation products. Nitrogen can move through ammonium, nitrite, nitrate, biomass, and dissolved nitrogen gas. Phosphorus can move between dissolved phosphate, microbial storage, and particulate forms. A change in one pathway can affect another through competition for oxygen or available carbon. Duan et al. (2022) discuss the range of microbial pathways involved in biological nitrogen removal. Wentzel et al. (1992) explain the connections among nitrogen removal and biological phosphorus removal. The process therefore requires a component representation that distinguishes substances with different roles.

The activated sludge model (ASM) family provides this representation through component balances and process rates (Henze et al., 1999, 2015). A model state contains more information than a few reported effluent measurements. Chemical oxygen demand (COD) measures oxygen equivalents associated with oxidizable material. Total nitrogen (TN), total phosphorus (TP), and total suspended solids (TSS) summarize other groups of constituents. These summaries are useful for treatment assessment, but several different component states can produce the same reported total. For example, two

effluents can have the same TN while one contains mainly ammonium and the other mainly nitrate. Their treatment implications differ because the underlying forms differ.

Operating conditions influence these transformations. Hydraulic retention time (HRT) relates reactor volume to the flow basis used in its definition. Aeration affects the availability of oxygen for biological reactions. In a connected plant, recycle and wasting also affect the material returned to the biological stages and the solids retained in the system (Henze et al., 2008). Reactor capacity, aeration, recycle, and wasting are consequently linked decisions. A prediction that represents only one effluent total cannot explain all of these connections.

Mechanistic simulation supplies a way to calculate the component response from specified inputs. Its equations express known reaction structure and hydraulic relationships. Repeated simulation can require substantial effort when an analysis explores many influent compositions and operating settings. Bhosekar and Ierapetritou (2018) explain how statistical surrogates can reduce the cost of repeated process evaluation. Durkin et al. (2024) develop this use in wastewater resource recovery. A surrogate learns an input–output relationship from simulated or measured observations and then approximates the response at new inputs. Its value depends on both the speed of this approximation and the properties of the state it returns.

1.2 The gap between approximation and a usable process state

Predictive accuracy describes agreement with a reference response. Physical admissibility describes compliance with stated physical requirements. Interpretability describes the ability to inspect how the model forms its prediction. Decision quality describes the performance of operating choices when they are assessed against the intended engineering problem. These properties are related, but each answers a different question.

A small average prediction error can hide a negative concentration in a small component. It can also hide a conservation error that is distributed across several components. Consider a hypothetical component whose reference concentration is 0.2 g m^{-3} . A prediction of -0.1 g m^{-3} has an absolute error of only 0.3 g m^{-3} , but it cannot describe a nonnegative concentration state. If larger components dominate an aggregate error measure, this defect can receive little attention. Conservation and nonnegativity therefore require direct assessment in the same component basis used by the mechanistic model (Hansen et al., 2024; Kircher & Votsmeier, 2025).

Conservation itself depends on the chosen system boundary. A biological reactor receives influent, discharges effluent, and may exchange oxygen with its environment. A plant also contains recycle streams and sludge withdrawal. Internal recirculation redistributes material without becoming a new external source. A valid conservation

equality must follow from the represented reactions, hydraulic terms, and external forcing. It cannot be inferred solely from the names of reported water quality measures (Henze et al., 2015; Sturm & Wexler, 2020).

Physical enforcement can be applied during learning or after prediction. A training penalty can encourage small violations, but a finite penalty does not establish exact compliance for every new input (Hansen et al., 2024; Raissi et al., 2019). An output correction can impose an equality directly, but that correction can still produce negative components unless nonnegativity is also included. Kircher and Votsmeier (2025) address these two requirements together through a correction of mechanistic kinetics surrogates. Their approach motivates a common enforcement procedure that can be applied to different activated sludge predictors.

An interpretable surrogate creates an additional opportunity. A structured regression can expose operating effects, influent effects, curvature, and interactions through named coefficient groups. Inspection becomes useful only when the meaning of those groups is clear. A coefficient in a raw regression is not automatically a causal effect or a derivative of the final corrected prediction (Brunton et al., 2016; Rudin, 2019). Physical correction can change the response and introduce boundaries at which the active constraints change. A useful interpretation must distinguish the learned regression from the complete prediction procedure.

The difficulty becomes greater when the surrogate supports optimization. An optimizer deliberately searches for favorable predicted behavior. It can select a point at which the surrogate is inaccurate even when average prediction error is acceptable. A projected state can satisfy imposed balances while missing the actual kinetic response of the plant. Approximation management and evaluation with the original mechanistic model are therefore necessary parts of decision assessment (Alexandrov et al., 1998; Pedrozo et al., 2025). Faster evaluation alone does not establish a better operating decision.

1.3 Research questions and objectives

The research examines how physical enforcement can support machine learning surrogates of activated sludge treatment without obscuring the difference between model agreement and engineering usefulness. Its central question concerns the conditions under which a fast statistical prediction can remain physically admissible, understandable, and suitable for operating analysis.

The first question concerns approximation of the complete reactor component response. It asks how different statistical learning methods behave when the available training data change and when predictions are requested beyond the sampled domain. The corresponding objective is to establish a common benchmark in which the prediction target, data partitions, and evaluation scales are defined independently of the model

family.

The second question concerns post prediction enforcement. It asks how a common correction can impose conservation and nonnegativity, and how correction affects component error, composite error, displacement, and computational cost. The objective is to evaluate raw and corrected states in paired comparisons. Feasibility is treated as a directly checked property rather than inferred from an error score.

The third question concerns an interpretable regression structure. It asks how second-order influent and operating terms can be coupled with invariant-aware fitting and checked deployment. The objective is to develop a transparent component predictor, explain the limits of its fitting problem, and separate interpretation of its raw response from interpretation of its final corrected output.

The fourth question concerns the connected treatment plant. It asks how a joint surrogate can preserve mixer, reactor, clarifier, and inventory relationships while supporting an operating search. The objective is to formulate common engineering priorities and compare surrogate-selected controls with direct mechanistic optimization. Both routes require independent mechanistic assessment at their selected controls.

1.4 Research logic and significance

The research logic begins with a mechanistic representation and a declared operating domain. Paired input and component responses provide the information needed to learn a statistical approximation. Reaction structure supplies the conservation relations. Nonnegativity defines an additional requirement on concentration states. Together, these elements permit separate assessment of approximation error and physical compliance.

This logic extends to interpretation and decision support. A structured predictor makes particular terms available for inspection. A checked prediction procedure makes stated constraints available for direct verification. A connected plant formulation makes hydraulic and inventory consistency available for assessment. An operating search then selects controls, and mechanistic evaluation determines their treatment and resource consequences. Each step provides evidence for a specific claim. No step substitutes for the evidence required by another.

The scientific significance lies in relating statistical approximation to the component space of biological wastewater treatment. The methodological significance lies in using shared data and paired diagnostics to distinguish the effect of learning from the effect of physical correction. The engineering significance lies in connecting surrogate predictions to feasible treatment states and to decisions that can be evaluated against the mechanistic reference. These purposes follow the broader use of surrogate models in process analysis and optimization (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019).

1.5 Scope and limits of the claims

The study concerns simulated steady-state activated sludge responses. It covers a standalone reactor and a connected plant with a reactor train, secondary clarification, and recycle streams. The mechanistic model supplies reference responses under fixed process definitions and specified parameter values. The standalone reactor and the connected plant use different operating domains because their hydraulic and solids relationships differ. These domains are stated separately in the methods.

Agreement with simulated data establishes agreement with the selected mechanistic model. It does not establish agreement with an operating treatment plant. Calibration uncertainty, sensor error, seasonal change, microbial adaptation, and unrepresented reactions can affect field performance (Machado et al., 2014; Newhart et al., 2019). The physical guarantees concern the specified balances, component bounds, and inventory relationships. They do not imply complete kinetic, thermodynamic, or dynamic feasibility.

The numerical examples used to explain conservation, correction, and decision assessment are hypothetical. They illustrate the logic of the formulations and do not report experimental or simulation findings. Empirical model rankings, treatment outcomes, optimization outcomes, and computational performance are outside the present presentation.

Chapter 2

Integrated Review of Related Literature

2.1 Treatment mechanisms and the meaning of model outputs

Water quality is shaped by pollutant sources, transport, exposure, and the receiving environment. Madhav et al. (2020) organize pollutants by their sources and effects, while Lin et al. (2022) examine the different health consequences associated with polluted water. These works support a treatment assessment that considers several pollutant groups. They also explain why reducing a single effluent quantity cannot describe every environmental consequence of a treatment decision.

Biological treatment makes this assessment a problem of interacting transformations. Duan et al. (2022) review heterotrophic nitrification and its role within biological nitrogen removal. The review shows that a process label can contain several microbial pathways. Wentzel et al. (1992) connect nitrification, denitrification, and excess biological phosphorus removal through process conditions and model structure. For a surrogate, the implication is that nitrogen and phosphorus predictions belong to a coupled treatment response. Learning each reported total independently can omit relationships that matter when operating conditions change.

The ASM family provides a systematic framework for these relationships. Henze et al. (1999) describe the component and process structure of the model that includes biological phosphorus removal and denitrification by phosphorus-accumulating organisms. The collected models of Henze et al. (2015) place this development alongside other activated sludge formulations. Henze et al. (2008) provide the broader treatment principles needed to relate biological models to engineering design. These sources anchor the separation among the component state, the reaction model, and reported treatment indicators.

Carbon availability provides a concrete example of coupling. Guerrero et al. (2011) examine competition between phosphorus-accumulating organisms and denitrifiers

under different carbon sources. Their work makes the identity and availability of carbon relevant to nutrient removal, rather than treating organic loading as a single interchangeable number. A component surrogate should therefore retain distinctions among fermentable material, readily available substrates, storage products, and biomass. The inclusion of interaction terms has a process rationale because the effect of one input can depend on another.

Oxygen supply has a similar role. Stenstrom and Song (1991) study the effect of oxygen transport limitations on nitrification. This distinguishes oxygen transfer from the biological capacity to consume oxygen. A model can include both and still give different responses when their relative limitations change. An educational explanation of aeration should therefore identify what is controlled, how oxygen enters the balance, and which reaction rates depend on dissolved oxygen.

The state of the biomass also depends on its operating history. Yilmaz et al. (2007) examine how alternating aerobic and anoxic or anaerobic conditions affect biomass activity during starvation. Their setting differs from a steady-state surrogate, but it identifies a limit of that surrogate. A mapping from current influent and operating conditions to one steady-state response does not represent all effects of prior exposure or prolonged substrate deprivation. Such evidence supports a precise boundary on claims about dynamic operation.

Model detail creates a calibration problem as well as a prediction problem. Machado et al. (2014) compare parameter identifiability with uncertainty in influent and operating conditions during full-scale model calibration. A close match to measurements does not uniquely identify every kinetic or stoichiometric parameter. This matters when interpreting a fitted surrogate. The surrogate approximates a response conditioned on the parameter values used to generate its training data. Its coefficients cannot be taken as newly identified biological constants.

2.2 Why treatment analysis calls for surrogate models

Treatment design often contains competing objectives and coupled decisions. Padrón-Páez et al. (2020) use multiobjective optimization to address sustainable wastewater treatment plant design. Aboagye et al. (2021) link wastewater network design to optimization and sustainability assessment. Their contributions show why treatment evaluation involves repeated comparisons among alternatives. The computational burden belongs to the decision problem as well as to an individual simulation.

Sensitivity analysis gives another reason for repeated evaluation. de Araujo et al. (2013) examine optimal operation and the selection of economic controlled variables in an activated sludge model. The work connects operating sensitivities to the quantities that are useful for control. A fast approximation can assist this analysis, but it must

retain the correct input definitions and reporting basis. A sensitivity with respect to reactor volume is different from a sensitivity with respect to a residence time that uses a recycle-adjusted flow.

The review of Bhosekar and Ierapetritou (2018) treats surrogate construction, feasibility analysis, and optimization as connected tasks. It emphasizes the importance of sampling, model selection, and validation for the intended application. McBride and Sundmacher (2019) place surrogate modeling within chemical process engineering and distinguish different ways of approximating an expensive response. These reviews support a broad comparison of learning methods. They do not imply that one statistical family is best for every response or data regime.

Fang et al. (2011) develop a simulation-based integrated approach to biological nutrient-removal optimization in a full-scale wastewater treatment plant. Their contribution links operating choices to the coupled nitrogen and phosphorus response through a process model. It also identifies the reference calculation that a statistical surrogate seeks to reduce. The surrogate can approximate repeated evaluations, but the original component and treatment relationships remain the basis for assessing an operating choice.

Wastewater resource recovery provides a direct application. Durkin et al. (2024) combine process surrogates with optimization of resource recovery from wastewater. The statistical approximation is useful because it participates in a larger constrained decision problem. He et al. (2023) use a Kriging surrogate in a papermaking wastewater treatment application with environmental objectives. Both studies connect faster response evaluation to engineering choice. Their relevance is methodological even when the treatment processes and outputs differ from an activated sludge component surrogate.

Recent network work extends this concern across treatment stages. Tang et al. (2026) develop surrogate-based design for a multistage coal chemical wastewater treatment network. A network surrogate must preserve the meaning of the connections among units as well as the response of each unit. Tsochatzidi et al. (2025) develop a surrogate optimization framework for pharmaceutical process systems. That work is outside wastewater treatment, but it illustrates the same distinction between approximation of process responses and optimization over the resulting model. Transfer to activated sludge requires the treatment-specific component and hydraulic constraints.

Reduced models can also be chosen around quantities that matter to operation. Straus and Skogestad (2018) generate surrogates using self-optimizing variables. Their contribution concerns the connection between reduced representation and decision purpose. A surrogate that is adequate for one operating objective can be inadequate for another. This supports retaining complete component predictions when the later objective may combine nutrients, organic matter, solids, and resource use.

2.3 From local approximations to nonlinear data-driven prediction

Local approximation has a clear engineering role. M. Li et al. (2020) use fuzzy clustering in dissolved oxygen model predictive control, while T. Yang et al. (2014) develop fuzzy model-based predictive control for the activated sludge process. Such approaches organize nonlinear behavior through local models or operating regimes. Their usefulness depends on how well the local representation covers the conditions encountered in operation. They provide an alternative to fitting one highly flexible global predictor.

Polynomial and response surface methods make nonlinear terms explicit. Kim et al. (2009) develop a dual optimization strategy for nitrogen and phosphorus removal. Lim et al. (2011) examine operating conditions for organic matter and nitrogen removal in a membrane bioreactor. Ahn et al. (2014) compare modeled and experimental operating factors for livestock wastewater treatment. These works demonstrate why transparent approximations remain useful in treatment analysis. The terms have an inspectable structure, but the selected polynomial degree and sampled domain limit the behaviors they can represent.

Nazlabadi et al. (2021) review the application of response surface methodology to biological wastewater treatment. The review places polynomial approximations within a wider design and analysis practice. A low-order surface can reveal curvature and input interaction without reproducing every mechanistic detail. It should therefore be assessed as an approximation over a stated domain. This perspective supports second-order regression while preventing a claim that a quadratic structure is a complete reaction model.

Partial least squares provides another transparent starting point. Wold et al. (2001) explain how latent variables relate predictors to responses in chemometric regression. Khatri et al. (2021) review its use in water quality analysis. The method is relevant when influent variables are correlated and when several outputs are modeled together. Its basic linear relationship still depends on whether the predictor representation captures the nonlinear behavior of interest.

Extensions address this limitation in different ways. Woo et al. (2009) use kernel partial least squares to estimate process variables in industrial wastewater treatment. Liu et al. (2021) use a dynamic concurrent kernel approach for wastewater process monitoring. C. Yang et al. (2021) embed a relevance vector model within a partial least squares framework for adaptive effluent prediction. These contributions show that latent structure, nonlinearity, and changing process behavior can be combined. They also separate dynamic monitoring from the steady-state input–output task considered here.

The wider evidence on wastewater data use is reviewed by Newhart et al. (2019).

Their discussion makes data quality, process knowledge, and validation important parts of performance analysis. Ba-Alawi et al. (2021) address influent sensor validation through stacked denoising autoencoders. Their contribution concerns the credibility of the inputs before a prediction is made. An accurate model cannot compensate for every error in a sensor signal or a mislabeled input quantity.

Bagherzadeh et al. (2021) examine total nitrogen prediction and feature selection in a wastewater treatment plant. Sadri Moghaddam and Mesghali (2023) develop a hybrid ensemble for total nitrogen prediction in a combined trickling filter and activated sludge system. These studies demonstrate the value of statistical prediction for an operational indicator. Their output target also clarifies a limitation. Evidence for one total nitrogen model does not establish conservation or nonnegativity of the complete component vector.

Recent studies connect prediction to other practical indicators. Wang et al. (2025) compare interpretable learning models with a mechanistic model for effluent nitrogen prediction. S. Yang et al. (2025) examine waste activated sludge generation for operational efficiency. Xiao et al. (2026) combine mechanistic dynamics and an interpretable surrogate in activated sludge management. Together, these applications broaden the role of learning from effluent estimation to sludge and operating analysis. Their different targets require careful comparison because an error measure over one indicator is not an error measure over the full state.

Asadi et al. (2017) use data mining in aeration optimization, and Mao et al. (2024) use artificial intelligence to balance effluent quality and aeration energy. These works give a concrete decision context for prediction. Oxygen delivery can support treatment while consuming resources. A useful operating surrogate must represent this relationship well enough at the selected decision, rather than only on average over randomly held-out inputs.

2.4 Learning families and what each comparison can reveal

Different learning methods offer different ways to approximate the same response. Hastie et al. (2009) provide the statistical foundations for regularization, ensembles, kernels, and model assessment. Shwartz-Ziv and Armon (2022) examine tabular data and caution against assuming that deep learning is necessary for every structured dataset. Borisov et al. (2024) review deep neural networks for tabular learning. These contributions motivate a comparison that includes simple, nonlinear, and neural methods under the same prediction target.

Tree ensembles form one important group. Breiman (2001) develop random forests by aggregating randomized trees. Averaging can reduce the instability of a single tree while retaining nonlinear input partitioning. Geurts et al. (2006) introduce extremely

randomized trees with stronger randomization of split choices. Their different construction makes both methods useful in a benchmark. Neither construction imposes a coupled material balance across outputs unless that relation is added explicitly.

Boosting builds an ensemble sequentially. Drucker (1997) adapt boosting to regression by increasing attention to difficult observations. Chen and Guestrin (2016) develop extreme gradient boosting (XGBoost) with regularized tree boosting and efficient treatment of large datasets. Ke et al. (2017) develop the light gradient boosting machine (LightGBM) with methods for efficient gradient boosting. Prokhorenkova et al. (2018) develop CatBoost with an ordered approach that addresses prediction shift and categorical variables. The component data considered here are numeric, but these algorithms still provide different nonlinear fitting procedures. Their names do not establish physical admissibility or superiority in advance.

Neighbor and kernel methods give complementary comparisons. Mack (1981) study the local properties of nearest-neighbor regression estimates. The prediction is strongly connected to the distribution of nearby training inputs. This makes domain distance relevant when the requested input moves outside the sampled range. Smola and Schölkopf (2004) explain support vector regression and its control of model complexity through an error-insensitive fitting objective and regularization. A kernel can represent nonlinear relationships, but the learned relationship still requires separate assessment in an extrapolation regime.

Multiresponse regularization addresses structure across outputs. Zou and Hastie (2005) combine shrinkage and variable selection through the elastic net. Yuan and Lin (2006) develop regression selection for grouped variables. Argyriou et al. (2008) treat feature learning across several tasks. Liu et al. (2009) provide blockwise procedures for the multi-task lasso, and Simon et al. (2013) develop blockwise descent for group-penalized multiresponse regression. These works justify comparisons that can share information across component outputs. A shared sparsity pattern is a statistical relation and should not be mistaken for stoichiometric conservation.

Neural regression offers a flexible function representation. Rumelhart et al. (1986) develop learning through back propagation, and Hornik et al. (1989) establish approximation properties of multilayer feedforward networks. Such approximation results concern representational capacity under stated conditions. They do not provide an automatic guarantee about finite-data accuracy, optimization success, extrapolation, or material balance. Curteanu and Cartwright (2011) discuss network topology in chemical applications, making architecture selection part of the modeling problem.

Arik and Pfister (2021) introduce TabNet with sequential attention over tabular inputs. Its feature masks can support inspection of which inputs are used. Attention-based selection still differs from a named algebraic coefficient or a conserved component balance. It is useful to compare such a model with an explicit second-order regression because the two forms of interpretability answer different questions.

2.5 Learning behavior, sampling, and fair assessment

Training error alone gives incomplete evidence of approximation quality. Geman et al. (1992) analyze the bias and variance dilemma in neural learning. A flexible model can reduce systematic approximation error while becoming more sensitive to the training sample. Belkin et al. (2019) show that modern learning behavior can be more complex than a single classical tradeoff curve. These contributions support direct learning-curve assessment rather than assigning a fixed sample-efficiency ranking to a model family.

Viering and Loog (2023) review the shapes of learning curves and the conditions that influence them. The curve relates available training data to predictive performance under an explicit evaluation design. Its interpretation depends on the error scale, spacing of sample sizes, and variability of the data partitions. A normalized area under such a curve is a useful summary only when those choices are common across the compared models.

Training inputs must also cover the intended process domain. McKay et al. (1979) compare sampling procedures and motivate stratified coverage of input ranges. Latin hypercube sampling can spread observations across each selected input range without enumerating every combination. It does not guarantee equal coverage of every joint response regime, nor does it make independently sampled influent components representative of every real wastewater composition. These limitations need to remain visible when simulated data define the benchmark.

Active learning raises a related question about the cost of obtaining informative targets. Guyon et al. (2011) organize an active learning challenge in which labels are limited. Cawley (2011) provide simple regularized baseline strategies and emphasize the value of comparing an elaborate selection method with simpler alternatives. Their contribution is relevant to simulator data because each reference response also has a cost. The present learning-curve design measures dependence on available labels. It does not claim to implement or validate an active selection strategy.

Hyperparameter selection is part of model fitting. Bergstra et al. (2011) develop model-based methods for searching hyperparameter spaces. Akiba et al. (2019) provide a framework for organized hyperparameter optimization. These methods can improve the use of a search budget, but they require separation between the observations used to select settings and the observations used to assess the selected model. The same requirement applies when choosing a correction threshold or a trust criterion.

Kaufman et al. (2012) formalize leakage in data mining and explain how information that should be unavailable can enter the fitted model. In this setting, scaling all observations before cross-validation would allow validation inputs to influence training transformations. Selecting hyperparameters against an outer assessment set would

allow the same set to guide and judge the model. Nested partitioning and training-only transformations are therefore part of the evidence design rather than minor computational details.

Pinheiro et al. (2025) examine how feature scaling affects learning tasks. Their contribution supports an explicit account of which methods use scaling and how those scales are fitted. Scaling also changes the numerical meaning of a coefficient. A standardized second-order term must be transformed back to physical input units before an engineering interpretation is attempted.

Comparison must account for variation across assessment problems. Demšar (2006) develop statistical comparisons for classifiers over multiple datasets. The present application concerns regression and repeated data partitions, so its recommendations require adaptation to the unit of independent evidence. Shevchenko et al. (2024) examine benchmarking variability in recommender systems. Their setting is also different, but the concern transfers. Reported rankings can depend on splits and evaluation choices. Paired model comparisons and transparent variability are more defensible than a universal ranking inferred from one split.

Error normalization deserves its own explanation. Tofallis (2015) examine relative prediction accuracy and the problems created by some relative measures. Near-zero component concentrations make ordinary percentage error unstable. A small absolute error can become an enormous percentage or remain undefined at zero. Componentwise normalization using a declared training reference provides a different comparison basis. It does not eliminate the need to report physically meaningful absolute errors and constraint violations.

Shahbazi et al. (2026) address imbalanced regression through data-level and algorithm-level balancing. This concern is relevant when rare operating conditions or low residual concentrations matter more to a decision than their frequency suggests. An average error over the sampled domain can conceal such behavior. Component-resolved errors and selected-decision evaluation are therefore needed alongside aggregate scores.

2.6 Physical constraints in scientific machine learning

Scientific machine learning combines data-based approximation with knowledge of physical structure. Karniadakis et al. (2021) review this combination across several forms of physics-informed learning. Kashinath et al. (2021) discuss applications in weather and climate modeling. These reviews show that physics can enter through inputs, architecture, training, or outputs. The practical question is which relation is enforced, at what stage, and under what conditions.

Raissi et al. (2019) develop physics-informed neural networks that include differential equation residuals in a learning objective. This approach can use governing equations

even when direct response data are limited. A finite residual penalty still permits a tradeoff with the data-fitting term and other penalties. Exact conservation of every returned component state requires an additional structural argument or an explicit feasibility procedure.

Wastewater-specific work makes the same distinction useful. H. Li et al. (2024) apply physics-informed learning to unit operations and processes, including an activated sludge reactor. The governing-equation information supports the approximation task. Its inclusion in training does not automatically establish every plant-wide hydraulic or clarifier balance. Those relations must be defined at the level where the coupled outputs are returned.

Architectural enforcement supplies one stronger route. Beucler et al. (2021) impose analytic constraints in neural models by arranging the output relationship so that the required equalities hold. Some outputs can be reconstructed from others to complete a constrained state. This depends on which quantities are independent and whether the reconstructed quantities remain within their allowed bounds. Equality completion alone does not settle nonnegativity.

Sturm and Wexler (2020) formulate a conserving approach for photochemical computations. Sturm and Wexler (2022) embed atom balance in a neural architecture through the stoichiometric relationship. These contributions connect learned transformations to reaction structure. Sturm et al. (2023) use mass-conserving dimensionality reduction for transport of organic aerosol. Their scaling can also maintain nonnegative concentrations in the reduced representation. This provides a precedent for addressing conservation and nonnegativity together while reducing the number of transported quantities. Its transport setting differs from steady-state effluent correction. In activated sludge, an analogous invariant construction must respect the native component basis and the external forcing terms.

Reconciliation provides another route from a statistical estimate to a consistent state. Hansen et al. (2024) develop learning models that respect conservation through a probabilistic reconciliation perspective. Conservation supplies information that can be combined with the learned prediction. This differs from assuming that a low training loss has already made the prediction conservative. The connection to activated sludge is the availability of known linear inventory relations. The probabilistic interpretation remains distinct from a deterministic correction that returns one state without a calibrated uncertainty distribution.

Ruckstuhl et al. (2021) learn a correction for a data assimilation procedure from the difference between an unconstrained analysis and a more expensive constrained analysis. The training target contains information about the correction that the physical constraints require. This is a different task from learning a reactor effluent directly from influent and operating conditions. It also illustrates a separate computational choice. One can solve a constrained problem at prediction time or train another

model to approximate its correction. If the correction is approximated, the returned state still requires a feasibility check. The information carried by constrained training examples should not be confused with exact enforcement of the constraints on every new prediction.

Geiss et al. (2022) address conservation when a neural model converts coarse atmospheric chemistry fields to a finer spatial grid. Their formulation accounts for unequal grid-cell areas and consistency between coarse inputs and fine outputs. It also treats the normalization of quantities whose distributions span large concentration ranges. This application clarifies why a physical operator must match the representation on which it acts. A spatial average needs area weights, while a treatment inventory needs native-unit constituent weights and a specified boundary. Directly borrowing a conserving layer without transforming those weights would enforce a different relation. The wastewater application therefore adopts the principle of explicit physical accounting while deriving its own component operator.

Harder et al. (2022) emulate aerosol microphysics by learning changes in state variables from mechanistic input–output pairs. They compare physical information in soft loss terms with completion and correction layers, and examine positivity enforcement. Their tendency target gives an additional lesson. A negative change can be physically meaningful when it represents consumption, even though the resulting concentration must remain nonnegative. The sign requirement belongs to the appropriate quantity. In activated sludge, driver coefficients and reaction-induced changes can have either sign, while the returned component state is subject to nonnegativity. These distinctions help specify what a constrained surrogate actually guarantees across different scientific tasks.

Output positivity is another physical requirement. Haddad et al. (2010) provide a mathematical treatment of nonnegative and compartmental dynamical systems. In a concentration model, nonnegativity belongs to the state description itself. Clipping a negative value can satisfy this bound while changing the total represented material. A correction that enforces conservation alone can redistribute material into a negative component. Simultaneous enforcement must therefore handle both requirements together.

Hard constraint layers also raise questions about expressive capacity and uncertainty. Min et al. (2024) study an input-dependent enforcement layer for affine and convex output constraints with neural approximation guarantees. These guarantees depend on the stated architecture and assumptions. They do not establish the approximation capacity of a finite second-order regression. Utkarsh et al. (2025) develop a probabilistic projection that uses prediction-variance information to impose hard constraints and assess uncertainty. These methods incorporate feasibility into the learning architecture or prediction distribution. The deterministic corrections used here have different weighting choices and do not inherit a probabilistic uncertainty guarantee.

Lakshminarayanan et al. (2017) develop deep ensembles for predictive uncertainty estimation. Spread among models can help describe uncertainty in a prediction. It does not certify a component balance or establish the validity of an extrapolated state. Likewise, a deterministic projection does not by itself produce a calibrated uncertainty distribution. Physical admissibility and uncertainty assessment are separate parts of model credibility.

2.7 Projection and the geometry of admissible states

Projection restricts a completed prediction to a set defined by physical relations. Sturm and Silva (2025) use a species-weighted correction for atom conservation in atmospheric composition. The weighting matters because species can differ greatly in concentration. An equal absolute adjustment can be small for a major component and large for a trace component. This motivates an explicit choice of correction geometry rather than treating every conservation correction as equivalent.

Valente et al. (2025) project predictions onto physical manifolds defined by conservation laws. Their work gives output correction a general geometric interpretation. A predicted point is moved to an admissible set according to a selected notion of distance. The interpretation of the correction depends on the distance, the set, and the existence of a feasible state. A closer point in one weighted distance need not have a smaller error in another evaluation metric.

Kircher and Votsmeier (2025) combine conservation and positivity for mechanistic kinetics surrogates. The procedure uses a positive provisional prediction to define relative weighting, obtains a conservative candidate, and restores nonnegativity through interpolation with an admissible reference when needed. This construction can be applied after different statistical predictors have been fitted. It is valuable for isolating the effect of physical enforcement from the effect of changing the learner.

The reference state has a practical meaning. If the influent is nonnegative and has the required conserved inventories, it lies in the feasible set for the standalone reactor invariant constraints. Interpolating toward this state can restore nonnegativity without changing those inventories. The resulting state is physically admissible under the imposed constraints. It need not reproduce the reference reactor kinetics. This distinction prevents a guarantee about conservation from becoming an unsupported claim about treatment accuracy.

The geometry also limits claims about accuracy improvement. Projection onto a closed convex set in a fixed Euclidean norm has useful distance properties when the true state lies in that set (Boyd & Vandenberghe, 2004). A relative-weighted conservative candidate followed by boundary interpolation is a different map. Its weights can depend on the prediction, and its final interpolation is not generally the nearest feasible point under the reported error metric. Paired empirical assessment is therefore necessary

even when feasibility follows from the construction.

2.8 Interpretable constrained regression

Rudin (2019) argue for interpretable models in high-consequence settings when such models are available. The argument concerns the ability to inspect the model itself, rather than relying only on an explanation added afterward. For treatment surrogates, an explicit regression can identify operating terms, loading terms, nonlinear terms, and component coupling. This creates a structure that an engineer can inspect in relation to the process representation.

Brunton et al. (2016) identify governing relationships through sparse candidate functions in dynamical systems. Their contribution shows how named terms can connect learned structure to scientific interpretation. A second-order treatment surrogate uses named terms for a different purpose. It approximates a steady-state response and does not claim to discover the biological reaction laws. Similar predictions can arise from different coefficient combinations when inputs are correlated or when several blocks compensate for one another.

Shen et al. (2023) discuss differentiable modeling that joins physical and learned components. This supports inspection of the entire prediction chain. For a coupled regression, the coefficients of the driver are transformed by the coupling solution. For a corrected model, the final output also depends on the active constraints. An explanation that ignores these transformations describes only part of the returned prediction.

The published formulation of Selerio (2026) supplies a specific activated sludge example. It combines an invariant-aware second-order regression with a checked output procedure that imposes conservation and component nonnegativity. The important theoretical distinction is between the unconstrained regression response, the fitted state variables used during training, and the final deployed state. Soft invariant-aware fitting guides the regression. The checked correction establishes the stated final-state constraints. These are different mechanisms and require different diagnostics.

Numerical optimization underlies both mechanisms. Boyd and Vandenberghe (2004) provide the convex optimization foundations for affine constraints and quadratic objectives. Razaviyayn et al. (2013) analyze block successive minimization under specified regularity conditions. A regression objective that jointly fits coupling and state variables can be nonconvex even when individual updates are convex. Decreasing the objective at each accepted update does not establish a globally optimal fit. Conditioning restrictions also need their own treatment because they are not ordinary box constraints.

Stellato et al. (2018) present an operator-splitting approach for quadratic programs, and Stellato et al. (2020) give its fuller mathematical and computational development. Huangfu and Hall (2018) develop a parallel dual revised simplex method for linear programming. These works support the numerical treatment of regression subproblems

and output corrections. Their role is to solve specified optimization problems. Numerical termination still needs explicit residual and bound checks before a predicted state is accepted.

2.9 Connected plants and the move from prediction to decisions

A connected activated sludge plant contains more physical structure than one reactor. Härtel and Pöpel (1992) develop a dynamic secondary clarifier model with sludge thickening. TAKACS (1991) model clarification and thickening through a settling relationship. Jeppsson and Diehl (1996) examine the propagation of biological components through the secondary clarifier. These contributions explain why clarification must preserve component relationships and why overflow, underflow, and stored solids cannot be predicted as unrelated quantities.

Alex et al. (2008) define a benchmark simulation setting that connects biological treatment, clarification, and recirculation. Its value is a common process context for comparing operating strategies. A plant with an extended component basis remains distinct from that benchmark, but can use the same discipline of declared hydraulics, unit connections, and assessment conditions. In particular, a recycle stream affects both concentration mixing and the flow passing through each reactor.

Solids retention adds another connection. Leu et al. (2012) examine the implications of long solids retention time for energy use, effluent quality, and stability. The retained solids depend on both reactor and clarifier inventories and on external solids losses. A surrogate that predicts effluent quality accurately but neglects inventory can therefore misrepresent a solids-age constraint. Miederer et al. (2025) develop an energy management model for wastewater treatment plants, further connecting operating choices to resource use.

The U.S. Environmental Protection Agency (2009) review provides engineering guidance for nutrient control technologies. Its role is to identify treatment considerations and design relationships. Site-specific effluent limits and safety thresholds still belong to the declared optimization problem. They should be supplied as explicit inputs rather than presented as universal regulatory limits.

Control studies show how a prediction model enters operation. Sadeghassadi et al. (2018) use neural models for setpoint design and predictive control. Bernardelli et al. (2020) combine a learning model with real-time predictive control of a wastewater treatment plant. Ren et al. (2026) study ammonium-based aeration scheduling under influent disturbances. Croll et al. (2023) review reinforcement learning for wastewater process control. These contributions demonstrate a range of operating applications. Their dynamic decisions remain distinct from a steady-state operating search, which evaluates equilibrium responses under fixed controls.

Joint output projection offers a way to preserve the specified connections in a statistical plant model. Mixer equations, reactor invariants, clarifier outlet balances, and inventory bounds can be imposed together. At fixed controls, the relevant relations can form a convex quadratic correction problem (Boyd & Vandenberghe, 2004; Stellato et al., 2020). The projection can remove contradictions among separately predicted unit responses. It does not impose every kinetic rate equation or every layerwise settling equation of the mechanistic plant.

This creates an optimization problem with a nested prediction procedure. Colson et al. (2007) review bilevel optimization, which helps explain the distinction between an outer control choice and an inner constrained response calculation. The plant correction has a strongly convex inner objective when the weights are positive and the feasible set is nonempty. Tøndel et al. (2003) analyze parametric quadratic programming and its relation to piecewise solution structure. Their results motivate attention to active constraints, but an operating problem with control-dependent balance coefficients need not have the same globally piecewise-affine structure.

2.10 Managing approximation during operating searches

An optimizer can exploit approximation error in a favorable region. Alexandrov et al. (1998) develop a trust-region framework for managing approximation models. The central idea is to judge the approximation where it is being used and control the search accordingly. Sahigara et al. (2012) compare applicability-domain methods for quantitative structure–activity models. Their application differs from treatment modeling, but it supports the general need to distinguish nearby training support from distant prediction requests.

Distance restrictions and correction size answer different questions. An input can be near the training observations while its raw output requires a large physical correction. Conversely, a distant input can produce an apparently admissible state. Both indicators can be useful screening criteria, but neither is a calibrated probability of prediction accuracy. Hameed et al. (2026) develop trust-region filter methods for gray-box optimization with derivative information and feasibility considerations. Their contribution reinforces the need to separate approximation agreement, feasibility, and progress in the original objective.

Direct optimization provides a meaningful comparator only when its numerical formulation is explicit. Buzzi-Ferraris and Manenti (2013) explain nonlinear systems and optimization in chemical engineering. Kraft (1988) develop sequential quadratic programming, while Wächter and Biegler (2006) develop an interior-point filter line-search method for nonlinear programming. These approaches support different numerical routes for the operating problem. A local termination condition does not establish the

global optimum of a nonconvex plant model.

The underlying steady-state solution also needs attention. Curtiss and Hirschfelder (1952) establish the problem of stiff differential equations. Biological treatment contains processes with different response scales, which can make steady-state initialization difficult. Kelley and Keyes (1998) analyze pseudo-transient continuation as a route toward a steady solution. Time relaxation followed by an algebraic solve can provide a better initial state, but acceptance still depends on checking the original equations and component bounds.

Audet and Dennis (2006) develop mesh adaptive direct search for constrained optimization. Ragonneau (2022) study model-based derivative-free methods. These works explain why an operating search can remain possible when the projected prediction map is nonsmooth or reliable derivatives are unavailable. The comparison with a smooth direct formulation should account for the differences in numerical structure and stopping rules.

Blik et al. (2023) benchmark surrogate optimization algorithms on expensive black-box functions. Their contribution makes assessment design and evaluation cost central to a performance comparison. Pedrozo et al. (2025) compare surrogate optimization with a reference process optimization setting. Both support evaluating the objective and constraints at the selected decisions using the original model. Search time, development time, and verification time should be reported separately because they answer different cost questions.

Zhang et al. (2026) develop an agile benchmarking framework for wastewater resource recovery technologies. Their emphasis on comparable assessment strengthens the need for common engineering priorities and clear process boundaries. Route-specific numerical scales can help solve the equations, but both routes must ultimately be judged using the same physical outputs and objective definition.

2.11 The integrated research gap

The literature provides strong foundations for component modeling, nonlinear statistical prediction, physical constraint enforcement, interpretable regression, and surrogate-assisted optimization. It also shows why these foundations cannot be reduced to one error score. A component surrogate must represent the response that is being assessed. A correction must enforce the relations that are being claimed. An interpretation must describe the prediction map that is actually returned. An operating decision must be evaluated against the original engineering problem.

The research gap concerns their integration in activated sludge analysis. A common reactor benchmark can compare raw and corrected predictions across learning families. A structured regression can make influent, operating, and coupling terms inspectable while preserving a separate feasibility procedure. A connected plant formulation can extend

enforcement to mixing, treatment stages, clarification, and solids inventory. Independent mechanistic evaluation can then assess the selected controls. This sequence supplies a coherent basis for studying predictive accuracy, physical admissibility, interpretability, and decision quality without treating any one of them as evidence for all the others.

Chapter 3

Mechanistic Foundations of Physically Admissible Prediction

3.1 The component state as the object of prediction

An activated sludge reactor changes the form and distribution of material. Organic substrate can become biomass, ammonium can become nitrite and nitrate, and soluble phosphate can enter biological storage or a precipitate. A description of treatment therefore needs more than one aggregate effluent concentration. The ASM framework separates material into components and connects their changes through biological and chemical processes (Henze et al., 2015). This separation provides the physical meaning of a surrogate prediction. The surrogate estimates a complete component state from the influent state and the operating conditions.

The mechanistic representation is Activated Sludge Model No. 2d with two-step nitrification (ASM2d-TSN). It combines the phosphorus-removal framework of Henze et al. (2015) with the nitrogen-transformation formulation used by Guerrero et al. (2011). The state contains 20 components and the reaction description contains 28 processes. Ten components are dissolved and ten are particulate. Their fixed ordering is

$$c = (S_O, S_F, S_A, S_{NH_4}, S_{NO_2}, S_{NO_3}, S_{N_2}, S_{PO_4}, S_I, S_{ALK}, X_I, X_S, X_H, X_{PAO}, X_{PP}, X_{PHA}, X_{AOB}, X_{NOB}, X_{MeP}, X_{MeOH})^T. \quad (3.1)$$

Every concentration is represented by its numerical value in the native unit assigned to that component. A vector of these values combines several constituent bases. An organic component expressed as g COD m⁻³ is not expressed on the same material basis as ammonium expressed as g N m⁻³. The coefficients of every physical operator must account for this distinction.

The symbols in Table 3.1 identify state variables. They do not introduce additional process abbreviations. For example, X_{AOB} is the concentration coordinate assigned to ammonia-oxidizing biomass. The numerical constituent contents and solids conversion factors are fixed assumptions of the adopted reporting map. They are not estimated from

Table 3.1: Component meanings, native units, and fixed coefficients for the four reported water quality quantities. Each coefficient converts the native component basis to the corresponding reported basis.

Component	Meaning	Native unit	COD	TN	TP	TSS
S_O	Dissolved oxygen	$\text{g O}_2 \text{ m}^{-3}$	0	0	0	0
S_F	Fermentable substrate	g COD m^{-3}	1	0	0	0
S_A	Acetate and fermentation products	g COD m^{-3}	1	0	0	0
S_{NH4}	Ammonium nitrogen	g N m^{-3}	0	1	0	0
S_{NO2}	Nitrite nitrogen	g N m^{-3}	0	1	0	0
S_{NO3}	Nitrate nitrogen	g N m^{-3}	0	1	0	0
S_{N2}	Dissolved dinitrogen	g N m^{-3}	0	0	0	0
S_{PO4}	Soluble phosphate	g P m^{-3}	0	0	1	0
S_I	Soluble inert organic matter	g COD m^{-3}	1	0.01	0	0
S_{ALK}	Alkalinity	mol m^{-3}	0	0	0	0
X_I	Particulate inert organic matter	g COD m^{-3}	1	0.02	0.01	0.75
X_S	Slowly biodegradable substrate	g COD m^{-3}	1	0	0	0.75
X_H	Heterotrophic biomass	g COD m^{-3}	1	0.07	0.02	0.90
X_{PAO}	Phosphate-accumulating biomass	g COD m^{-3}	1	0.07	0.02	0.90
X_{PP}	Stored polyphosphate	g P m^{-3}	0	0	1	3.23
X_{PHA}	Stored polyhydroxyalkanoates	g COD m^{-3}	1	0	0	0.60
X_{AOB}	Ammonia-oxidizing biomass	g COD m^{-3}	1	0.07	0.02	0.90
X_{NOB}	Nitrite-oxidizing biomass	g COD m^{-3}	1	0.07	0.02	0.90
X_{MeP}	Precipitated metal phosphate	g TSS m^{-3}	0	0	1/4.87	1
X_{MeOH}	Metal hydroxide	g TSS m^{-3}	0	0	0	1

the surrogate fit. Their meaning follows the component accounting used in activated sludge modeling (Henze et al., 2015).

The full state preserves distinctions that aggregate quantities cannot recover. Ammonium, nitrite, and nitrate contribute to the same reported nitrogen total, but their conversion involves different microbial populations and oxygen requirements. Dissolved dinitrogen remains a component of the modeled nitrogen inventory even though it has zero weight in reported TN. Similarly, soluble phosphate, stored polyphosphate, and metal phosphate have different process roles even when all contribute to TP. These differences matter when a predicted state is used to interpret an operating decision.

3.2 Process transformations and their rates

Hydrolysis transfers slowly biodegradable material toward forms that microorganisms can use more readily. Heterotrophic metabolism links substrate consumption to biomass growth and electron-acceptor use. Under aerobic conditions oxygen acts as an electron acceptor. Under anoxic conditions nitrate or nitrite can support the corresponding modeled transformations. Fermentation supplies acetate from suitable organic material. Biomass lysis redistributes material from active populations to substrate and inert pools. Biological phosphorus removal adds storage, growth, phosphate uptake, and release

mechanisms. Chemical precipitation and redissolution redistribute phosphorus between soluble and particulate forms (Guerrero et al., 2011; Henze et al., 2015).

Two-step nitrification represents ammonium oxidation to nitrite separately from nitrite oxidation to nitrate. It therefore retains both ammonia-oxidizing and nitrite-oxidizing biomass. A one-step description can reproduce an overall ammonium-to-nitrate conversion while hiding a nitrite intermediate. Retaining the intermediate allows the state to distinguish a low-ammonium effluent that contains nitrite from one in which nitrite has been converted further. The distinction concerns the reaction representation rather than an assurance that every nitrogen pathway occurring in a real plant is included. The broader diversity of microbial nitrogen-removal pathways reviewed by Duan et al. (2022) sets that limitation.

The stoichiometric matrix, also called the Petersen matrix, records which components are consumed and produced by each process. Its orientation is fixed as

$$\nu \in \mathbb{R}^{P \times F}, \quad P = 28, \quad F = 20. \quad (3.2)$$

Row p describes process p and column j describes component j . A negative entry ν_{pj} represents consumption and a positive entry represents production. A zero entry means that the process has no direct stoichiometric contribution to that component. Coefficients contain yields and constituent conversions appropriate to the native units.

For a rate vector $\rho(c, u) \in \mathbb{R}^{28}$, the reaction contribution to the component balance is

$$r_{\text{reaction}}(c, u) = \nu^\top \rho(c, u), \quad r_{\text{reaction},j}(c, u) = \sum_{p=1}^{28} \nu_{pj} \rho_p(c, u). \quad (3.3)$$

Stoichiometry specifies the proportions of a transformation. Kinetics specifies its speed at a particular state. These are different responsibilities. Increasing the concentration of biomass can change a process rate while leaving its stoichiometric row unchanged. Likewise, changing aeration can alter the oxygen limitation of a process without changing the material accounting of that process.

A simple saturation expression makes this distinction concrete. If a limiting dissolved substrate has concentration S and saturation constant $K_S > 0$, the factor

$$\frac{S}{K_S + S} \quad (3.4)$$

equals one half at $S = K_S$, approaches zero as S approaches zero, and approaches one as S becomes large. A representative growth expression can combine substrate and oxygen limitation with biomass abundance,

$$\rho_{\text{growth}} = \mu X \frac{S}{K_S + S} \frac{S_O}{K_O + S_O}. \quad (3.5)$$

Here μ is a maximum specific growth rate, X is the associated biomass coordinate, and K_O is an oxygen saturation constant. Equation (3.5) illustrates the role of multiplicative

limitation in activated sludge kinetics. It does not replace the complete 28-process rate formulation, whose process-specific yields, inhibition terms, and storage dependencies follow Henze et al. (2015) and Guerrero et al. (2011).

A hypothetical substrate concentration of $K_S/9$ gives a substrate factor of 0.1. Raising the concentration to $9K_S$ gives a factor of 0.9. The growth rate changes by a factor of nine if all other factors stay fixed, although the substrate concentration changes by a factor of 81. This calculation explains why a linear relationship between influent load and effluent state is not generally sufficient. Several limiting factors and competing populations can act together. It also explains why an accurate surrogate needs to represent interactions without confusing them with a new stoichiometric law.

3.3 The steady-state reactor boundary

The single-reactor prediction problem uses a completely mixed reactor at steady state. Perfect mixing makes the reactor concentration equal to the effluent concentration. The standalone boundary contains hydraulic inflow and outflow, the modeled reaction processes, and dissolved-oxygen transfer. It excludes a clarifier, solids return, and other external additions. These exclusions define the balance being enforced. They do not imply that those processes are absent from wastewater treatment plants.

Let c_{in} and c_{out} be the influent and effluent component vectors. Let $D > 0$ be the hydraulic dilution rate. Let $e_{S_O} \in \mathbb{R}^{20}$ select the dissolved-oxygen coordinate. The steady balance is

$$0 = D(c_{in} - c_{out}) + \nu^\top \rho(c_{out}, u) + \omega e_{S_O}, \quad (3.6)$$

or equivalently

$$D(c_{out} - c_{in}) = \nu^\top \rho(c_{out}, u) + \omega e_{S_O}. \quad (3.7)$$

The hydraulic term has the same native component-unit-per-day basis as the corresponding reaction and transfer terms. Treating the balance as a numerical vector equation requires a fixed unit convention throughout.

The operating inputs are HRT and a dimensionless aeration coordinate u_{air} . The adopted parameterization is

$$D = \frac{24 \text{ h d}^{-1}}{\text{HRT}}, \quad (3.8)$$

$$k_{La} = (2 + 18u_{air}) \text{ d}^{-1}, \quad (3.9)$$

$$\omega = k_{La}(S_{O,\text{sat}} - S_{O,\text{out}}), \quad S_{O,\text{sat}} = 8.5 \text{ g O}_2 \text{ m}^{-3}. \quad (3.10)$$

The dilution rate connects the time basis of the hydraulic specification to that of the process rates. The oxygen-transfer term depends on both the aeration setting and the difference between saturation and the reactor oxygen concentration.

For a hypothetical HRT of 12 h, Equation (3.8) gives $D = 2 \text{ d}^{-1}$. A component difference of 5 g m^{-3} then corresponds to a net transformation or transfer of 10 g m^{-3}

d^{-1} in that coordinate. For a hypothetical aeration setting of 1, the transfer coefficient is 20 d^{-1} . If dissolved oxygen is $2 \text{ g O}_2 \text{ m}^{-3}$, the oxygen-transfer rate is $130 \text{ g O}_2 \text{ m}^{-3} \text{ d}^{-1}$. These calculations illustrate the accounting. They are not simulated treatment outcomes.

Steady state means that each modeled inventory has zero time derivative. It does not establish a unique steady state, dynamic stability, or a field-calibrated description. Sensitivity of operating choices to the process representation remains relevant even when a balance closes (de Araujo et al., 2013). In particular, a small numerical residual certifies agreement with the stated equations at the returned state. It cannot certify the completeness of the equations for a real wastewater system.

3.4 From stoichiometry to conservation invariants

A stoichiometric invariant is a linear combination of component concentrations that the modeled reactions cannot change. If $a \in \mathbb{R}^{20}$ satisfies $\nu a = 0$, then

$$a^\top \nu^\top \rho = (\nu a)^\top \rho = 0 \quad (3.11)$$

for every process-rate vector. This identity does not depend on the accuracy of an estimated rate or on the value of an operating input. It follows from the reaction accounting alone. Conservation-aware surrogate constructions use this separation to preserve balances without solving all of the original rate equations during inference (Hansen et al., 2024; Sturm & Wexler, 2020, 2022).

An independent invariant basis is obtained by singular-value decomposition of the 28×20 matrix ν . With

$$\nu = U \Sigma V^\top, \quad (3.12)$$

the numerical rank counts singular values exceeding

$$\tau_{\text{rank}} = \max(P, F) \epsilon_{\text{mach}} \|\nu\|_2. \quad (3.13)$$

Here ϵ_{mach} is the machine precision used for the calculation and $\|\nu\|_2$ is the largest singular value. The adopted stoichiometric representation has rank 15. The rank–nullity identity therefore gives five independent invariants. This is a structural property of the specified reaction matrix.

Let $V_0 \in \mathbb{R}^{20 \times 5}$ contain the right singular vectors spanning $\text{null}(\nu)$. Define

$$A = V_0^\top \in \mathbb{R}^{5 \times 20}. \quad (3.14)$$

The resulting operator obeys

$$A \nu^\top = 0, \quad A A^\top = I_5. \quad (3.15)$$

Each row represents an independent conserved combination. The orthonormal rows need not correspond individually to named elemental inventories. Their purpose is to provide a numerically specified basis for the same invariant space.

The matrix orientation matters. Because processes are rows of ν , the conserved component combinations lie in its right null space. Taking the right null space of $\nu^\top$ instead produces vectors in process coordinates. Those vectors describe dependencies among process contributions and have a different interpretation. Dimension checks prevent the two constructions from being confused.

Conservation of reactions does not by itself establish conservation across an open reactor boundary. Premultiplying Equation (3.7) by A gives

$$DA(c_{out} - c_{in}) = \omega Ae_{SO}. \quad (3.16)$$

The aeration forcing must therefore be compatible with the selected invariants. For the adopted matrix, the oxygen selector lies in $\text{range}(\nu^\top)$. There is a process-coordinate vector $\lambda_O \in \mathbb{R}^{28}$ such that

$$e_{SO} = \nu^\top \lambda_O, \quad Ae_{SO} = A\nu^\top \lambda_O = 0. \quad (3.17)$$

Since $D > 0$, the invariant boundary relation reduces to

$$Ac_{out} = Ac_{in}. \quad (3.18)$$

The relation is justified jointly by stoichiometry, the forcing direction, and the reactor boundary. It should not be imposed on a system with an unaccounted external phosphorus dose, sludge withdrawal, or another source that changes the selected inventories.

More generally, an additional component source g gives

$$A(c_{out} - c_{in}) = D^{-1}Ag \quad (3.19)$$

when aeration remains compatible. A hypothetical phosphate addition with a nonzero phosphorus-inventory contribution changes the right-hand side. Forcing equality with the original influent would then remove a real external contribution. This example shows why a physical constraint must be derived from the actual system boundary rather than chosen solely because it improves a statistical score.

The full-precision operator is used for all calculations. Its construction is checked through $A\nu^\top$, $AA^\top - I_5$, and Ae_{SO} . Rounding coefficients for display cannot replace the original operator in a numerical calculation. Any nonsingular row transformation T defines $A' = TA$ and the same equality set. An arbitrary transformation can change the magnitude of a residual. The orthonormal construction fixes the basis and scaling used for the conservation diagnostic.

3.5 Reaction progress and the feasible state geometry

The invariant equation can also be understood through reaction progress. Rearranging the steady balance and using Equation (3.17) gives

$$c_{out} - c_{in} = \nu^\top r, \quad r = D^{-1}(\rho + \omega\lambda_O). \quad (3.20)$$

The vector r gives one set of process coordinates for the total component change. It is not a unique reconstruction of the biological rates because $\nu^\top$ has rank 15 and 28 columns. It also includes the algebraic representation of oxygen transfer. Individual entries of r therefore should not be interpreted as independently measured reaction extents or microbial activity.

Let $Z \in \mathbb{R}^{20 \times 15}$ have orthonormal columns spanning $\text{null}(A)$. The fundamental subspace relation gives

$$\text{range}(\nu^\top) = \text{null}(A), \quad AZ = 0, \quad Z^\top Z = I_{15}. \quad (3.21)$$

Every invariant-consistent component change can consequently be written as Zq for a unique $q \in \mathbb{R}^{15}$ in the selected orthonormal basis. The coordinates q describe an admissible direction of material redistribution. They do not identify a unique sequence of the 28 kinetic processes.

For sample i , define the target $b_i = Ac_{in,i}$ and the feasible set

$$\mathcal{F}_i = \{c \in \mathbb{R}^{20} \mid Ac = b_i, c \geq 0\}. \quad (3.22)$$

The equality describes a 15-dimensional affine space before concentration bounds are applied. Nonnegativity intersects that space with the nonnegative orthant. The intersection is closed and convex. A nonnegative influent belongs to the set, so the set is nonempty and the influent supplies a feasible anchor.

A two-component hypothetical system with conserved sum $c_1 + c_2 = 10$ illustrates the geometry. The equality permits states such as $(-1, 11)$, $(4, 6)$, and $(12, -2)$. Nonnegativity retains only the line segment between $(0, 10)$ and $(10, 0)$. Clipping $(-1, 11)$ to $(0, 11)$ repairs its negative coordinate but changes the sum to 11. Enforcing the equality alone and enforcing nonnegativity alone therefore solve different problems. Simultaneous enforcement requires the intersection.

The geometry also explains why a feasible prediction need not be kinetically correct. Several nonnegative states can share all five invariant values. A state with the wrong distribution between ammonium and nitrate may satisfy the invariant equality. A state with an implausible biomass concentration may do the same. The feasible set establishes the declared material accounting and sign restrictions. It does not establish that the 28 process rates balance hydraulics and aeration at that state. This distinction is central to the constraint methods of Kircher and Votsmeier (2025) and to the broader physical-consistency perspective of Valente et al. (2025).

3.6 Reported water quality quantities and hidden component errors

Reported water quality quantities are calculated after a component state has been predicted. With the coefficients in Table 3.1,

$$y = I_{comp}c, \quad y = (\text{COD}, \text{TN}, \text{TP}, \text{TSS})^\top, \quad I_{comp} \in \mathbb{R}^{4 \times 20}. \quad (3.23)$$

The four rows convert the component vector to g COD m⁻³, g N m⁻³, g P m⁻³, and g TSS m⁻³, respectively. All coefficients are nonnegative. A nonnegative component state therefore gives nonnegative reported quantities.

The nitrogen map excludes dissolved dinitrogen. This reporting convention makes TN different from a conserved inventory that includes nitrogen gas. In a hypothetical redistribution of 3 g N m⁻³ from nitrate to dissolved dinitrogen, the reported TN decreases by 3 g N m⁻³ while the complete nitrogen inventory can remain unchanged. Conversely, a phosphorus inventory that includes every modeled phosphorus pool can coincide with a reporting row. The relationship between a reported total and an invariant must be checked from their coefficients rather than inferred from the word “total.”

The same coefficient table explains constituent conversion. A hypothetical X_H concentration of 100 g COD m⁻³ contributes 100 g COD m⁻³ to COD, 7 g N m⁻³ to TN, 2 g P m⁻³ to TP, and 90 g TSS m⁻³ to TSS under the adopted map. A hypothetical X_{PP} concentration of 2 g P m⁻³ contributes 2 g P m⁻³ to TP and 6.46 g TSS m⁻³ to TSS. These examples illustrate fixed accounting factors rather than evidence about a treatment outcome.

Aggregation can hide prediction errors. Equal and opposite errors in ammonium and nitrate cancel in reported TN because both have unit nitrogen weights. A negative ammonium value can even be hidden by a sufficiently large nitrate prediction. Likewise, different biomass and substrate errors can compensate in COD while producing distinct errors in TSS because their solids factors differ. Four accurate reported quantities therefore cannot establish an accurate or physically admissible 20-component state.

The reporting map and the invariant operator serve different purposes. The map provides engineering summaries. The invariant operator tests reaction-compatible material accounting at the declared boundary. Neither can recover all of the component information lost by aggregation. Component prediction, composite interpretation, conservation, and nonnegativity therefore remain separate objects of assessment.

Chapter 4

Learning the Reactor Response and Enforcing Physical Admissibility

4.1 Accuracy and physical admissibility as distinct questions

A surrogate learns the relationship between a reactor input and its steady-state effluent. Its statistical task is to reproduce component concentrations from examples. Its physical task is to return a nonnegative component vector with the invariant values appropriate to the supplied influent. These tasks are related because the mechanistic targets satisfy the reactor equations. They remain distinct because an ordinary regression loss does not explicitly restrict every prediction to the sample-specific feasible set (Hansen et al., 2024; Kashinath et al., 2021).

The complete input for sample i is

$$x_i = (u_i^\top, c_{in,i}^\top)^\top \in \mathbb{R}^{22}, \quad u_i = (\text{HRT}_i, u_{\text{air},i})^\top. \quad (4.1)$$

A fitted surrogate f_m produces the raw component prediction

$$c_{\text{raw},i}^{(m)} = f_m(x_i) \in \mathbb{R}^{20}. \quad (4.2)$$

The raw prediction is assessed in its returned physical units. A post prediction projection then produces a second prediction for the same sample. Pairing the two states permits the effect of physical enforcement to be assessed separately from the choice of learner, data partition, and preprocessing.

The comparison includes 13 statistical surrogates. The set spans boosting, randomized ensembles, kernel regression, neighborhood regression, latent-variable regression, sparse multiresponse regression, and neural networks. This breadth addresses the model-selection concern raised by Bhosekar and Ierapetritou (2018). The surveys of Borisov et al. (2024) and Shwartz-Ziv and Armon (2022) further explain why a neural architecture alone should not determine the comparator set for a tabular prediction problem. The

purpose of the common protocol is to compare explicitly specified predictors for the same physical state.

4.2 Constructing a mechanistic learning domain

The single-reactor dataset is designed to contain 10,000 accepted steady states. Each candidate varies all 22 input coordinates. A randomized, scrambled, strength-one Latin hypercube places one sample in each stratum of every marginal input distribution. The construction uses seed 42 and no subsequent design optimization. Linear scaling maps the unit-hypercube samples to the bounds in Table 4.1. Latin hypercube sampling distributes coverage across marginal ranges without requiring a full Cartesian grid (McKay et al., 1979). It does not guarantee uniform coverage of every nonlinear response feature or every high-dimensional interaction.

The standalone domain is a declared simulation design. It is not a distribution estimated from plant measurements. In particular, its alkalinity interval is retained in the stated native coordinate. The operating and influent domains of a coupled plant are specified independently. A common component symbol does not establish that two evaluation designs use the same bounds.

For each candidate input, nonlinear least squares solves Equation (3.6) with numerical state bounds $[10^{-8}, 1500]$ in each component's native unit. The cost-change, step-size, and gradient convergence tolerances are 10^{-9} . Each solve allows at most 1200 function evaluations. The generation procedure has an overall attempt cap of 25 times the target retained count. A state is retained only if the solver converges and the maximum absolute component-balance residual is no greater than 10^{-9} on the fixed native component-unit-per-day coordinates. The residual criterion is a numerical acceptance rule rather than a dimensionless physical error measure.

Two physically informed starting states are tried before dynamic relaxation. The first blends the sampled influent with a preceding accepted effluent when one is available. It lowers readily consumed substrate coordinates, retains biomass pools, and estimates dissolved oxygen from dilution and transfer. The second applies stronger substrate depletion and biomass enrichment. Starting vectors are restricted to $[10^{-6}, 1500]$. If both steady solves fail, a stiff backward differentiation formula integrates the dynamic balance for 20 days and supplies a new initial state. The state bounds and starting-state restrictions apply to concentrations. The 28 kinetic process rates are evaluated without an artificial ceiling or clipping.

Solver acceptance, invariant residual, and minimum concentration are recorded separately. A converged bounded least-squares calculation can terminate without meeting the required balance residual. A positive state can also fail conservation. Retaining these separate diagnostics prevents one successful numerical check from standing in for all others. The surrogate inputs contain the sampled operating and

Table 4.1: Design bounds for the standalone reactor learning domain. These intervals specify sampling limits and do not report realized sample extrema.

Input	Meaning	Interval	Unit
HRT	Hydraulic operating input	[6, 36]	h
u_{air}	Dimensionless aeration input	[0.5, 2.5]	1
S_O	Influent dissolved oxygen	[0, 0.5]	g O ₂ m ⁻³
S_F	Influent fermentable substrate	[20, 180]	g COD m ⁻³
S_A	Influent acetate	[5, 80]	g COD m ⁻³
S_{NH_4}	Influent ammonium nitrogen	[12, 55]	g N m ⁻³
S_{NO_2}	Influent nitrite nitrogen	[0, 3]	g N m ⁻³
S_{NO_3}	Influent nitrate nitrogen	[0, 8]	g N m ⁻³
S_{N_2}	Influent dissolved dinitrogen	[0, 2]	g N m ⁻³
S_{PO_4}	Influent soluble phosphate	[2, 18]	g P m ⁻³
S_I	Influent soluble inert organic matter	[10, 90]	g COD m ⁻³
S_{ALK}	Influent alkalinity	[80, 260]	mol m ⁻³
X_I	Influent particulate inert matter	[20, 120]	g COD m ⁻³
X_S	Influent slowly biodegradable substrate	[60, 280]	g COD m ⁻³
X_H	Influent heterotrophic biomass	[15, 100]	g COD m ⁻³
X_{PAO}	Influent phosphate-accumulating biomass	[5, 60]	g COD m ⁻³
X_{PP}	Influent stored polyphosphate	[2, 20]	g P m ⁻³
X_{PHA}	Influent stored polyhydroxyalkanoates	[1, 30]	g COD m ⁻³
X_{AOB}	Influent ammonia-oxidizing biomass	[0.5, 8]	g COD m ⁻³
X_{NOB}	Influent nitrite-oxidizing biomass	[0.5, 8]	g COD m ⁻³
X_{MeP}	Influent precipitated metal phosphate	[0, 12]	g TSS m ⁻³
X_{MeOH}	Influent metal hydroxide	[0, 12]	g TSS m ⁻³

influent coordinates. Initialization choices and solver history are not predictive features.

4.3 How the thirteen predictors represent the response

4.3.1 Boosting and randomized tree ensembles

XGBoost builds an additive sequence of regression trees. Each new tree addresses the error left by the current ensemble while regularization discourages an unnecessarily complex fit. Its split structure represents changes in response across input regions, and repeated additions permit interactions and nonlinear response shapes. The computational design and regularized objective of Chen and Guestrin (2016) motivate its inclusion as a flexible tabular baseline.

LightGBM also uses gradient-boosted trees, but grows trees through leaf-wise expansion. This strategy can allocate more detail to regions where an additional split offers a larger loss reduction. The efficiency-oriented design of Ke et al. (2017) makes the method relevant when each candidate fit must be repeated over several validation partitions. Its learning rate, leaf count, and minimum leaf support jointly control how closely it follows the training response.

CatBoost uses symmetric boosted trees and a Bayesian bootstrap in the specified regression setting. The work of Prokhorenkova et al. (2018) addresses bias associated with categorical-feature handling and boosting. The reactor benchmark has continuous inputs, so its inclusion evaluates the corresponding regression learner without claiming a categorical-data advantage. Each level of a symmetric tree uses the same splitting condition across nodes at that level. This structural restriction gives a different approximation class from the leaf-wise trees of LightGBM.

Adaptive boosting (AdaBoost) regression combines shallow trees through adaptive emphasis on observations that are difficult to predict. The base trees have depth three. The estimator count, learning rate, and loss choice control how strongly the ensemble responds to poorly fitted examples. Drucker (1997) develops the regression extension of boosting that supports this comparator. These four boosting methods fit one regressor for each effluent component. All 20 regressors within a method use one shared selected hyperparameter vector.

Random Forest averages randomized regression trees. Variation in sampled observations and candidate features reduces dependence among trees, so the average is less sensitive to a particular tree fit (Breiman, 2001). Extra Trees adds randomization of split thresholds (Geurts et al., 2006). Both methods fit the 20 outputs jointly in native multiresponse ensembles. Joint output handling lets the same tree partition support several component predictions. It does not impose a stoichiometric relationship among those outputs.

Tree averaging can preserve nonnegativity when all leaf responses are nonnegative. It cannot generally preserve the balance target for a new influent. For a hypothetical two-component system, two training responses with totals of 10 and 20 have an equally weighted average with total 15. If a new influent requires total 12, the average is positive but not conservative for that sample. This distinction is caused by the sample-specific target b_i , rather than by whether the predictor averages positive observations.

4.3.2 Kernel, neighborhood, and latent-variable regression

Support vector regression fits a function with an error-insensitive region around each training response. Predictions within that region incur no corresponding error penalty. A linear kernel gives a globally linear input mapping. A radial kernel allows a nonlinear mapping through similarity to training inputs. The penalty parameter balances fit and regularity, while the insensitive width sets the tolerated response discrepancy. Smola and Schölkopf (2004) explains these mechanisms and the importance of their parameterization. The benchmark fits one regressor per component with one shared setting vector.

The k -nearest-neighbor predictor finds training observations close to a new input and combines their 20-component responses. The neighborhood size controls the degree of smoothing. Uniform weights give every selected neighbor equal influence, whereas distance weights give closer observations greater influence. The local behavior analyzed by Mack (1981) explains why input scaling and distance choice matter. An unscaled alkalinity coordinate spanning hundreds of numerical units could dominate a much smaller aeration coordinate even though both influence the reactor. Standardization avoids using native numerical range as an implicit measure of scientific importance.

Partial least squares regression represents the inputs and joint response through latent directions chosen to capture their covariance. It can summarize strongly related input variables with a relatively small number of components. The chemometric formulation described by Wold et al. (2001) provides an inspectable linear baseline. Its wastewater use is illustrated by Woo et al. (2009), who employs a kernel extension for process-variable estimation, and by C. Yang et al. (2021), who combines partial least squares with another predictor for adaptive effluent-quality prediction. The benchmark retains ordinary partial least squares with one to six latent components. The nonlinear extensions in those applications do not make the present baseline nonlinear.

4.3.3 Shared sparse coefficient models

Multi-task Lasso fits one coefficient matrix for all component outputs. If row $B_{j,\cdot}$ contains the coefficients of input j across those outputs, the row-group penalty is

$$\|B\|_{2,1} = \sum_j \|B_{j,\cdot}\|_2. \quad (4.3)$$

This penalty can set an entire input row to zero. It therefore promotes a shared selection of inputs rather than 20 unrelated selections. The feature-sharing principle of Argyriou et al. (2008), the multi-task Lasso algorithm of Liu et al. (2009), and the grouped-variable framework of Yuan and Lin (2006) provide the relevant foundations.

Multi-task Elastic Net combines the row-group penalty with a squared Frobenius penalty. The row-group part encourages selection and the squared part shrinks the retained coefficients. This extends the elastic-net principle of Zou and Hastie (2005) to a joint response structure, with multiresponse computation described by Simon et al. (2013). Both methods use standardized inputs and targets. Their global coefficient matrices can be inspected to identify fitted associations. A nonzero row shows that an input contributes to the fitted prediction, but it does not establish a causal effect or a mechanistic reaction rate.

These two predictors and partial least squares are grouped as structurally inspectable surrogates. The grouping identifies their globally available coefficient structures. It is not a measured claim that users understand them more easily or that their coefficients are scientifically correct. It also does not establish physical admissibility. A global linear relation can still predict a negative concentration or violate the influent-specific balance.

4.3.4 Neural predictors

The multilayer perceptron uses three fully connected hidden layers with 128, 64, and 32 units and one joint 20-output layer. Each hidden layer combines its inputs and applies a nonlinear activation. Backpropagation updates the weights according to the prediction loss (Rumelhart et al., 1986). The approximation result of Hornik et al. (1989) motivates the expressive capacity of feedforward networks. It does not guarantee that this finite architecture learns the reactor response from a given sample size. The topology considerations discussed by Curteanu and Cartwright (2011) likewise motivate explicit architectural and training limits rather than an unspecified neural comparator.

TabNet applies sequential attentive feature selection. Different decision steps can emphasize different subsets of input features before their contributions enter a joint output layer. Arik and Pfister (2021) develops this approach for tabular learning. The attention masks provide information about the model's use of inputs, but they do not impose conservation or make the component outputs causal explanations. TabNet remains a neural comparator even though its internal masks can be inspected.

Both neural predictors start from random initialization and use only the simulated training partition available to that fit. No external learned representation is used. The multilayer perceptron uses all available training rows and stops on training-loss convergence or at 250 epochs. Validation-based early stopping is disabled. Its training-loss patience is 15 epochs. TabNet's epoch count is selected in the inner search and ranges from 20 to 50. Its patience is zero, so a refit runs the selected epoch count

without reserving an additional response-based stopping partition.

4.4 Preprocessing and validation without response leakage

The evaluation uses nested subsets at

$$N \in \{500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, 10000\}. \quad (4.4)$$

A seed-42 permutation determines the nested row order and persistent master fold identifiers before subsets are formed. Every smaller subset is contained in each larger subset. A row retains its outer-fold role at every scale. Five outer folds train on $0.8N$ observations and score the remaining $0.2N$. The same rows and fold identifiers are used by every surrogate and both prediction types.

Hyperparameters are selected using four inner folds within the largest outer-training partition. The scored outer fold is absent from the search. For each outer fold, its selected setting is retained over the smaller nested subsets. Learning curves therefore examine the response to sample size under a fixed fold-specific setting. They do not represent a new optimum obtained through a separate hyperparameter search at every scarce-data scale. The bias–variance discussion of Belkin et al. (2019) cautions against assuming that error must vary monotonically with model capacity or available data. The learning-curve review of Viering and Loog (2023) supports treating the complete curve as an object of assessment.

Every transformation is fitted within the applicable training partition. During inner selection, that partition is the inner-training split. During outer scoring, it is the outer-training subset at the current size. During extrapolation assessment, it is the full in-domain training pool. The mean and population standard deviation of a training coordinate define its z-score transformation. Validation inputs are transformed with those training quantities, and target-scaled predictions are restored to physical units before projection or physical diagnostics.

Tree partitions depend on coordinate ordering, so a positive affine input rescaling preserves candidate threshold partitions. Distance-based, covariance-based, and regularized models require more deliberate scaling because relative numerical magnitudes affect their objectives. Separate-output tree regressors do not combine different component units within one joint fitting loss. The target-only scaling of Extra Trees and AdaBoost is retained as a specified benchmark choice. Independent of those training choices, every scored component error is standardized with the designated training reference. Training target scaling and evaluation normalization are distinct operations.

No missing-value imputation, response-based outlier removal, or target filtering is applied. No validation or extrapolation response is used to estimate a scaling parameter. A hypothetical decision to standardize a validation target with the standard deviation

Table 4.2: Output handling and preprocessing for the statistical comparators. Scaling uses only the training partition applicable to each fit.

Surrogates	Output handling	Scaling
XGBoost, LightGBM, Cat-Boost	Separate component regressors	None
AdaBoost	Separate component regressors	Targets only
Random Forest	Joint tree response	None
Extra Trees	Joint tree response	Targets only
Support vector regression	Separate component regressors	Inputs and targets
k -nearest neighbors	Joint neighborhood response	Inputs and targets
Partial least squares, Multi-task Lasso, Multi-task Elastic Net	Joint coefficient structure	Inputs and targets
Multilayer perceptron, TabNet	Joint neural output	Inputs and targets

of its own fold would change the scoring reference using data outside training. Retaining a training reference instead asks how large an error is relative to the variability the learner was allowed to observe.

4.5 Hyperparameter selection and finite comparison scope

Each surrogate uses a seeded tree-structured Parzen search with 100 trials per study and no pruning or timeout. The search mechanism is supported by the optimization framework of Akiba et al. (2019). The objective is the mean squared standardized component error over the inner validation folds. Five studies select the settings for the five largest-scale outer-training partitions. A sixth, independent four-fold study selects the setting for a final full-domain fit. The design specifies 600 trials per surrogate and 7800 across all 13 surrogates. These are allocated search budgets rather than performance findings.

The finite ranges define the actual comparison. A linear or radial support vector model does not represent every possible kernel configuration. A neural predictor limited to 250 epochs does not represent every training budget or architecture. The shared 100-trial allocation does not guarantee equal difficulty of tuning across model families. The study therefore assesses these specified approximation classes under a common protocol rather than claiming universal superiority of one family.

Table 4.3: Search domains and fixed controls for tree ensemble surrogates. Integer limits are inclusive. “Log” denotes a log-uniform search.

Surrogate	Search domain and fixed controls
XGBoost	80–320 trees. Learning rate 0.01–0.2 (log). Depth 3–8. Minimum child weight 0.5–8 (log). Row and feature fractions 0.6–1. Absolute-value penalty 10^{-6} –1 (log). Squared penalty 10^{-3} –10 (log). Histogram or approximate trees. Squared-error objective.
LightGBM	80–320 trees. Learning rate 0.01–0.2 (log). 15–63 leaves. Depth unbounded or in $\{4, 6, 8, 10\}$. Minimum child rows 5–40. Row and feature fractions 0.6–1 with subsampling frequency 1. Absolute-value penalty 10^{-6} –1 (log). Squared penalty 10^{-6} –10 (log). Regression objective.
CatBoost	80–320 iterations. Learning rate 0.01–0.2 (log). Depth 4–8. Squared penalty 0.01–20 (log). Bagging temperature 0–5. Random strength 10^{-3} –5 (log). Root mean squared error loss and Bayesian bootstrap.
AdaBoost	50–250 depth-three trees. Learning rate 0.01–1 (log). Linear, squared, or exponential regression loss.
Random Forest	100–400 trees. Depth unbounded or in $\{6, 10, 14, 18\}$. Minimum split size 2–10. Minimum leaf size 1–4. Feature fraction 0.5–1. Bootstrap enabled or disabled.
Extra Trees	100–400 trees. Depth unbounded or in $\{6, 10, 14, 18\}$. Minimum split size 2–10. Minimum leaf size 1–4. Feature fraction 0.5–1. Bootstrap enabled or disabled.

Table 4.4: Search domains and fixed controls for kernel, neighbor, multiresponse, and neural surrogates. Integer limits are inclusive. “Log” denotes a log-uniform search.

Surrogate	Search domain and fixed controls
Support vector regression	Penalty 0.1–10 (log). Error-insensitive width 0.01–0.5 (log). Linear or radial kernel. Radial coefficient set from standardized input variance or inverse input dimension. Tolerance 10^{-3} – 10^{-2} (log). Shrinking enabled. At most 20,000 iterations. Kernel cache of 256 megabytes per output regressor.
k -nearest neighbors	1–15 neighbors. Uniform or distance weights. Automatic, ball-tree, k -dimensional tree, or brute-force search. Leaf size 15–60. Minkowski, Euclidean, or Manhattan distance. Minkowski order in $\{1, 2, 3\}$.
Partial least squares	1–6 latent components. 200–1000 maximum iterations. Tolerance 10^{-8} – 10^{-4} (log). Internal scaling disabled because external training-partition scaling is applied.
Multi-task Elastic Net	Penalty strength 10^{-6} –10 (log). Row-group mixing ratio 0.05–0.95. Cyclic coordinate descent. At most 20,000 iterations and tolerance 10^{-6} .
Multi-task Lasso	Penalty strength 10^{-6} –10 (log). Cyclic coordinate descent. At most 20,000 iterations and tolerance 10^{-6} .
Multilayer perceptron	Hidden widths (128, 64, 32). Rectified linear, hyperbolic tangent, or logistic activation. Adaptive moment optimization. Squared-weight penalty 10^{-6} – 10^{-2} (log). Batch size in $\{64, 128, 256\}$. Learning rate 5×10^{-4} – 10^{-2} (log). Training-loss tolerance 10^{-4} – 10^{-3} (log). Shuffling enabled. Training-loss patience 15 epochs. At most 250 epochs.
TabNet	Equal decision and attention widths in $\{8, 16, 24\}$. Three or four attention steps. Relaxation coefficient 1–2. Sparsity coefficient 10^{-6} – 10^{-2} (log). Adaptive moment learning rate 0.003–0.03 (log). Sparsemax or entmax-1.5 masks. 20–50 selected epochs. Batch size in $\{512, 1024\}$. Virtual batch size 128. Patience zero.

4.6 The Kircher–Votsmeier conservative projection

4.6.1 A positive provisional target and relative-error geometry

Kircher and Votsmeier (2025) develop a correction that combines conservation with positivity for mechanistic-kinetics surrogates. Its transfer to the reactor prediction problem uses the invariant operator A , the basis Z , and the feasible influent anchor. The projection depends on the completed component prediction rather than on predictor internals. It can therefore be applied to every comparator without changing its training objective or architecture. It still requires the full component vector, the corresponding nonnegative influent, and the physical operators.

The first step forms a strictly positive provisional target,

$$c_{+,i}^{(m)} = \max(c_{raw,i}^{(m)}, \epsilon_c), \quad W_i^{(m)} = \text{Diag}(\mathbf{1} \oslash c_{+,i}^{(m)}). \quad (4.5)$$

The maximum and division are componentwise. Each entry of ϵ_c is 10^{-8} in its component's native unit. The provisional target supplies both a reference state and finite positive weights. It is an intermediate construction, so its clipped coordinates are not the final physical prediction.

For a component change h_j , the weighted discrepancy is $h_j/c_{+,ij}^{(m)}$. The geometry therefore penalizes relative movement. A hypothetical movement of 1 g m^{-3} corresponds to a relative discrepancy of 0.01 at a component value of 100 g m^{-3} and 10 at a component value of 0.1 g m^{-3} . Small coordinates receive strong protection. Large coordinates can absorb larger absolute changes at the same relative cost. This explains why relative-error correction and variance-standardized accuracy need not favor the same redistribution.

4.6.2 Selecting an invariant-consistent candidate

All invariant-consistent states for sample i have the form

$$c = c_{in,i} + Zq, \quad q \in \mathbb{R}^{15}. \quad (4.6)$$

The projection selects the conservative change closest to the provisional target change in the relative-error geometry,

$$q_i^{*,(m)} = \underset{q \in \mathbb{R}^{15}}{\text{argmin}} \left\| W_i^{(m)} [Zq - (c_{+,i}^{(m)} - c_{in,i})] \right\|_2^2. \quad (4.7)$$

The squared objective is a quadratic function of q . Differentiation gives

$$Z^\top (W_i^{(m)})^2 Z q_i^{*,(m)} = Z^\top (W_i^{(m)})^2 (c_{+,i}^{(m)} - c_{in,i}). \quad (4.8)$$

For any nonzero vector v , the associated quadratic form equals $\|W_i^{(m)} Z v\|_2^2 > 0$. The matrix is therefore positive definite and the minimizer is unique.

The conservative candidate is

$$c_{cand,i}^{(m)} = c_{in,i} + Zq_i^{*,(m)}. \quad (4.9)$$

Since $AZ = 0$, it satisfies $Ac_{cand,i}^{(m)} = Ac_{in,i} = b_i$ in exact arithmetic. If the provisional target is already invariant-consistent, its change lies in $\text{null}(A)$. The least-squares residual can then be zero and the candidate reproduces that target to numerical tolerance.

Equation (4.8) provides the derivation rather than the preferred numerical solution. The weighted design $W_i^{(m)}Z$ is solved through an orthogonal–triangular factorization. Singular-value decomposition with relative cutoff 10^{-12} is the fallback. No explicit inverse of the normal-equation matrix is formed. Near-zero provisional targets can produce very unequal weights, so solving the original weighted system avoids the extra conditioning penalty associated with forming a normal-equation inverse.

4.6.3 The largest feasible interpolation toward the candidate

Conservation of the candidate does not establish its nonnegativity. Define

$$\delta_i^{(m)} = c_{cand,i}^{(m)} - c_{in,i} = Zq_i^{*,(m)}. \quad (4.10)$$

The segment $c_{in,i} + \alpha\delta_i^{(m)}$, with $0 \leq \alpha \leq 1$, is conservative because $A\delta_i^{(m)} = 0$. For every decreasing coordinate, nonnegativity requires

$$\alpha \leq \frac{c_{in,ij}}{-\delta_{ij}^{(m)}} \quad \text{when } \delta_{ij}^{(m)} < 0. \quad (4.11)$$

The minimum of these bounds is the first point at which the segment reaches the nonnegative-orthant boundary.

With numerical safety margin $\eta = 10^{-12}$, the accepted interpolation factor and final state are

$$\alpha_i^{(m)} = \begin{cases} 1, & c_{cand,i}^{(m)} \geq 0, \\ (1 - \eta) \min_{j \mid \delta_{ij}^{(m)} < 0} \frac{c_{in,ij}}{-\delta_{ij}^{(m)}}, & \text{otherwise,} \end{cases} \quad (4.12)$$

$$\tilde{c}_i^{(m)} = c_{in,i} + \alpha_i^{(m)} \delta_i^{(m)}.$$

If the candidate is negative somewhere, at least one bound lies below one. Nonnegativity of the influent makes every bound nonnegative. The factor is restricted to $[0, 1]$ against roundoff. Decreasing coordinates satisfy $\tilde{c}_{ij}^{(m)} \geq \eta c_{in,ij} \geq 0$, and increasing coordinates remain nonnegative. Conservation follows from

$$A\tilde{c}_i^{(m)} = Ac_{in,i} + \alpha_i^{(m)} AZq_i^{*,(m)} = b_i. \quad (4.13)$$

The unrestricted boundary factor is the largest feasible interpolation factor. The safety margin makes the returned factor slightly smaller when correction is necessary.

A hypothetical three-component system with conserved sum 10 illustrates this safeguard. Let the feasible influent be $(8, 1, 1)^\top$ and a conservative candidate be $(4, -0.5, 6.5)^\top$. The direction is $(-4, -1.5, 5.5)^\top$. Its decreasing-coordinate bounds are 2 and $2/3$, so the largest feasible factor is $2/3$. Ignoring the numerical safety margin only for this illustration gives $(16/3, 0, 14/3)^\top$. The sum remains 10 and every component is nonnegative. The first coordinate changes even though the second coordinate determines the boundary. One scalar shortens the complete conservative direction.

If an influent coordinate is zero and the candidate direction decreases that coordinate, its bound is zero. The final state then returns to the influent anchor. This is a valid feasible fallback, but it can remove useful information from the raw prediction. Likewise, a component that is absent from the invariant rows can still move during interpolation because the complete direction is shortened. Recording the factor and activation frequency is therefore necessary for interpreting the correction.

4.6.4 Acceptance checks and limits of the guarantee

A returned projection is accepted only when

$$\|A\tilde{c}_i^{(m)} - b_i\|_2 \leq \tau_{mc}, \quad \tilde{c}_{ij}^{(m)} \geq -\tau_{nn,j} \quad \text{for every } j, \quad (4.14)$$

with $\tau_{mc} = 10^{-10}$ in the fixed invariant-coordinate convention and $\tau_{nn,j} = 10^{-10}$ in the native unit of component j . These checks distinguish the exact mathematical construction from its finite-precision realization. A finite raw vector is required. The method does not define an admissible correction of a nonfinite prediction through this algebra.

The final state is feasible for the declared invariants and sign restrictions. It need not minimize the weighted distance over the full intersection $\mathcal{F}_i$. The least-squares stage minimizes distance over an affine space, and the interpolation stage searches one line segment. A general constrained nearest-point problem searches a larger set. The construction also does not minimize the variance-standardized prediction error, whose denominator differs from c_+ . A reduction in conservation error therefore does not imply a reduction in statistical error.

No kinetic, equilibrium, stability, or field-performance guarantee follows from the projection. A conservative nonnegative state can have a large residual in Equation (3.6). The distinction between physical reconciliation and generalization under a changed input distribution is consistent with Shen et al. (2023) and Valente et al. (2025). The projection also assumes that the influent anchor is feasible for the correct boundary. An unmodeled external source invalidates that assumption even if the numerical correction succeeds.

Raw and projected composites are calculated with the same map,

$$y_{raw,i}^{(m)} = I_{comp} c_{raw,i}^{(m)}, \quad \tilde{y}_i^{(m)} = I_{comp} \tilde{c}_i^{(m)}. \quad (4.15)$$

Feasibility is checked before this aggregation. Composite values cannot reveal every negative coordinate or every inconsistent component redistribution.

4.7 Evaluation of error, feasibility, and correction size

4.7.1 Component errors on a training-defined scale

Let $\mathcal{T}$ denote the applicable training reference and let $\sigma_j^{(\mathcal{T})}$ be the population standard deviation of target component j in that reference. For prediction type t , which is either raw or projected, define

$$\xi_{ij}^{(m,t)} = \frac{c_{ij}^{(m,t)} - c_{out,ij}}{\sigma_j^{(\mathcal{T})}}. \quad (4.16)$$

The normalized mean squared error (nMSE), normalized root mean squared error (nRMSE), and normalized mean absolute error (nMAE) are

$$\text{nMSE}_{m,t} = \frac{1}{20n} \sum_{i=1}^n \sum_{j=1}^{20} (\xi_{ij}^{(m,t)})^2, \quad (4.17)$$

$$\text{nRMSE}_{m,t} = \sqrt{\text{nMSE}_{m,t}}, \quad (4.18)$$

$$\text{nMAE}_{m,t} = \frac{1}{20n} \sum_{i=1}^n \sum_{j=1}^{20} |\xi_{ij}^{(m,t)}|. \quad (4.19)$$

The component state is the primary scoring space because it is the prediction target and the space in which constraints are imposed. The root mean squared score emphasizes larger standardized errors. The absolute score gives a complementary average deviation. A component with zero training standard deviation makes these normalized measures undefined and invalidates the run.

A hypothetical error of 1 g m^{-3} has standardized magnitude 0.1 for a component with training standard deviation 10 g m^{-3} . The same physical error has magnitude 10 for a component with training standard deviation 0.1 g m^{-3} . The score gives equal numerical weight to standardized component variation. It does not claim that every component has equal regulatory importance or equal measurement uncertainty.

For a scored partition with true mean $\bar{c}_{out,j}$, the component coefficient of determination and its equal-weight average are

$$R_{j,m,t}^2 = 1 - \frac{\sum_i (c_{ij}^{(m,t)} - c_{out,ij})^2}{\sum_i (c_{out,ij} - \bar{c}_{out,j})^2}, \quad \bar{R}_{m,t}^2 = \frac{1}{20} \sum_{j=1}^{20} R_{j,m,t}^2. \quad (4.20)$$

An undefined component denominator makes the corresponding macro-average undefined. The coefficient can be negative when the prediction is worse than the scored-partition

mean. It complements the standardized error but uses a different response-variability reference.

Physical-unit errors are retained for individual components. Secondary composite error is calculated separately for COD, TN, TP, and TSS. The normalized root mean squared error of composite q is its physical-unit root mean squared error divided by $\sigma_q^{(\mathcal{T})}$, the training standard deviation of that composite. This is not the primary component-average nRMSE. Composite standard deviations are calculated after applying the common reporting map to the training targets.

4.7.2 Violation frequencies and their magnitudes

The conservation residual and standardized total negative magnitude of a state c are

$$v_{mc,i}(c) = \|Ac - b_i\|_2, \quad v_{nn,i}^{\text{std}}(c) = \left\| \min(c \oslash \boldsymbol{\sigma}^{(\mathcal{T})}, 0) \right\|_1. \quad (4.21)$$

The minimum is componentwise. The conservation residual uses the same full-precision orthonormal basis for every sample. It is a numerical norm of the adopted invariant coordinates rather than a direct mass concentration in one constituent unit. The negative magnitude uses training standard deviations to compare different component bases without summing incompatible physical units.

The two row-level violation frequencies are

$$\pi_{mc} = \frac{1}{n} \sum_i \mathbb{I}[v_{mc,i}(c_i) > \tau_{mc}], \quad \pi_{nn} = \frac{1}{n} \sum_i \mathbb{I}[\min_j(c_{ij} + \tau_{nn,j}) < 0]. \quad (4.22)$$

A state is counted as nonnegative only when every component passes its tolerance. Component-specific violation frequencies identify which coordinates cross the boundary. The joint feasible fraction is

$$\pi_{\text{feasible}} = \frac{1}{n} \sum_i \mathbb{I}[v_{mc,i}(c_i) \leq \tau_{mc} \text{ and } \min_j(c_{ij} + \tau_{nn,j}) \geq 0]. \quad (4.23)$$

Conservation and nonnegativity frequencies are not added to calculate this fraction because the same state can violate both requirements.

Mean, median, and maximum violation magnitudes are retained along with the frequencies. A hypothetical set of many concentrations just below zero can have a high violation frequency but small total negative magnitude. Another set with one strongly negative state can have low frequency and a large maximum magnitude. The two summaries answer different questions. Physical-unit negative magnitudes are also retained separately for each component.

4.7.3 Paired accuracy effects and displacement

For each outer fold, the effect on the primary error score is

$$\Delta \text{nRMSE}_{m,N,g} = \text{nRMSE}_{m,\text{projected},N,g} - \text{nRMSE}_{m,\text{raw},N,g}. \quad (4.24)$$

A negative difference indicates lower error after projection, and a positive difference indicates higher error. Differences are calculated within paired folds before summary. Component and composite differences are retained because a single aggregate score can conceal opposite changes among outputs.

The sample-level standardized displacement is

$$d_{c,i}^{\text{std},(m)} = \left\| (\tilde{c}_i^{(m)} - c_{\text{raw},i}^{(m)}) \oslash \boldsymbol{\sigma}^{(\mathcal{T})} \right\|_2. \quad (4.25)$$

This norm measures how far the correction moves the state relative to training variability. It measures movement rather than accuracy. A large displacement can repair a large physical inconsistency while moving away from the true kinetic response. Mean, median, and maximum displacement, componentwise physical-unit movement, the fraction requiring backtracking, and the distribution of α are therefore assessed together.

Learning curves retain the full error trajectory over Equation (4.4). As a descriptive summary, the normalized area under an error curve is calculated by trapezoidal integration,

$$\mathcal{A}_{m,t} = \frac{1}{N_{11} - N_1} \sum_{\ell=1}^{10} \frac{E_{m,t}(N_\ell) + E_{m,t}(N_{\ell+1})}{2} (N_{\ell+1} - N_\ell), \quad (4.26)$$

where $E_{m,t}(N)$ is the mean held-out error at size N . This quantity has the same scale as the selected error measure. It summarizes behavior across the declared size interval and does not replace the individual size scores. Training and validation error at the largest size provide a separate descriptive gap. Neither summary establishes an optimal future sample size without an additional learning-curve model (Viering & Loog, 2023).

Each in-domain metric is summarized as the arithmetic mean and sample standard deviation of five outer-fold values. The standard deviation uses denominator four and describes partition-to-partition variation. No hypothesis test or inferential model ranking follows from these summaries. They also do not supply calibrated predictive uncertainty for a new state. The ensemble uncertainty approach of Lakshminarayanan et al. (2017) addresses a different task, and no such uncertainty calibration is assumed here.

4.8 Testing inputs beyond the learning domain

Extrapolation assessment uses separately simulated inputs outside the training hyperrectangle. The final setting for each surrogate is selected with four folds on all 10,000 in-domain states. Its preprocessing and predictor are then refitted on that complete pool. No extrapolation response enters selection, fitting, scaling, or calibration. This separation addresses the domain-shift concern discussed in scientific machine learning by Karniadakis et al. (2021) and Shen et al. (2023).

Four inputs can exceed their upper learning bounds. For input ℓ with bounds $[L_\ell, U_\ell]$, define its normalized exceedance as

$$e_{i\ell} = \frac{x_{i\ell} - U_\ell}{U_\ell - L_\ell}. \quad (4.27)$$

The mild design uses $e_{i\ell} \in [0.10, 0.25]$ and the severe design uses $e_{i\ell} \in [0.35, 0.60]$. The generator targets a mixture of 70% one-input excursions and 30% two-input excursions. All remaining inputs remain within the original bounds. Every candidate therefore lies outside the training hyperrectangle and consequently outside the training-point convex hull.

Table 4.5: Fixed exterior intervals for the two extrapolation designs. The intervals specify input conditions rather than prediction outcomes.

Input	Learning interval	Mild interval	Severe interval	Unit
HRT	[6, 36]	[39, 43.5]	[46.5, 54]	h
u_{air}	[0.5, 2.5]	[2.7, 3.0]	[3.2, 3.7]	1
S_{NH4}	[12, 55]	[59.3, 65.8]	[70.1, 80.8]	g N m ⁻³
X_S	[60, 280]	[302, 335]	[357, 412]	g COD m ⁻³

Each severity design targets 600 accepted steady states and retains the same solver, initialization sequence, and balance-residual criterion. Candidate, accepted, failed, valid-but-unselected, and retained counts are distinguished. The realized one-input and two-input proportions are also distinguished from their design targets. The intervals alone do not establish which exterior conditions are hardest to predict because reactor kinetics can respond differently to the four inputs.

Extrapolation errors are standardized with population component standard deviations from all 10,000 in-domain targets. Composite normalization uses the corresponding full-domain composite standard deviations. The nMSE, nRMSE, nMAE, conservation and nonnegativity diagnostics, displacement, and backtracking diagnostics are calculated on identical raw and projected cases. The coefficient of determination is secondary because the deliberately shifted response distribution changes its denominator.

Each exterior metric is a single descriptive summary over 600 cases from one full-domain refit. It is not an average over the five outer folds and has no fold standard deviation. Satisfaction of the projected constraints is interpreted separately from the predictive accuracy of extrapolation. A state can remain nonnegative and conservative while its estimated nitrogen distribution or biomass abundance is inaccurate.

4.9 Computational cost and the meaning of a fast prediction

Computation is partitioned into hyperparameter search, preprocessing and fitting, raw inference, projection alone, and complete projected inference. Search time is reported once for each distinct study because a largest-scale outer setting is reused at all eleven sizes. Summing that same search duration over the nested size rows would count the selection cost repeatedly. Fitting time excludes the search and reflects the preparation required for the specified training partition.

Inference uses a deterministic batch of 512 rows constructed by cycling through the relevant validation inputs. The batch size stays fixed even when fewer than 512 unique validation observations are available. Five warm-up calls precede 30 timed repetitions for each path. The median batch latency is divided by 512 and expressed as milliseconds per sample. For in-domain assessment, the five fold medians are summarized by their arithmetic mean and sample standard deviation. Exterior timing uses the same batching and repetitions for one final fit and has no fold uncertainty.

Projection timing includes positive-target construction, relative weighting, the weighted solve, conservation and nonnegativity checks, and any positivity backtracking. Complete projected inference includes both raw prediction and correction. All physical operators, returned component states, preprocessing, and metrics use double precision. TabNet’s internal tensors use single precision before its predictions return to the common physical-coordinate calculations. Pseudorandom generation and stochastic fitting use streams derived from seed 42. Deterministic coordinate-descent settings are retained for the multi-task models.

These timing measures describe the declared batch protocol and computational environment. Dividing a batch duration by its row count does not establish the latency of a single isolated query. Likewise, low projection latency would not establish field readiness. A useful predictor must be assessed through its component accuracy, physical diagnostics, domain of use, preparation cost, and the behavior of downstream calculations. The post prediction correction provides a defined admissibility layer within that assessment.

Chapter 5

Interpretable regression with checked physical deployment

5.1 An inspectable model of component transformations

A component prediction is useful for engineering analysis when its dependence on the inputs can be examined and its material accounting can be checked. These requirements address different questions. A transparent regression equation explains how the predicted state depends on operating conditions and influent loads. A conservation test determines whether that state preserves the inventories represented by the reaction network. A nonnegativity test determines whether each predicted concentration is physically meaningful. An explicit equation can fail either physical test. A physically corrected prediction can also remain difficult to interpret if the underlying regression is opaque.

Invariant-constrained second-order regression (ICSOR) combines an explicit polynomial driver, learned dependence among effluent components, and checked deployment. The driver retains separate terms for operating conditions, influent composition, curvature, and interactions. The coupling relation expresses how the driver contributes to the predicted component state. Deployment then checks that state and applies a constrained correction when necessary. The interpretation of the regression and the guarantee provided by deployment therefore remain distinct.

The preference for an inspectable regression follows the argument of Rudin (2019) that transparent model structures can support decisions that require scrutiny. It also follows the broader development of models that combine learned relations with physical structure discussed by Shen et al. (2023). Transparency does not establish that a fitted coefficient is a biological mechanism. It makes the relation available for examination, comparison with process knowledge, and testing across operating conditions.

Activated sludge nutrient removal provides a strong reason to retain interactions.

Carbon availability influences denitrification and the activity of organisms that store phosphorus. Oxygen availability influences both carbon oxidation and nitrification. Wentzel et al. (1992) describe these connected pathways, and Guerrero et al. (2011) examine the role of carbon source in the competition between phosphorus-storing organisms and denitrifiers. A regression that represents an aeration effect independently of loading can overlook such dependencies. An interaction term can describe an association between an operating action and a loading condition while keeping that association visible.

The prediction target is the effluent component vector $c_{out} \in \mathbb{R}^F$. The adopted process basis has $F = 20$ components and $P = 28$ reactions. The operating vector $u \in \mathbb{R}^{M_{op}}$ has $M_{op} = 2$ entries for HRT and aeration level. The full influent vector is $c_{in} \in \mathbb{R}^F$. Component concentrations retain their physical units. The four reported composites are obtained from a fixed map $I_{comp} \in \mathbb{R}^{4 \times F}$. That map is applied after component prediction.

5.2 Conservation in the adopted state space

The stoichiometric matrix $\nu \in \mathbb{R}^{P \times F}$ records reactions as rows and components as columns. Let $V_0 \in \mathbb{R}^{F \times K}$ span its right null space, where $K = 5$. The invariant operator is

$$A = V_0^\top \in \mathbb{R}^{K \times F}, \quad \nu V_0 = 0, \quad A\nu^\top = 0. \quad (5.1)$$

The columns of V_0 are independent, so A has full row rank. With an orthonormal null-space basis, $AA^\top = I_K$. A different full-rank basis describes the same conservation conditions, although its row scaling changes the numerical size of a soft invariant penalty.

For the control volume and boundary accounting represented by the adopted process model, a reaction-induced change has the form

$$c_{out} - c_{in} = \nu^\top \xi, \quad \xi \in \mathbb{R}^P. \quad (5.2)$$

The reaction-progress vector ξ expresses the net contribution of the reactions in concentration-equivalent units. Multiplication by A removes that unknown contribution and gives

$$A(c_{out} - c_{in}) = 0. \quad (5.3)$$

This construction uses the conservation logic of the photochemical surrogate framework of Sturm and Wexler (2020). Sturm and Wexler (2022) further show how a stoichiometric transformation can connect learned reaction information to conservative state changes. The same linear algebra applies here, but the conserved combinations are those of the activated sludge reaction network.

The equality in Equation (5.3) applies only when the modeled boundary supports the change relation in Equation (5.2). External gas transfer, chemical dosing, solids

withdrawal, or another boundary flux must be included in the balance where it occurs. If an explicit boundary contribution s is present, the relevant relation becomes $A(c_{out} - c_{in} - s) = 0$. Using $Ac_{out} = Ac_{in}$ while omitting a boundary contribution would impose the wrong constraint. The invariant operator does not make an incomplete material balance correct.

For a new influent state, the component set used by checked deployment is

$$\mathcal{F}(c_{in}) = \{c \in \mathbb{R}^F \mid Ac = Ac_{in}, c \geq \mathbf{0}\}. \quad (5.4)$$

When $c_{in} \geq \mathbf{0}$, this set contains c_{in} and is therefore nonempty. Its members satisfy the modeled linear invariants and component nonnegativity. Membership does not show that a state solves the nonlinear kinetic equations, is dynamically stable, or is reachable at the stated operating point. The set gives necessary physical conditions within the declared boundary.

The reported composites provide another reason to work in component space. Distinct component states can have the same COD, TN, TP, and TSS values. A negative component can also be hidden by positive contributions to an aggregate. When I_{comp} is entrywise nonnegative, a nonnegative component state produces nonnegative composites. The reverse implication does not generally hold. Conservation and nonnegativity are therefore tested before the component state is collapsed into the four reported quantities.

5.3 A partitioned second-order driver

5.3.1 Separating operation from loading

Operating conditions and influent composition play different engineering roles. HRT and aeration define aspects of the reactor environment. Influent components define the material presented to that environment. This distinction is consistent with the operating-sensitivity analysis of de Araujo et al. (2013) and the separation of operational and influent uncertainty examined by Machado et al. (2014).

The feature vector retains both roles through the ordered expansion

$$\phi(u, c_{in}) = \begin{bmatrix} 1 \\ u \\ c_{in} \\ u \otimes u \\ c_{in} \otimes c_{in} \\ u \otimes c_{in} \end{bmatrix} \in \mathbb{R}^D, \quad D = 1 + M_{op} + F + M_{op}^2 + F^2 + M_{op}F. \quad (5.5)$$

The symbol $\otimes$ denotes the Kronecker product. The adopted dimensions give $D = 467$. The ordered expansion includes both $u_i u_j$ and $u_j u_i$, and both $c_{in,i} c_{in,j}$ and $c_{in,j} c_{in,i}$. These pairs are equal as numerical features. Their interpretation requires the symmetry convention developed below.

The features produce a driver $r \in \mathbb{R}^F$ through

$$r(u, c_{in}) = B\phi(u, c_{in}), \quad B \in \mathbb{R}^{F \times D}. \quad (5.6)$$

The driver is an intermediate regression quantity. It is not a vector of mechanistic reaction rates and is not the final deployed state. Its coefficients have no sign restriction, so the regression can represent both increasing and decreasing effects. Nonnegativity is imposed on component states where it has a physical meaning. The driver block form is

$$\begin{aligned} r(u, c_{in}) = & b + W_u u + W_{in} c_{in} + \Theta_{uu}(u \otimes u) \\ & + \Theta_{cc}(c_{in} \otimes c_{in}) + \Theta_{uc}(u \otimes c_{in}). \end{aligned} \quad (5.7)$$

Here $b \in \mathbb{R}^F$ is a baseline term. The matrices $W_u \in \mathbb{R}^{F \times M_{op}}$ and $W_{in} \in \mathbb{R}^{F \times F}$ contain first-order operating and influent terms. The matrices $\Theta_{uu} \in \mathbb{R}^{F \times M_{op}^2}$ and $\Theta_{cc} \in \mathbb{R}^{F \times F^2}$ contain curvature within each input group. The matrix $\Theta_{uc} \in \mathbb{R}^{F \times M_{op} F}$ contains operating-load interactions.

5.3.2 Illustrative operating and loading effects

Consider an illustrative driver for one component with aeration level a and influent ammonium concentration n . A term of the form

$$r_f = \beta_0 + \beta_a a + \beta_n n + \beta_{aa} a^2 + \beta_{an} a n \quad (5.8)$$

has an aeration derivative $\partial r_f / \partial a = \beta_a + 2\beta_{aa} a + \beta_{an} n$. The effect described by this derivative changes with both aeration level and ammonium loading. This example uses a hypothetical polynomial and reports no fitted relation. It explains why a single linear aeration coefficient cannot summarize the response represented by the entire polynomial.

The same reasoning applies to simultaneous influent loads. A carbon-ammonium interaction allows the modeled association with carbon availability to depend on nitrogen loading. A phosphorus-storage interaction allows the modeled association with dissolved phosphate to depend on a particulate storage pool. These terms provide a transparent approximation to a nonlinear response. They do not replace the microbial pathways described by Henze et al. (1999), and their signs do not establish causation.

The choice of a second-order library limits the model deliberately. It can represent smooth local curvature and pairwise input interactions. It cannot reproduce every threshold, saturation relation, or higher-order interaction exactly. Explicit nonlinear libraries are useful for inspecting candidate structure, as illustrated by Brunton et al. (2016). Their usefulness depends on the sampling domain, the adequacy of the library, and the stability of the fitted terms.

5.3.3 Symmetry and the meaning of cross terms

For component f , reshape the corresponding rows of Θ_{uu} and Θ_{cc} into square matrices $Q_{uu}^{(f)}$ and $Q_{cc}^{(f)}$. A quadratic term can then be written as $x^\top Qx$. For any real square matrix,

$$x^\top Qx = x^\top \text{sym}(Q)x, \quad \text{sym}(Q) = \frac{Q + Q^\top}{2}. \quad (5.9)$$

The skew-symmetric part contributes nothing to the prediction. It can nevertheless increase the coefficient norm because

$$\|Q\|_F^2 = \|\text{sym}(Q)\|_F^2 + \|\text{skew}(Q)\|_F^2. \quad (5.10)$$

Replacing each square slice by its symmetric part preserves every fitted value and cannot increase a positive ridge penalty. The projection is applied after each driver update. With a strictly positive driver ridge weight, the exact minimum-norm solution has no skew-symmetric part.

Symmetry also determines how a coefficient display is read. If $q_{ij} = q_{ji}$, the two displayed off-diagonal cells together contribute $2q_{ij}x_i x_j$. A displayed cell is therefore half the coefficient of the corresponding unordered cross term. For an illustrative matrix with $q_{12} = q_{21} = 0.3$, the polynomial contribution is $0.6x_1 x_2$. Reading either cell as the entire interaction coefficient would halve its contribution.

Each component has $M_{op}(M_{op} + 1)/2 = 3$ distinct operating quadratic terms and $F(F + 1)/2 = 210$ distinct influent quadratic terms. Together with the baseline, linear terms, and operating-load interactions, the library has 276 distinct polynomial terms per component. These are structural dimensions of the model. They do not describe how many coefficients are statistically important or biologically meaningful.

5.4 Learned coupling among effluent components

The regression includes a matrix $\Gamma \in \mathbb{R}^{F \times F}$ that describes fitted dependence among the effluent components. Define

$$R = I_F - \Gamma, \quad Rc = r(u, c_{in}). \quad (5.11)$$

Equivalently, the equation for component f is $c_f = r_f + \sum_{j \neq f} \Gamma_{fj} c_j$. This form separates the input-dependent driver from dependence among the predicted component channels. The diagonal of Γ is fixed to zero to avoid a redundant self-coupling coefficient.

The sign of Γ_{fj} describes reinforcement or compensation within this fitted simultaneous relation. It does not describe a one-way biochemical reaction from component j to component f . Activated sludge reactions involve several components at once, and the stoichiometric matrix already specifies those transformations. The coupling matrix is learned from state patterns. Its entries can reflect correlated transformations, loading patterns, and the allocation of effects between the driver and coupling blocks.

When R is nonsingular, the raw component prediction is $R^{-1}r$. The dependence on R^{-1} means that the sign of a driver coefficient alone does not determine the response of the raw component prediction. Coupling can redistribute or amplify that response. For interpretation, the driver, coupling matrix, and complete raw mapping are examined together.

The adopted coupling constraints require

$$\Gamma_{ff} = 0, \quad |\Gamma_{fj}| \leq \gamma_{abs} \quad (f \neq j), \quad \kappa_2(I_F - \Gamma) \leq \kappa_{max}. \quad (5.12)$$

The condition number is the ratio of the largest to smallest singular value of the coupled-system matrix. A nearly singular matrix can amplify small driver perturbations and numerical errors. Bounding individual entries does not by itself guarantee a well-conditioned matrix, so the condition check is separate from the box constraints.

Shrinking a candidate coupling matrix toward zero can provide a safe candidate because $I_F - 0 = I_F$. A zero coupling matrix gives a condition number of one. A prescribed $\kappa_{max} > 1$ therefore permits this fallback. The admissible coupling set contains a nonlinear condition-number requirement. It is not generally a convex set, even though the box-constrained estimation problem used to propose a coupling update is convex.

5.5 Fitting in prediction space

5.5.1 The coupled training objective

For N training observations, $\Phi \in \mathbb{R}^{N \times D}$ contains feature vectors as rows. The matrices C_{in} and C_{out} contain influent and reference effluent states in the same row convention. An auxiliary fitted-state matrix $\hat{C} \in \mathbb{R}_+^{N \times F}$ separates the predicted training states from the regression coefficients. The training criterion is

$$\begin{aligned} J(B, \Gamma, \hat{C}) = & \|C_{out} - \hat{C}\|_F^2 + \lambda_{inv} \|(\hat{C} - C_{in})A^\top\|_F^2 \\ & + \lambda_{sys} \|\hat{C}(I_F - \Gamma)^\top - \Phi B^\top\|_F^2 \\ & + \lambda_B \|B\|_F^2 + \lambda_\Gamma \|\Gamma\|_F^2. \end{aligned} \quad (5.13)$$

The estimation problem minimizes this criterion subject to nonnegative fitted training states, zero coupling diagonal, coupling box bounds, and the conditioning safeguard. All penalty weights are nonnegative. The adopted ridge weights and coupled-system weight are strictly positive.

The first term measures component prediction error. The second term discourages disagreement with the modeled invariants. The third term encourages agreement between the fitted states and the coupled regression equation. The final two terms limit coefficient magnitude. A nonnegative fitted training state can therefore balance prediction error, invariant mismatch, and coupled-system mismatch. It need not satisfy either equality exactly.

This distinction matters for a new input. Neither a nonnegative $\hat{C}$ nor a finite invariant penalty guarantees that $R^{-1}B\phi$ is nonnegative or conservative. The auxiliary state is a variable in the fitting problem, while the raw prediction is obtained by solving the learned coupled relation. A soft residual penalty encourages agreement but does not impose equality. Physics-informed residual training uses this general penalty principle, as developed by Raissi et al. (2019). Exact feasibility requires the separate deployment conditions below.

The objective is expressed in physical component coordinates. Components have different units and numerical scales, so the magnitudes of their residuals and coefficients reflect those conventions. The four penalty weights also depend on the fixed feature and invariant scaling. They cannot be transferred unchanged to arbitrary unit systems. Any rescaling must transform the corresponding features, coefficients, invariant operator, and reporting map consistently. Pinheiro et al. (2025) discuss the influence of feature scaling on learning. For physical interpretation, unit consistency is as necessary as numerical conditioning.

5.5.2 Deterministic block updates

The term $\hat{C}\Gamma^\top$ makes the joint objective nonconvex. Holding two variable blocks fixed gives a simpler problem in the remaining block. The fitting procedure updates the driver, coupling matrix, and fitted states in that order within each cycle.

For fixed Γ and $\hat{C}$, the driver update is a ridge-regularized linear least-squares problem. Its exact solution satisfies

$$B^\top = \left(\lambda_{sys} \Phi^\top \Phi + \lambda_B I_D \right)^{-1} \lambda_{sys} \Phi^\top \hat{C} R^\top. \quad (5.14)$$

The positive driver ridge weight makes the matrix in this expression positive definite, even when ordered quadratic columns are duplicated. In computation, the corresponding linear system is solved. An explicit matrix inverse is unnecessary. The square quadratic slices are then symmetrized without changing their predictions.

For fixed B and $\hat{C}$, define $E = \hat{C} - \Phi B^\top$. The coupling proposal solves

$$\min_{\Gamma} \lambda_{sys} \|E - \hat{C}\Gamma^\top\|_F^2 + \lambda_\Gamma \|\Gamma\|_F^2 \quad (5.15)$$

subject to its zero diagonal and box bounds. This is a convex quadratic program. Its candidate is subsequently screened using Equation (5.12). A candidate that fails the condition check is shrunk toward zero or replaced by an admissible fallback. Screening preserves numerical admissibility, but it need not preserve the minimum of the unrestricted convex subproblem.

For fixed B and Γ , the fitted-state update separates over the training rows. For sample i , write $\hat{c}_i$ as a column vector and define

$$\begin{aligned} H_C &= I_F + \lambda_{inv} A^\top A + \lambda_{sys} R^\top R, \\ h_i &= c_{out,i} + \lambda_{inv} A^\top A c_{in,i} + \lambda_{sys} R^\top B \phi_i. \end{aligned} \quad (5.16)$$

Each row minimizes $\hat{c}_i^\top H_C \hat{c}_i - 2h_i^\top \hat{c}_i$ over $\hat{c}_i \geq \mathbf{0}$. The identity term makes H_C positive definite, so the row minimizer is unique. Clipping the unconstrained solution componentwise generally does not solve this problem because the off-diagonal terms in H_C connect component adjustments.

The quadratic-program method of Stellato et al. (2018) provides an algorithmic foundation for the constrained coupling and fitted-state blocks. These blocks use their quadratic structure directly. A generic stochastic first-order method could also optimize a related criterion, but it would require separate mechanisms to control nonnegativity, coupling bounds, and conditioning. The block formulation keeps each requirement attached to an explicit subproblem.

5.5.3 Descent, stopping, and numerical safeguards

Exact minimization over a block cannot increase the objective when its admissible set contains the previous block value. Since $J \geq 0$, an objective sequence that is monotone non-increasing converges in value. This argument does not establish a global minimum of the joint problem. The convergence analysis of Razaviyayn et al. (2013) specifies additional conditions needed to relate blockwise minimization to stationary solutions.

The conditioning screen requires particular care. Shrinking a coupling proposal can increase the training criterion relative to the previous admissible coupling matrix. The procedure therefore evaluates the objective after screening and retains the previous admissible matrix when a screened candidate cannot provide an acceptable non-increasing update. Solver tolerances also require an explicit allowance for numerical error. A claim of monotone descent applies to accepted objective-decreasing updates. It does not follow from box-constrained minimization alone.

Each restart begins with an admissible coupling matrix and nonnegative fitted states. A ridge solve supplies the initial driver. Full cycles continue until an iteration limit is reached or the slope of the trailing objective history is sufficiently small. A running minimum records the best admissible state reached. When several restarts are used, the admissible state with the smallest recorded objective is retained. A nearly flat objective indicates limited further progress under that stopping rule. It does not certify parameter uniqueness or global optimality.

Table 5.1 specifies numerical fitting choices. The condition threshold κ_{max} remains an additional prescribed numerical safeguard and is checked independently of the coefficient bound. Hyperparameter assessment uses validation observations within the training pool. Held-out test observations do not determine penalty weights or stopping choices. Fitting settings remain fixed within a sample-size comparison so that changes in the learning curve reflect changing sample availability under a common model specification.

Table 5.1: Specified fitting settings for the structured regression.

Setting	Value
Baseline feature	Included
Invariant penalty λ_{inv}	8.68×10^{-3}
Coupled-system penalty λ_{sys}	1.27×10^{-3}
Driver ridge penalty λ_B	6.09×10^{-5}
Coupling ridge penalty λ_Γ	6.55×10^{-4}
Off-diagonal bound γ_{abs}	0.368
Maximum full fitting cycles	60
Restarts	1
Trailing objective window	12 cycles
Objective slope tolerance	2.85×10^{-7}

5.6 Checked deployment

5.6.1 The raw coupled state

For a new pair $(u_*, c_{in,*})$, the fitted parameters give

$$\hat{R} = I_F - \hat{\Gamma}, \quad d_* = \hat{B}\phi(u_*, c_{in,*}), \quad \hat{R}c_{raw} = d_*. \quad (5.17)$$

The notation $c_{raw} = \hat{R}^{-1}d_*$ describes the mapping mathematically. A linear-system solve evaluates it numerically. The accepted conditioning bound limits the sensitivity of that solve. The raw state is returned only when it satisfies both the invariant and nonnegativity checks.

For any candidate c , define the diagnostic residuals

$$v_{mc}(c) = \|A(c - c_{in,*})\|_2, \quad v_{nn}(c) = \|\min(c, \mathbf{0})\|_1. \quad (5.18)$$

The minimum is taken componentwise. The adopted diagnostic tolerance is 10^{-10} . Algebraic equality and nonnegativity define the mathematical requirement. Residuals below the numerical tolerance define its computational assessment. Invariant residual magnitudes are meaningful only under the stated scaling of A .

5.6.2 Affine restoration of the invariants

When the raw state fails its checks, the first correction solves

$$c_{aff} = \arg \min_{c \in \mathbb{R}^F} \|c - c_{raw}\|_2^2 \quad \text{subject to} \quad Ac = Ac_{in,*}. \quad (5.19)$$

Introduce a Lagrange multiplier $\eta \in \mathbb{R}^K$. Stationarity gives $c - c_{raw} + A^\top \eta = 0$. Substitution into the equality gives $AA^\top \eta = A(c_{raw} - c_{in,*})$. The full row rank of A

therefore gives the unique solution

$$c_{aff} = c_{raw} - A^\top (AA^\top)^{-1} A(c_{raw} - c_{in,*}). \quad (5.20)$$

The fixed matrix $A^\top (AA^\top)^{-1} A$ can be prepared once. The affine correction is then a matrix-vector operation for each new prediction.

This correction is an orthogonal projection onto the invariant-consistent affine set in the chosen numerical component coordinates. It removes the part of the prediction error normal to that set. It retains the part tangent to the set, which can still be large. The correction therefore enforces bookkeeping without establishing predictive accuracy. Its distance also depends on the component unit convention. Sturm and Silva (2025) develop related conservation-restoring corrections and explain why species-dependent weights matter when concentration scales differ.

The affine state is checked for both residual conservation and nonnegativity before deployment. The equality follows analytically from Equation (5.20), but checking its residual catches numerical or operator inconsistency. A conservative affine state can still contain a negative concentration. That possibility requires a correction with inequality constraints.

5.6.3 Joint conservation and nonnegativity

When the affine state fails the nonnegativity check, the final correction solves

$$\begin{aligned} \hat{c} &= \arg \min_{c, \rho \in \mathbb{R}^F} w_c^\top \rho \\ \text{subject to } & Ac = Ac_{in,*}, \\ & c \geq \mathbf{0}, \\ & -\rho \leq c - c_{aff} \leq \rho, \\ & \rho \geq \mathbf{0}. \end{aligned} \quad (5.21)$$

For strictly positive fixed weights w_c , minimization makes $\rho_f = |c_f - c_{aff,f}|$ at an optimum. The objective is therefore the weighted absolute adjustment $\sum_f w_{c,f} |c_f - c_{aff,f}|$. The reference is the affine state. The program does not minimize distance to the original raw state or re-estimate the regression coefficients.

The program is linear because the absolute values are represented by auxiliary variables and linear inequalities. The revised-simplex method described by Huangfu and Hall (2018) provides an algorithmic foundation for solving this type of correction. The mathematical feasible set is nonempty whenever the supplied influent state is nonnegative under the adopted boundary relation. The positive weights make the adjustment objective bounded below and discourage unconstrained movement of any component. A minimizer exists, although several minimizing states can exist when the absolute-distance objective has a flat face.

Large weights make changes to a component more costly. Small weights permit larger changes when those changes help restore feasibility. Weights are fixed using the intended component scaling and scientific priorities. They are not chosen after seeing the reference target for a deployed observation. A sensitivity analysis of the weights can examine how much the returned state depends on that modeling choice. Equal numerical weights also embody a choice because component units and typical magnitudes differ.

Kircher and Votsmeier (2025) show why a conservation correction needs simultaneous positivity control in reaction-kinetics surrogates. Their scaled-space treatment protects low-concentration species during correction. Equation (5.21) addresses the same joint requirement through a weighted absolute-distance problem for activated sludge components. It imposes the adopted linear invariants and nonnegativity on the returned state.

5.6.4 Illustrative correction and the failure of clipping

Consider an illustrative two-component system with $A = [1 \ 1]$, a total influent inventory of 10, and a raw prediction $c_{raw} = (12, 1)^\top$. The raw total is 13. The affine formula subtracts 1.5 from each component and gives $c_{aff} = (10.5, -0.5)^\top$. The total is now 10, but the second component is negative.

Clipping that negative value to zero produces $(10.5, 0)^\top$, which violates the total inventory. With equal correction weights, Equation (5.21) instead returns $(10, 0)^\top$. Both constraints hold. These values are a hypothetical illustration of the correction geometry. They are not activated sludge predictions or experimental findings.

The deployment order uses this geometry directly. A feasible raw state needs no adjustment. An infeasible raw state is first projected onto the conservation set. A feasible affine state can then be returned. The linear program is required only when the affine state still fails nonnegativity. Every returned candidate is checked under the declared numerical tolerances. An unsuccessful numerical solve does not authorize an unchecked prediction.

The component state returned by this rule is the deployed output. Its reported quantities are

$$\hat{y}_{\text{comp}} = I_{\text{comp}} \hat{c}. \quad (5.22)$$

The deployment constraint is responsible for admissibility. The soft training penalties can reduce the need for correction, but that possibility is an evaluation question rather than a mathematical guarantee.

5.7 Interpreting the complete prediction map

5.7.1 Driver coefficients and back projection

The rows of $\hat{B}$ refer to component drivers. They cannot be labeled directly as COD, TN, TP, or TSS coefficients. The raw component coefficient matrix is

$$B_{\text{raw}} = \hat{R}^{-1} \hat{B}, \quad c_{\text{raw}} = B_{\text{raw}} \phi. \quad (5.23)$$

The corresponding map in reported-composite space is

$$B_{\text{raw,comp}} = I_{\text{comp}} \hat{R}^{-1} \hat{B}, \quad y_{\text{raw,comp}} = B_{\text{raw,comp}} \phi. \quad (5.24)$$

This transformation carries the driver coefficients through the learned coupling and then through the reporting operator. Inspection of a COD quadratic block uses its row in $B_{\text{raw,comp}}$, rather than an arbitrary row of $\hat{B}$.

The named blocks can be displayed as coefficient maps. Operating terms are grouped separately from influent terms. Symmetric quadratic maps display both ordered cells while reporting unique cross terms consistently. Operating-load maps show which operating variables interact with which influent components. The coupling matrix is displayed separately because it has a different interpretation from the input-driver terms.

A coefficient-display threshold is used only to simplify reporting after fitting. For each reported-composite raw-head map, the threshold is 10^{-4} times its largest absolute coefficient. For the coupling matrix, the threshold is 10^{-4} times its largest absolute off-diagonal entry. The threshold does not modify the fitted mapping, select hyperparameters, or impose sparsity during training. Symmetric quadratic summaries count unique upper-triangular terms. A small coefficient can still have a large contribution when multiplied by a large feature. Interpretation therefore examines coefficient magnitude, input scale, and the contribution of the corresponding term over the declared domain.

5.7.2 Analytical jacobians of the raw head

A coefficient describes a term in the polynomial. A Jacobian describes the change in the whole mapping at an operating point. For component f , let $H_{uc}^{(f)} \in \mathbb{R}^{M_{op} \times F}$ reshape its interaction coefficients so that the cross term is $u^\top H_{uc}^{(f)} c_{in}$. The driver can then be expressed as

$$\begin{aligned} r_f = & b_f + (W_u)_{f,:} u + (W_{in})_{f,:} c_{in} + u^\top Q_{uu}^{(f)} u \\ & + c_{in}^\top Q_{cc}^{(f)} c_{in} + u^\top H_{uc}^{(f)} c_{in}. \end{aligned} \quad (5.25)$$

With symmetric square slices, its gradients are

$$\begin{aligned} \nabla_u r_f &= (W_u)_{f,:}^\top + 2Q_{uu}^{(f)} u + H_{uc}^{(f)} c_{in}, \\ \nabla_{c_{in}} r_f &= (W_{in})_{f,:}^\top + 2Q_{cc}^{(f)} c_{in} + (H_{uc}^{(f)})^\top u. \end{aligned} \quad (5.26)$$

Stacking the transposed gradients gives $J_{r,u} \in \mathbb{R}^{F \times M_{op}}$ and $J_{r,c} \in \mathbb{R}^{F \times F}$. The raw-state and raw-composite Jacobians follow through the same back projection as the coefficients,

$$\begin{aligned} J_{raw,u} &= \hat{R}^{-1} J_{r,u}, & J_{raw,c} &= \hat{R}^{-1} J_{r,c}, \\ J_{raw,comp,u} &= I_{comp} J_{raw,u}, & J_{raw,comp,c} &= I_{comp} J_{raw,c}. \end{aligned} \quad (5.27)$$

These derivatives describe the complete fitted raw head at the specified inputs. They include curvature and interactions. They do not include the physical correction applied during deployment.

An illustrative aeration sensitivity can be inspected at low and high ammonium loading while holding the remaining inputs fixed. A difference in the two derivatives identifies a loading-dependent association within the fitted model. Oxygen-transfer limitations and heterotrophic competition provide process context for assessing such a pattern, as discussed by Stenstrom and Song (1991). Agreement with that context is a plausibility check. It is not evidence that changing aeration in a plant will cause exactly the fitted response.

5.7.3 Sensitivity after the affine correction

Define the complementary projection matrices

$$Q_A = A^\top (A A^\top)^{-1} A, \quad P_A = I_F - Q_A. \quad (5.28)$$

Then $c_{aff} = P_A c_{raw} + Q_A c_{in}$. With the fitted coefficients and invariant operator held fixed, its Jacobians are

$$J_{aff,u} = P_A J_{raw,u}, \quad J_{aff,c} = P_A J_{raw,c} + Q_A. \quad (5.29)$$

The second expression includes the change in the conservation reference when the influent changes. Omitting Q_A would treat that reference as fixed and give the wrong influent sensitivity.

Multiplication by A gives $A J_{aff,u} = 0$ and $A J_{aff,c} = A$. An operating perturbation changes the affine prediction only along directions that preserve its fixed influent inventories. An influent perturbation changes those inventories consistently with the influent reference. These relations provide analytical checks on the sensitivity calculation.

When the affine branch is used over a neighborhood of inputs, its coefficients and Jacobians describe that branch. The complete deployment rule still includes transitions between raw, affine, and linear-program branches. A single coefficient map does not describe every branch at once.

5.7.4 Active constraints in the final correction

The linear program can make a component concentration equal to zero or bind one of the absolute-adjustment inequalities. These binding constraints define an active set.

If the optimal solution is unique and its nondegenerate optimal basis remains unchanged over a local neighborhood, the solution varies affinely with the relevant right-hand sides in that neighborhood. A branch-specific sensitivity can then be derived from that fixed basis.

At an active-set transition, the derivative can change abruptly. When several optimal corrected states exist, the returned solution can also depend on the rule used to select among them. A global raw-head coefficient display cannot resolve these effects. Local deployment sensitivity is assessed by differentiating a verified fixed basis where appropriate or by perturbing admissible inputs and resolving the entire deployment problem. The perturbation size and any branch changes are recorded as part of that sensitivity assessment.

This distinction prevents three different quantities from being conflated. A driver coefficient belongs to $\hat{B}$. A raw-state sensitivity passes through $\hat{R}^{-1}$. A deployed-state sensitivity also depends on the conservation reference and active correction constraints. Each quantity answers a different question about the model.

5.8 Assessment design for the structured regression

5.8.1 Common inputs and comparator models

The structured-regression assessment uses a design collection of 10,000 accepted steady-state reactor cases. Every method receives the same 22 physical inputs and predicts the same 20 effluent components. Its reported composites are obtained from the common composition matrix. The input domain, reaction model, simulation bounds, and acceptance criterion remain those of the declared reactor benchmark. The comparison therefore changes the predictor while holding the modeled process and prediction task fixed.

The nine unconstrained comparators are XGBoost, LightGBM, CatBoost, AdaBoost, random forest, support vector regression, k -nearest neighbors, partial least squares, and a fully connected multilayer perceptron. These methods represent boosted and randomized tree ensembles, kernel regression, local neighbor averaging, latent linear regression, and nonlinear neural regression. Their foundations include the ensemble construction of Breiman (2001), the boosting formulations of Chen and Guestrin (2016); Drucker (1997); Ke et al. (2017); Prokhorenkova et al. (2018), and the latent regression of Wold et al. (2001). The comparators have no invariant penalty, auxiliary nonnegative fitted-state matrix, or learned component-coupling equation during fitting.

The neural comparator has hidden layers with 128, 64, and 32 units and logistic activation. Dense connections map the 22 inputs through these layers to 20 component outputs. Its capacity is evaluated under the same samples and reported-composite map as the other predictors. A transparent polynomial and a neural network can therefore

differ in their internal structure without changing the supervised target.

5.8.2 Training partitions and hyperparameter selection

The primary assessment uses a row-wise 80% training and 20% held-out split with random seed 42. The specified collection gives a training pool of 8000 rows and a held-out set of 2000 rows. Hyperparameter selection uses only the training pool. Half of that pool forms a 4000-row tuning subset. A further 80% and 20% split gives 3200 tuning-training rows and 800 tuning-validation rows. This is one internal validation split rather than a cross-validation design.

Each predictor receives 100 hyperparameter trials under a seeded tree-structured Parzen estimator with seed 42. No trial pruning or wall-clock timeout is imposed. Bergstra et al. (2011) develop this search approach, while Akiba et al. (2019) provide the associated hyperparameter optimization framework. The selection criterion is mean squared error in the shared physical component space on the internal validation set. Selected settings are held fixed for the primary assessment and the repeated sample-size design. The final primary fit uses the full 8000-row training pool.

For the unconstrained comparators, input and output standardization parameters are estimated from the relevant training partition. Predictions are transformed back to physical component coordinates before any invariant check, reporting map, or error assessment. ICSOR remains in physical coordinates so that its driver, invariant operator, and deployed constraints use the same component convention. Internal-validation transformations are estimated from tuning-training rows only. Held-out observations do not determine the means, standard deviations, hyperparameters, or correction weights.

Table 5.2 gives the specified numerical configurations. These values define fitted model settings rather than measured predictive performance. The internal search compares candidate configurations according to the stated validation objective.

5.8.3 Scoring states and physical diagnostics

The scored ICSOR output is the final checked component state. For comparator accuracy assessment, the raw component prediction receives the same equality-only affine alignment in Equation (5.20). The composition map is then applied to the aligned state. This comparison describes the accuracy of a checked structured prediction and equality-aligned comparator predictions. The comparator alignment imposes neither component nonnegativity nor the weighted linear correction.

Physical diagnostics use the direct raw states of the comparators. For ICSOR, diagnostics distinguish the raw coupled state, affine state, and final deployed state. This stage separation prevents equality alignment used during scoring from being attributed

Table 5.2: Specified configurations of the nine unconstrained comparators.

Comparator	Configuration
XGBoost	309 trees, learning rate 0.125, maximum depth 3, minimum child weight 6.84, row fraction 0.915, feature fraction 0.828, absolute penalty 1.18×10^{-3} , and squared penalty 0.382.
LightGBM	302 trees, learning rate 0.0809, 16 leaves, maximum depth 8, minimum leaf count 14, row fraction 0.845, feature fraction 0.869, absolute penalty 3.17×10^{-3} , and squared penalty 3.36×10^{-4} .
CatBoost	236 boosting iterations, learning rate 0.172, depth 4, leaf squared penalty 0.0266, bagging temperature 0.0277, and random strength 0.0657.
AdaBoost	236 estimators, learning rate 0.999, and squared regression loss.
Random forest	390 trees, maximum depth 14, minimum split count 3, minimum leaf count 1, feature fraction 0.500, and no bootstrap resampling.
Support vector regression	Polynomial kernel of degree 4, regularization parameter 0.118, insensitive-error width 0.0458, constant offset 0.889, and solver tolerance 8.96×10^{-4} . The kernel scale is the reciprocal of the product of input dimension and training input variance.
k -nearest neighbors	15 neighbors, inverse-distance weighting, Manhattan distance, and a k -dimensional search tree with leaf size 40.
Partial least squares	6 latent components, maximum 887 fitting iterations, and tolerance 9.91×10^{-5} .
Multilayer perceptron	Hidden widths (128, 64, 32), logistic activation, adaptive moment estimation, squared penalty 6.33×10^{-4} , batch size 64, initial learning rate 1.08×10^{-3} , maximum 500 iterations, and tolerance 1.63×10^{-5} . Training rows are not shuffled, and early stopping is disabled.

to the learner. At each stage, the violation frequency for diagnostic v is

$$f_v = \frac{100}{n} \sum_{i=1}^n \mathbf{1}\{v(c_i) > 10^{-10}\}, \quad (5.30)$$

where n is the number of assessed observations. The diagnostics v_{mc} and v_{nn} are defined in Equation (5.18). Frequencies, residual magnitudes, adjustment distances, and deployment-branch use describe different aspects of the same prediction procedure. None is treated as a substitute for predictive error.

5.8.4 Composite error measures

For reported quantity $q \in \{1, 2, 3, 4\}$, define the prediction residual $e_{iq} = \hat{y}_{iq} - y_{iq}$. The five measures are mean squared error, root mean squared error, mean absolute error, mean absolute relative error, and the coefficient of determination. They are computed per reported quantity and summarized across the four quantities. The squared and absolute measures are

$$E_{2,q} = \frac{1}{n} \sum_i e_{iq}^2, \quad E_{root,q} = \sqrt{E_{2,q}}, \quad E_{1,q} = \frac{1}{n} \sum_i |e_{iq}|. \quad (5.31)$$

For strictly nonzero reference values, the relative measure is

$$E_{rel,q} = \frac{1}{n} \sum_i \frac{|e_{iq}|}{|y_{iq}|}. \quad (5.32)$$

It is reported as a ratio. Multiplication by 100 converts it to a percentage. A zero reference value makes this expression undefined, and a very small value can dominate it. Any zero-reference convention is declared explicitly before comparing predictors. A finite relative-error summary is not assigned to a target for which the unmodified expression is undefined.

The per-target coefficient of determination is

$$R_q^2 = 1 - \frac{\sum_i e_{iq}^2}{\sum_i (y_{iq} - \bar{y}_q)^2}. \quad (5.33)$$

A target with zero reference variance makes this expression undefined. A negative value is possible when the predictor has a greater squared error than the reference-mean predictor. This coefficient measures relative prediction error and is not a correlation coefficient.

The aggregate squared error pools the four composite residuals. Its root is

$$E_{root} = \left(\frac{1}{4n} \sum_{q=1}^4 \sum_{i=1}^n e_{iq}^2 \right)^{1/2}. \quad (5.34)$$

Aggregate absolute and relative errors use the corresponding mean over observations and composites. Aggregate R^2 uses the mean of the four target-specific coefficients.

The composite coordinates retain their own physical units. Their pooled error is a numerical benchmark summary under that coordinate convention, rather than a single common material quantity. Per-target scores remain necessary to show which reported quantities contribute to an aggregate comparison.

The primary ordering measure is aggregate root mean squared error. The other measures provide complementary assessments. Squared error emphasizes larger residuals, absolute error gives average deviation, relative error emphasizes deviations in relation to the reference magnitude, and R^2 compares the squared residual with reference variation. A predictor need not receive the same ordering under all these measures.

5.8.5 Repeated sample-size assessment

The sample-size design uses

$$n \in \{500, 1450, 2400, 3350, 4300, 5250, 6200, 7150, 8100, 9050, 10000\}. \quad (5.35)$$

At each size, ten subsamples are drawn without replacement from the design collection. Deterministic seeds are derived from seed 42. Each run uses a row-wise 80% training and 20% held-out split. All ten predictors receive the same sampled rows and partitions. The design specifies 110 fits per predictor across the size schedule. This is repeated random splitting rather than ten-fold cross-validation.

The fixed hyperparameters isolate sensitivity to available observations under one model specification. The mean and spread of each held-out score are summarized across the ten runs. Learning curves are assessed over the entire schedule because behavior at one sample size does not establish behavior at other sizes. Geman et al. (1992) explain the relation between flexibility and the bias-variance problem, and Viering and Loog (2023) review the varied shapes of learning curves.

Let $\overline{M}_{m,j}(n)$ denote the mean held-out measure j for predictor m at sample size n . The normalized area under its learning curve is

$$L_{m,j} = \frac{1}{n_{max} - n_{min}} \int_{n_{min}}^{n_{max}} \overline{M}_{m,j}(n) \, dn. \quad (5.36)$$

The trapezoidal rule evaluates this integral over the stated sample sizes. Normalization by the sample-size interval gives an average height of the learning curve and preserves the units of measure j . It does not normalize the prediction error by a target standard deviation. Cawley (2011); Guyon et al. (2011) provide related trajectory summaries in active learning. The present design changes the number of sampled observations without choosing observations through an active acquisition rule.

For each measure j , reported quantity q , and retained size n_s , predictors receive a dense rank $r_{m,j,q}(n_s)$ from their mean held-out score. Lower squared, root, absolute, and relative errors receive better ranks. Larger R^2 values receive better ranks. Equal

scores receive equal ranks, and the next distinct score receives the next consecutive rank. With $S = 11$ sizes and five measures, the summaries are

$$\bar{r}_{m,j} = \frac{1}{4S} \sum_{q=1}^4 \sum_{s=1}^S r_{m,j,q}(n_s), \quad \bar{r}_m = \frac{1}{5} \sum_{j=1}^5 \bar{r}_{m,j}. \quad (5.37)$$

These are descriptive summaries of relative ordering. They do not establish statistical significance. Demšar (2006) develop broader rank-based model comparisons, while Shevchenko et al. (2024) emphasize variability in benchmarking. Their contributions support careful comparative assessment without equating a mean rank with an absolute error or a significance test.

At the largest sample size, the generalization gap is

$$\Delta E_{root,m} = E_{root,m}^{held-out} - E_{root,m}^{training}. \quad (5.38)$$

A positive gap indicates greater held-out error. A small gap can accompany either accurate prediction or high-error underfitting, so it is interpreted with the corresponding absolute scores. Training time is recorded for every repeated fit and summarized as a distribution across runs. Deployment effort is assessed separately through prediction time, correction-branch use, and constrained-solve effort. This separation makes the cost of learning distinct from the cost of returning a checked state.

5.9 Interpretability assessment and modeling limits

The interpretability assessment examines the six driver blocks, the coupling matrix, and the corresponding component and composite raw-head maps. It evaluates whether important terms remain stable across repeated training partitions and whether the associated response patterns agree with the declared process domain. Stability assessment is particularly relevant because an explicit polynomial library does not remove identifiability problems. Brunton et al. (2016) connect interpretable nonlinear terms to the problem of recovering structure from data, while Machado et al. (2014) show how operational and influent uncertainty complicate parameter identification in activated sludge models.

The driver and coupling parameters can trade off in the relation $Rc = B\phi$. Different joint stationary solutions can therefore predict similar component states with different coefficients. Symmetry removes the redundancy of ordered self-quadratic pairs. It does not establish uniqueness of B and Γ together. Coefficient signs and magnitudes are interpreted as properties of a specified fitted representation. Repeated stability checks and response comparisons provide stronger evidence than an isolated coefficient value.

Prediction error and physical diagnostics are assessed separately for the raw head, the affine state, and the final deployed state. The comparison distinguishes what the regression learns from what the correction enforces. The magnitude of the correction

is also examined because a feasible deployed state can lie far from its raw prediction. Computational assessment records fitting effort and deployment effort separately. These measures establish the cost of the structured regression and the cost of enforcing its physical conditions without assuming an accuracy benefit.

The scope remains steady-state component prediction within the stated input domain and control volume. The corrected state preserves the adopted invariants and nonnegativity when the optimization and numerical checks succeed. It does not inherit the kinetic equations, dynamic behavior, or thermodynamic conditions of the full process model. The raw head supplies an inspectable approximation. The deployment constraints supply defined material-accounting conditions. Their combined use is evaluated as an engineering surrogate under these limits.

Chapter 6

Connected plant surrogates and operating optimization

6.1 From an admissible reactor prediction to a plant decision

An activated sludge plant repeatedly moves material between biological reactors and a secondary clarifier. Settled biomass returns to the biological treatment train. Mixed liquor may return to an upstream stage to support nitrogen removal. A separate waste stream removes accumulated solids. Aeration, recycle, and wasting therefore affect more than one treatment unit. The account of biological treatment given by Henze et al. (2008) explains why these decisions need a common plant balance. The clarification and thickening model of Härtel and Pöpel (1992) adds the storage and separation processes that determine which solids return to the reactors.

The operating question is how to choose these settings while balancing treated-water quality and resource use. de Araujo et al. (2013) formulate activated sludge operation as a constrained economic problem and examine the sensitivity of the resulting decisions. Padrón-Páez et al. (2020) examine wastewater treatment design through multiple objectives. Aboagye et al. (2021) organize treatment-network design and sustainability assessment within mathematical programming. These studies make a common point. A preferred decision depends on the objective, the feasible operating region, and the quantities used to measure treatment and resource use.

A process surrogate can reduce the cost of evaluating the same plant at many candidate settings. McBride and Sundmacher (2019) describe surrogate construction from experiments and process simulations. Bhosekar and Ierapetritou (2018) connect surrogate prediction to feasibility analysis and optimization. Applications include the Kriging treatment model of He et al. (2023), the machine learning process model within predictive control developed by Bernardelli et al. (2020), and the unit surrogates used for wastewater resource-recovery optimization by Durkin et al. (2024). Each application

links prediction to a decision problem. Its objective and constraints determine which prediction errors matter most.

A physically admissible prediction for one reactor does not by itself establish a consistent plant prediction. The predicted clarifier underflow may disagree with the predicted return-sludge contribution at the mixer. Separate predictions may also assign different amounts of a conserved inventory to connected streams. ICSOR supplies a useful starting point because it separates second-order statistical prediction from enforcement of declared physical relations (Selerio, 2026). Extending this principle to a plant requires a joint response and a joint correction. The correction must account for the mixer, all reactors, both clarifier outlets, and the solids stored in the clarifier.

Three questions remain distinct. A network prediction can satisfy the imposed balances. It can also approximate a mechanistic reference accurately. Finally, it can lead an optimizer to a useful operating decision. None of these properties is a substitute for the others. The surrogate comparisons studied by Pedrozo et al. (2025) make optimization performance a separate endpoint from predictive accuracy. The gray-box methods of Hameed et al. (2026) also retain checks against the original model when managing approximation error. The connected formulation developed here therefore includes independent mechanistic evaluation of selected controls.

6.2 Hydraulic connections and reactor inventories

6.2.1 Flow through the treatment train

Let Q_0 denote the fresh-influent flow in $\text{m}^3 \text{d}^{-1}$ and let $c_{\text{in}} \in \mathbb{R}_+^F$ denote its component concentrations. The plant has n_r completely mixed reactors and an L -layer clarifier. The reactor concentrations are $c_1, \dots, c_{n_r}$. The mixer concentration is m . Clarifier overflow and underflow concentrations are c_E and c_U . The overflow is the final plant outlet, so $c_{\text{out}} = c_E$.

The dimensionless ratios r_I , r_R , and w describe mixed-liquor recycle, return activated sludge, and waste activated sludge flow relative to Q_0 . Mixed-liquor recycle leaves the final reactor before the clarifier. Return and waste sludge share the clarifier underflow concentration. Dividing each hydraulic balance by Q_0 gives

$$q_P = 1 + r_I + r_R, \quad q_C = 1 + r_R, \quad q_U = r_R + w, \quad q_E = 1 - w. \quad (6.1)$$

Here q_P is the reactor-train flow ratio and q_C is the clarifier-feed flow ratio. The identity $q_C = q_E + q_U$ confirms the clarifier water balance. The adopted domain requires $r_I \geq 0$, $r_R \geq 0$, $0 \leq w < 1$, and $q_U > 0$. The outlet concentrations are then defined by positive outlet flows.

The mixer balance is a component balance rather than a balance of a single water-

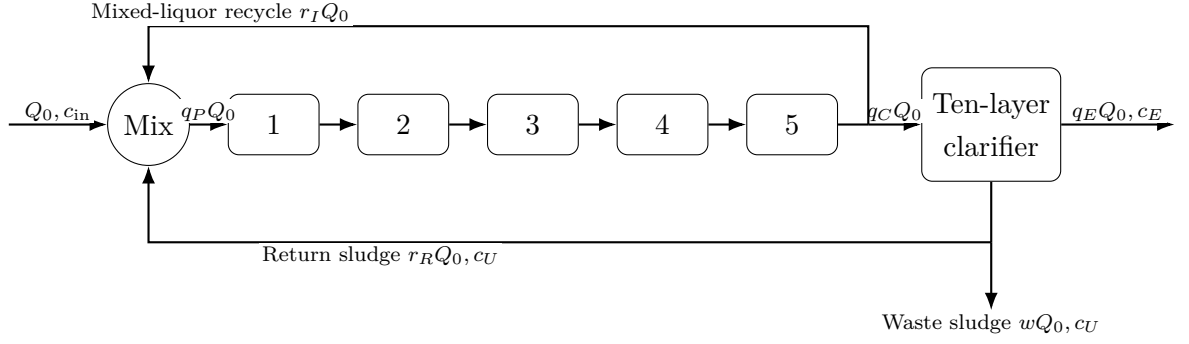


Figure 6.1: Connected plant configuration used for the operating study. Reactors 1 and 2 are unaerated. Reactors 3–5 have independent aeration controls.

quality total. Its concentration satisfies

$$q_P m = c_{\text{in}} + r_I c_{n_r} + r_R c_U. \quad (6.2)$$

A high mixer concentration can therefore result from internal circulation. It does not imply that an external pollutant load has been created. The component-resolved framework of Zhang et al. (2026) treats these recirculation and discharge streams explicitly.

The nominal stage HRT is measured on the fresh-flow basis. With reactor volume V_i ,

$$h_i = \frac{24V_i}{Q_0}, \quad H = \sum_{i=1}^{n_r} h_i, \quad d_i = \frac{Q_0 q_P}{V_i} = \frac{24q_P}{h_i}. \quad (6.3)$$

The factors h_i and H are in hours. The dilution rate d_i is in d^{-1} . The actual residence time during one passage through stage i is h_i/q_P . For a hypothetical plant with $H = 24$ h, $r_I = 2$, and $r_R = 0.5$, the reactor-train flow ratio is 3.5. Its train residence time per passage is about 6.86 h. Equal tanks then have a residence time per passage of about 1.37 h. These illustrative numbers explain why nominal HRT and recycle flow must be interpreted together.

Fixed stage fractions can be represented by $h_i = f_i^V H$, where $f_i^V > 0$ and $\sum_i f_i^V = 1$. The case study uses equal fractions. Varying H at fixed Q_0 varies biological reactor volume. The resulting problem combines capacity selection and operating decisions. For an existing plant with installed tanks, the volumes would be fixed and H would be removed from the free-control vector.

Let ϑ collect only settings that vary freely. A constant setting remains a parameter. Treating it as a varying regression coordinate would create a zero-variance feature. A carbon or chemical addition may also enter the control vector. An addition represented only by a source term has negligible liquid carrier flow. An appreciable carrier flow requires a new hydraulic stream and revised downstream dilution rates.

6.2.2 Biological conversion and nonnegativity

The reaction network uses the component and process representation of Chapter 3. Its stoichiometric matrix is $\nu \in \mathbb{R}^{P \times F}$ and its process-rate vector in stage i is $\rho_i \in \mathbb{R}_+^P$. With $c_0 = m$, the steady reactor equation is

$$0 = d_i(c_{i-1} - c_i) + \nu^\top \rho_i(c_i, \vartheta) + k_L a_i(a_i)(S_O^* - c_{i,S_O})e_{S_O} + \xi_i(\vartheta). \quad (6.4)$$

The vector e_{S_O} selects dissolved oxygen. The parameter S_O^* is its saturation concentration. The vector ξ_i represents declared stage additions. Each term has units of component concentration per day. The treatment-model framework of Henze et al. (2015) separates hydraulic transport, stoichiometric conversion, and external sources in this way.

At a zero component concentration, the combined reaction and external-source terms must not direct that component below zero. With nonnegative inflow and a locally Lipschitz balance function defined on the boundary, this condition preserves the nonnegative state space of the dynamic model (Haddad et al., 2010). A numerical integrator does not inherit this property automatically. Acceptance therefore checks concentrations and reaction rates independently of the solver's stopping message.

The same distinction matters for a surrogate. A fitted quadratic equation can predict a negative concentration even when every training state is nonnegative. Polynomial approximation imposes no sign restriction. The work of Kircher and Votsmeier (2025) addresses the simultaneous requirements of conservation and positivity in kinetic surrogates. Both requirements need explicit treatment.

6.2.3 Stage invariants and the external plant balance

Let $A \in \mathbb{R}^{K \times F}$ contain independent inventory rows such that

$$A\nu^\top = 0, \quad Ae_{S_O} = 0, \quad \text{rank}(A) = K. \quad (6.5)$$

The rows are normalized to unit Euclidean length. Unit row normalization does not make them orthogonal. Physically named inventories and an orthonormal basis can span the same conservative subspace while using different coordinates. The stoichiometric fixed-layer formulation of Sturm and Wexler (2022) illustrates the value of separating conserved quantities from the rates that redistribute them.

Multiplying Equation (6.4) by A removes the reaction and oxygen-transfer terms. The remaining stage relation is

$$A(c_i - c_{i-1}) = d_i^{-1} A\xi_i(\vartheta). \quad (6.6)$$

If there is no stage addition, $Ac_i = Ac_{i-1}$. For example, nitrification can transfer nitrogen from ammonium to oxidized forms while denitrification transfers it to dissolved nitrogen gas. These changes do not remove nitrogen from the model's complete nitrogen

inventory. A reporting quantity that excludes dissolved nitrogen gas can decrease while the conserved inventory remains unchanged. Nutrient-form interpretation in U.S. Environmental Protection Agency (2009) supports the need to inspect individual species as well as reported totals.

Internal recycle streams also cancel from an external invariant balance. Summing the stage transitions, using $d_i^{-1} = V_i/(Q_0 q_P)$, and combining the mixer and clarifier balances gives

$$Ac_{\text{in}} + \sum_{i=1}^{n_r} \frac{V_i}{Q_0} A\xi_i = q_E Ac_E + w Ac_U. \quad (6.7)$$

Only fresh feed, stage additions, effluent, and waste sludge remain. The recycle ratios still influence the state and treatment processes. Their cancellation means that they are internal transfers rather than external sources or sinks.

For ASM2d-TSN, $F = 20$, $P = 28$, and $K = 5$. The five inventories concern soluble inert material, total nitrogen including dissolved nitrogen gas, total phosphorus, an alkalinity-related combination, and a metal-related combination. The last two unnormalized combinations are

$$I_{\text{alk}} = S_{ALK} - \frac{S_{NH4}}{14} + \frac{S_{NO2}}{14} + \frac{S_{NO3}}{14} - \frac{S_{PO4}}{31}, \quad (6.8)$$

$$I_{\text{metal}} = 3.45X_{MeP} + 4.87X_{MeOH}. \quad (6.9)$$

The nitrogen and phosphorus rows use the constituent weights in Table 3.1. The nitrogen inventory also gives dissolved nitrogen gas unit weight. The stoichiometric rank is 15 under the stated parameterization. The inventory identities are audited rather than inferred from a reporting label such as TN.

6.3 Clarification, component separation, and stored solids

6.3.1 A nonreactive separator with two outlet flows

The clarifier uses a nonreactive settling representation. Soluble components follow the liquid. Particulate components can settle and concentrate in the underflow. TAKACS (1991) provide the settling and thickening formulation used for the aggregate solids profile. Jeppsson and Diehl (1996) explain why component propagation through a clarifier requires attention beyond a single aggregate solids concentration.

Let $E_S \in \mathbb{R}^{F_S \times F}$ and $E_X \in \mathbb{R}^{F_X \times F}$ select soluble and particulate components. They satisfy

$$E_S^\top E_S + E_X^\top E_X = I_F, \quad E_S E_X^\top = 0. \quad (6.10)$$

The column t_X maps component concentrations to TSS. Its entries are nonnegative and its soluble entries are zero.

The statistical response uses component flows divided by Q_0 ,

$$g_E = q_E c_E, \quad g_U = q_U c_U. \quad (6.11)$$

Their units are the same as concentration because the flow ratios are dimensionless. The actual component discharge rates are $Q_0 g_E$ and $Q_0 g_U$. Writing balances in these coordinates avoids dividing by a small underflow ratio inside the equality system. The component balance and soluble pass-through are

$$g_E + g_U = q_C c_{n_r}, \quad E_S g_U = q_U E_S c_{n_r}. \quad (6.12)$$

The two relations imply $E_S g_E = q_E E_S c_{n_r}$. Settling therefore cannot remove a soluble nutrient directly in this model.

The particulate densification requirement is

$$E_X g_U \geq q_U E_X c_{n_r}. \quad (6.13)$$

After division by q_U , each underflow particulate concentration is at least its feed concentration. Nonnegative overflow and the total component balance also prevent the underflow from recovering more than the incoming particulate flow.

For an illustrative particulate component, take $q_C = 1.5$, $q_U = 0.52$, and feed concentration 100 g m^{-3} . Densification requires $g_U \geq 52 \text{ g m}^{-3}$ on the fresh-flow basis. Conservation and nonnegative overflow require $g_U \leq 150 \text{ g m}^{-3}$. Thus the underflow recovery fraction lies between $0.52/1.5$ and one. A soluble component with the same feed concentration instead has $g_U = 52$ and unchanged soluble outlet concentrations. These examples concern the constraints alone and do not predict a settling-model outcome.

Applying t_X gives feed and outlet solids concentrations

$$X_N = t_X^\top c_{n_r}, \quad X_E = \frac{t_X^\top g_E}{q_E}, \quad X_U = \frac{t_X^\top g_U}{q_U}. \quad (6.14)$$

The declared constraints imply $X_E \leq X_N \leq X_U$. They bound separation but do not determine its magnitude. The nonlinear settling calculation and an empirical endpoint model supply additional information through different routes.

6.3.2 The layer-resolved mechanistic model

Clarifier layers are numbered from the surface to the hopper. Their TSS concentrations are $s_1, \dots, s_L$, with $s_1 = X_E$ and $s_L = X_U$. The exact settling velocity at concentration s is

$$v(s, X_N) = \max \left\{ 0, \min \left[v_{\max}, v_0 \left(e^{-r_h(s - f_{\text{ns}} X_N)} - e^{-r_p(s - f_{\text{ns}} X_N)} \right) \right] \right\}. \quad (6.15)$$

The gravitational solids flux is $sv(s, X_N)$. The nonsettleable fraction f_{ns} makes the settling offset depend on the feed concentration. At an interface, a receiver concentration

above X_t activates a limitation by the smaller adjacent gravitational flux. Otherwise, the upper-layer gravitational flux passes to the receiver. Hydraulic transport uses the overflow velocity above the feed layer and the underflow velocity below it.

Each finite-volume layer equation is its incoming solids flux minus its outgoing solids flux, multiplied by $A_{cl}/V_{cl,\ell}$. The feed layer also receives $Q_0 q_C X_N / V_{cl,\ell_F}$. Internal interface fluxes cancel when these equations are summed. At steady state, the resulting total solids balance agrees with Equation (6.12) after application of t_X .

The mechanistic state retains one solids concentration per layer. It does not retain a separate transient concentration for every particulate species in every layer. At a steady state, particulate outlet components are reconstructed with the final-reactor particulate composition,

$$E_X c_E = \frac{s_1}{X_N} E_X c_{n_r}, \quad E_X c_U = \frac{s_L}{X_N} E_X c_{n_r}. \quad (6.16)$$

Soluble outlet components equal their feed concentrations. This reduced representation defines the physical scope of the mechanistic reference. Its steady component mapping is well defined when $X_N > 0$. It does not establish the dynamic propagation of each particulate species considered in more detailed component-transport models (Jeppsson & Diehl, 1996).

6.3.3 An exact interval for aggregate inventory under the endpoint envelope

The clarifier's stored solids mass is

$$M_{cl} = \sum_{\ell=1}^L V_{cl,\ell} s_\ell, \quad V_{cl} = \sum_{\ell=1}^L V_{cl,\ell}. \quad (6.17)$$

An outlet concentration cannot determine this mass. Two clarifiers can have the same endpoints and different internal storage. Whole-plant solids retention time therefore requires the inventory as well as the two outlet solids flows. The benchmark formulation of Alex et al. (2008) includes reactor and settler inventories in its sludge-age calculation.

The mechanistic acceptance envelope requires

$$X_E \leq s_\ell \leq X_U, \quad \ell = 2, \dots, L-1. \quad (6.18)$$

This is an endpoint envelope. It does not require monotonic concentration from one layer to the next. With positive layer volumes, minimizing the inventory assigns every internal layer X_E . Maximizing it assigns every internal layer X_U . Consequently,

$$M_{cl}^- = (V_{cl} - V_{cl,L}) X_E + V_{cl,L} X_U, \quad (6.19)$$

$$M_{cl}^+ = V_{cl,1} X_E + (V_{cl} - V_{cl,1}) X_U, \quad (6.20)$$

$$M_{cl}^- \leq M_{cl} \leq M_{cl}^+. \quad (6.21)$$

These bounds are necessary and sufficient for a profile with the prescribed endpoints and envelope to have the stated inventory. To see sufficiency, give every internal layer a common concentration s_* between X_E and X_U . The inventory is then

$$V_{\text{cl},1}X_E + V_{\text{cl},L}X_U + (V_{\text{cl}} - V_{\text{cl},1} - V_{\text{cl},L})s_*. \quad (6.22)$$

This expression moves continuously through the full interval as s_* varies. For equal-volume layers, it also gives a simple constructive profile for any admitted inventory.

As an illustrative calculation, take ten layers of 600 m^3 , $X_E = 10 \text{ g m}^{-3}$, and $X_U = 5,000 \text{ g m}^{-3}$. The allowable inventory interval is 3,054,000–27,006,000 g. The broad interval reflects how little the endpoints alone say about internal storage. An inventory inside it has an envelope-compatible profile. It may still have no profile that satisfies the nonlinear settling equations. Joint projection enforces the interval, while the layer-resolved mechanistic evaluation checks the flux equations.

6.4 Statistical prediction of the connected response

6.4.1 A response that preserves network and storage information

The connected surrogate predicts

$$\chi = \text{col}(m, c_1, \dots, c_{n_r}, g_E, g_U, M_{\text{cl}}) \in \mathbb{R}^{n_\chi}, \quad n_\chi = (n_r + 3)F + 1. \quad (6.23)$$

Here col stacks its arguments into one column. The response preserves concentrations needed for stage inventories, outlet flows needed for material balances, and clarifier storage needed for solids retention time. It omits the internal layer profile. The full mechanistic state is

$$y = \text{col}(c_1, \dots, c_{n_r}, s_1, \dots, s_L) \in \mathbb{R}_+^{n_y}, \quad n_y = n_r F + L. \quad (6.24)$$

The mapping from y to χ reconstructs the mixer, both outlets, and inventory deterministically. This mapping lets two different response dimensions support the same engineering calculations.

6.4.2 Complete second-order features and ridge estimation

The statistical input includes every free control and every fresh-influent component. For fitting-set centers and positive diagonal scales, define

$$z = \text{col} \left[D_{\vartheta, \text{fit}}^{-1}(\vartheta - \bar{\vartheta}), D_{\text{in}, \text{fit}}^{-1}(c_{\text{in}} - \bar{c}_{\text{in}}) \right] \in \mathbb{R}^{d_z}, \quad d_z = d_\vartheta + F. \quad (6.25)$$

The complete quadratic feature family is

$$r(z) = \text{col} \left[z, \text{vech}(zz^\top) \right], \quad (6.26)$$

$$\phi(\vartheta, c_{\text{in}}) = \text{col} \left[1, D_{r, \text{fit}}^{-1}(r(z) - \bar{r}) \right], \quad (6.27)$$

$$n_\phi = 1 + d_z + \frac{d_z(d_z + 1)}{2}. \quad (6.28)$$

The operator vech retains one triangular copy of the symmetric matrix. It includes every square and every pairwise product once. A product between ammonium loading and aeration allows their joint effect to differ from the sum of their separate effects. A square of the recycle coordinate permits local curvature. These terms are statistical approximations. They are not a replacement for kinetic rate laws.

Let $\Phi_{\mathcal{I}}$ contain the feature rows for fitting indices $\mathcal{I}$ and let $\tilde{Y}_{\mathcal{I}}$ contain standardized joint responses. With $R_\phi = \text{diag}(0, 1, \dots, 1)$, the coefficient matrix solves

$$\hat{C}_\gamma = \arg \min_C \left\{ \frac{\|\tilde{Y}_{\mathcal{I}} - \Phi_{\mathcal{I}} C^\top\|_F^2}{|\mathcal{I}| n_\chi} + \frac{\gamma}{n_\chi} \|C R_\phi\|_F^2 \right\}. \quad (6.29)$$

The intercept is unpenalized. A positive γ regularizes correlated quadratic features. Hastie et al. (2009) explain this stabilization and the corresponding trade-off between model flexibility and coefficient shrinkage. The raw prediction is

$$\chi_{\text{raw}} = \bar{\chi} + D_\chi \hat{C}_\gamma \phi(\vartheta, c_{\text{in}}). \quad (6.30)$$

The response scale D_χ is fixed from fitting data. It places the inventory, component concentrations, and outlet flows on comparable statistical scales. The raw prediction still has no enforced physical constraints.

The quadratic feature family is retained from ICSOR (Selerio, 2026). Its coefficients describe the raw statistical relation. If projection is active, a raw coefficient alone does not give the derivative of the final projected response. That derivative also depends on the constraints, their control dependence, and the active inequalities.

6.4.3 A separate positive model for overflow solids

Low overflow TSS calls for a different error scale from large reactor solids concentrations. An absolute error of 1 g m^{-3} is 1% of 100 g m^{-3} and 100% of 1 g m^{-3} . A joint standardized response score may not adequately describe this distinction at the discharge point. The relative-error argument of Tofallis (2015) motivates a separate logarithmic model.

For positive mechanistic targets, define

$$\ell_E = \log(X_E/X_\star), \quad X_\star = 1 \text{ mg L}^{-1}. \quad (6.31)$$

The ratio inside the logarithm is dimensionless. The scalar ridge fit uses the same complete quadratic features,

$$\hat{\beta}_{E, \gamma_E} = \arg \min_\beta \left\{ \frac{\|\ell_{E, \mathcal{I}} - \Phi_{\mathcal{I}} \beta\|_2^2}{|\mathcal{I}|} + \gamma_E \|R_\phi \beta\|_2^2 \right\}. \quad (6.32)$$

Its concentration-scale prediction is

$$\widehat{X}_{E,\log} = X_\star \exp\left(\widehat{\beta}_{E,\gamma_E}^\top \phi\right) > 0. \quad (6.33)$$

A twofold concentration error has the same logarithmic discrepancy at low and high concentrations. Exponentiation guarantees a positive prediction. It does not guarantee an accurate concentration, a mechanistic minimum, or a conservative effluent estimate.

Under squared logarithmic error, direct exponentiation represents a conditional geometric center rather than an unbiased concentration-scale mean. No concentration-scale bias adjustment is applied. The separate model also avoids imposing a constant solids-removal fraction across all influents and controls. Such a fraction would disregard variation in feed composition, hydraulics, recycle, and biological state. Its fitted endpoint remains an empirical closure rather than a conservation law.

6.5 Joint projection onto the declared plant constraints

6.5.1 Equalities and inequalities at fixed controls

The projection adjusts every connected response block together. The physical equality operator $\mathcal{H}(\vartheta)\chi = b(\vartheta, c_{\text{in}})$ collects

$$\text{col} \begin{bmatrix} q_P m - r_I c_{n_r} - (r_R/q_U) g_U - c_{\text{in}} \\ \{A(c_i - c_{i-1}) - d_i^{-1} A \xi_i\}_{i=1}^{n_r} \\ g_E + g_U - q_C c_{n_r} \\ E_S g_U - q_U E_S c_{n_r} \end{bmatrix} = 0. \quad (6.34)$$

The empirical overflow relation is appended as

$$t_X^\top g_E = q_E \widehat{X}_{E,\log}(\vartheta, c_{\text{in}}). \quad (6.35)$$

The complete operator is denoted $\mathcal{H}_{\log}$ with right-hand side $b_{\log}$. It has $2F + n_r K + F_S + 1$ rows. The last row fixes total overflow TSS. It does not prescribe its particulate composition.

The physical inequality operator is

$$\mathcal{G}(\vartheta)\chi = \text{col} \begin{bmatrix} q_U E_X c_{n_r} - E_X g_U \\ q_U (V_{\text{cl}} - V_{\text{cl},L}) t_X^\top g_E + q_E V_{\text{cl},L} t_X^\top g_U - q_E q_U M_{\text{cl}} \\ q_E q_U M_{\text{cl}} - q_U V_{\text{cl},1} t_X^\top g_E - q_E (V_{\text{cl}} - V_{\text{cl},1}) t_X^\top g_U \end{bmatrix} \leq 0. \quad (6.36)$$

The first block enforces particulate densification. The last two rows are the inventory bounds multiplied by the positive product $q_E q_U$. This multiplication removes divisions without changing feasibility. Nonnegativity requires $\chi \geq 0$ separately.

At fixed $(\vartheta, c_{\text{in}})$, every constraint is affine in χ . This remains true for the overflow closure because its fitted right-hand side is then a known scalar. The endpoint-envelope inequalities also remain affine even though their coefficients depend on the controls. The distinction between fixed-control linearity and control dependence is central to the numerical formulation.

6.5.2 The scaled nearest-response problem

Let D_b and D_g be positive diagonal row scales. The projected standardized displacement solves

$$\hat{u} = \arg \min_{u \in \mathbb{R}^{n_\chi}} \frac{1}{2} u^\top u, \quad (6.37)$$

$$\begin{aligned} \text{subject to } & D_b^{-1} \{ \mathcal{H}_{\log}(\vartheta)(\chi_{\text{raw}} + D_\chi u) - b_{\log}(\vartheta, c_{\text{in}}) \} = 0, \\ & \chi_{\text{raw}} + D_\chi u \geq 0, \\ & D_g^{-1} \mathcal{G}(\vartheta)(\chi_{\text{raw}} + D_\chi u) \leq 0, \\ & \hat{\chi} = \chi_{\text{raw}} + D_\chi \hat{u}. \end{aligned} \quad (6.38)$$

The identity Hessian defines a strictly convex quadratic program. In physical coordinates, its metric is $D_\chi^{-\top} D_\chi^{-1}$. Response scaling determines which physical changes count as large corrections. Positive constraint-row scaling changes numerical conditioning while leaving the feasible set unchanged. Sturm and Silva (2025) motivate species-sensitive weighting for physical correction. Boyd and Vandenberghe (2004) supply the convex optimization principles used for feasibility, uniqueness, and optimality.

The physically constrained surrogate combines the original ICSOR separation of prediction and projection with a connected plant response (Selerio, 2026). The manifold projection developed by Valente et al. (2025) provides related output-correction context. Analytic neural-output constraints in Beucler et al. (2021), conserved-flux learning in Sturm and Wexler (2020), and constrained training in Ruckstuhl et al. (2021) provide alternative locations for physical enforcement. H. Li et al. (2024) include governing-equation residuals in physics-informed unit-operation models, including an activated sludge reactor. A residual penalty makes disagreement costly during fitting. It differs from an equality that each admitted prediction must satisfy. These studies do not supply the present plant balances or inventory interval. Here those constraints follow from the declared process model.

For a nonempty feasible set, closedness, coercivity, and strict convexity give a unique projected response. Nonemptiness is an assumption that must be checked. Positive outlet flows and a full-rank inventory operator alone do not guarantee it. The fitted endpoint may conflict with conservation, nonnegativity, or densification. For instance, demanding overflow solids above every feed solids concentration allowed by the other balances contradicts $X_E \leq X_N$. The empirical closure is therefore capable of making

an otherwise feasible physical polyhedron empty. A failed projection is retained as a failure. It is not changed by clipping the endpoint or relaxing a balance.

6.5.3 An independently auditable optimality certificate

At fixed inputs, write the standardized problem as

$$\min_u \frac{1}{2} u^\top u, \quad Eu = e, \quad G_u u \leq g. \quad (6.39)$$

Let $\lambda^\top$ be unrestricted equality multipliers and $\lambda^\leq \geq 0$ inequality multipliers. The optimality conditions are primal feasibility, dual feasibility, stationarity, and complementarity,

$$u + E^\top \lambda^\top + G_u^\top \lambda^\leq = 0, \quad (6.40)$$

$$\lambda_j^\leq (g_j - (G_u u)_j) = 0 \quad \text{for every } j. \quad (6.41)$$

The dual value is

$$d(\lambda^\top, \lambda^\leq) = -\frac{1}{2} \|E^\top \lambda^\top + G_u^\top \lambda^\leq\|_2^2 - e^\top \lambda^\top - g^\top \lambda^\leq. \quad (6.42)$$

For primal- and dual-feasible quantities, the gap is

$$\Gamma = \frac{1}{2} \|u + E^\top \lambda^\top + G_u^\top \lambda^\leq\|_2^2 + (\lambda^\leq)^\top (g - G_u u) \geq 0. \quad (6.43)$$

It vanishes exactly when stationarity and complementarity hold. This gives a mathematical certificate of the unique projection (Boyd & Vandenberghe, 2004). Numerical acceptance checks the individual residuals independently. A small reported gap does not replace a nonnegativity or equality check.

Every distinct trial receives a projection solve from an independent initialization. An operator-splitting quadratic solver of the type studied by Stellato et al. (2020) supplies a numerical candidate. Multipliers are reconstructed independently through bounded-variable least squares for the audit. Numerical retries change the solver settings while preserving the mathematical problem.

6.5.4 The conditional error guarantee

Let $\mathcal{C}$ be the fixed-input feasible set and use the norm $\|v\|_{D_\chi^{-1}} = \|D_\chi^{-1} v\|_2$. If a mechanistic reference χ_{ref} belongs exactly to $\mathcal{C}$, the projection variational inequality gives

$$\|\hat{\chi} - \chi_{\text{ref}}\|_{D_\chi^{-1}}^2 \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_\chi^{-1}}^2 - \|\hat{u}\|_2^2. \quad (6.44)$$

Thus the correction cannot increase aggregate squared error in this particular metric to an exactly feasible target. It does not guarantee improvement in every component, an unweighted physical-unit score, or the operating objective.

The fitted overflow equality usually differs from the mechanistic endpoint. A physically consistent mechanistic target can therefore lie outside $\mathcal{C}$. Define

$$\delta_{\text{feas}} = \|P_{\mathcal{C}}(\chi_{\text{ref}}) - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}. \quad (6.45)$$

Let $r_{\text{corr}} = \|\chi_{\text{raw}} - P_{\mathcal{C}}(\chi_{\text{raw}})\|_{D_{\chi}^{-1}}$. Apply the projection variational inequality with the feasible comparison point $P_{\mathcal{C}}(\chi_{\text{ref}})$. Expanding the squared error and bounding the remaining inner product gives

$$\|P_{\mathcal{C}}(\chi_{\text{raw}}) - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}^2 \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}^2 - r_{\text{corr}}^2 + 2r_{\text{corr}}\delta_{\text{feas}} \quad (6.46)$$

$$\leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_{\chi}^{-1}}^2 + \delta_{\text{feas}}^2. \quad (6.47)$$

Taking square roots therefore gives the more easily interpreted bound

$$\|P_{\mathcal{C}}(\chi_{\text{raw}}) - \chi_{\text{ref}}\|_{D_{\chi}^{-1}} \leq \|\chi_{\text{raw}} - \chi_{\text{ref}}\|_{D_{\chi}^{-1}} + \delta_{\text{feas}}. \quad (6.48)$$

The distance includes any finite numerical infeasibility of the reference. The guarantee cannot be invoked by checking only the physical identities and ignoring the empirical equality. The weighted-correction discussion in Sturm and Silva (2025) likewise separates conservation from predictive improvement. Biological kinetics, nonlinear settling, and dynamic stability remain outside the projected set.

6.5.5 Control-dependent sensitivities

With fixed constraint matrices and an affine parameter-dependent right-hand side, a parametric quadratic projection can have a piecewise affine solution (Tøndel et al., 2003). That more restrictive setting is not assumed here. Hydraulic coefficients, inverse flow ratios, the input features, and the exponential endpoint all depend on ϑ . The projected plant response is generally piecewise smooth where regularity holds. It need not be globally piecewise affine or differentiable at an active-set change.

For a stable active set, stack equality rows and active inequality rows into $B(\vartheta)$ and their right-hand sides into $b_a(\vartheta)$. The stationary system is

$$\begin{bmatrix} I & B^{\top} \\ B & 0 \end{bmatrix} \begin{bmatrix} u \\ \lambda_a \end{bmatrix} = \begin{bmatrix} 0 \\ b_a \end{bmatrix}. \quad (6.49)$$

For a control ϑ_j , differentiation gives

$$\begin{bmatrix} I & B^{\top} \\ B & 0 \end{bmatrix} \begin{bmatrix} \partial_j u \\ \partial_j \lambda_a \end{bmatrix} = \begin{bmatrix} -(\partial_j B)^{\top} \lambda_a \\ \partial_j b_a - (\partial_j B)u \end{bmatrix}. \quad (6.50)$$

The response derivative also includes the raw regression derivative. The endpoint derivative includes

$$\partial_j(q_E \widehat{X}_{E,\log}) = (\partial_j q_E) \widehat{X}_{E,\log} + q_E \widehat{X}_{E,\log} \widehat{\beta}_{E,\gamma_E}^{\top} \partial_j \phi. \quad (6.51)$$

Ignoring this term would differentiate a different optimization problem.

These derivatives are used only when active rows are independent, the stationary matrix is well conditioned, active inequality multipliers are strictly positive, and independent perturbation checks pass. Near degeneracy, different active descriptions can correspond to the same unique response. Uniqueness of the projection therefore does not ensure a reliable classical derivative. A value-only outer method remains available when the derivative chain cannot be validated.

6.6 Treatment priorities, resource measures, and operating safeguards

6.6.1 A common engineering objective

The effluent-quality map in Table 3.1 gives

$$z_E = I_{\text{comp}} c_E = \text{col}(\text{COD}, \text{TN}, \text{TP}, \text{TSS}). \quad (6.52)$$

For positive fixed normalization D_T and quality priorities $\varpi_Q \geq 0$ with $\mathbf{1}^\top \varpi_Q = 1$, define

$$j_Q = \varpi_Q^\top D_T^{-1} z_E. \quad (6.53)$$

The four totals describe different treatment concerns. They do not identify which individual nutrient transformations occur. A nearly constant TN across a reactor can accompany conversion of ammonium to nitrite. A decrease in overflow TP can arise from particulate separation rather than an additional clarifier reaction. The nutrient-control guidance of U.S. Environmental Protection Agency (2009) and component-resolved treatment framework of Zhang et al. (2026) support inspecting species and solids flows alongside aggregate quality measures.

The resource terms are

$$j_H = \frac{H - H^L}{H^U - H^L}, \quad j_A = \frac{\sum_{i \in \mathcal{A}} V_i k_{La_i}}{E_A^{\text{ref}}}, \quad (6.54)$$

$$j_I = \frac{r_I - r_I^L}{r_I^U - r_I^L}, \quad j_R = \frac{r_R - r_R^L}{r_R^U - r_R^L}, \quad (6.55)$$

$$j_W = \frac{w X_U}{w^U X_U^{\text{ref}}}. \quad (6.56)$$

The set $\mathcal{A}$ contains stages that can be aerated. The aeration reference $E_A^{\text{ref}} > 0$ is fixed before optimization. It is the maximum transfer-capacity numerator over the declared operating box. A fixed reference prevents an increase in selected volume from being canceled by a changing denominator.

The aeration proxy measures transfer capacity. Actual oxygen transfer also depends on the dissolved-oxygen deficit in Equation (6.4). The energy-management model of

Miederer et al. (2025) distinguishes aeration and other energy demands. The treatment model of Zhang et al. (2026) makes the oxygen-transfer driving force explicit. The present dimensionless terms do not constitute a monetary cost calculation or a measurement of energy consumption.

The complete objective is

$$J(\vartheta, \chi, \varpi) = \varpi^\top \text{col}(j_Q, j_H, j_A, j_I, j_R, j_W), \quad \varpi \geq 0, \quad \mathbf{1}^\top \varpi = 1. \quad (6.57)$$

The overall vector ϖ separates quality from capacity and resource priorities. The vector ϖ_Q separates priorities among the quality measures. These coefficients are choices about trade-offs. They are not percentages of the objective attained at an operating point.

For the equal-volume case,

$$j_A = \frac{H}{36 \text{ h}} \frac{a_3 + a_4 + a_5}{3}. \quad (6.58)$$

This follows from $V_i = Q_0 H / (24 n_r)$ and $k_L a_i = 47 a_i$. For hypothetical settings with all three aeration coordinates equal to one, reducing H from 36 h to 12 h reduces this proxy from one to one-third. The normalized aeration settings have stayed the same, but the total transfer capacity has changed with volume. A comparison based only on the aeration coordinates would miss this effect.

The baseline priority vectors are

$$\varpi = (0.50, 0.15, 0.20, 0.05, 0.05, 0.05)^\top, \quad \varpi_Q = \frac{1}{4} \mathbf{1}_4. \quad (6.59)$$

Quality has the largest individual coefficient. Aeration has the largest resource coefficient. The importance of oxygen supply and solids age to treatment and energy use is discussed by Leu et al. (2012). The exact coefficients remain case-study preferences. Changes in those preferences define separate operating questions and require separately reported decisions.

6.6.2 Solids retention time and clarifier loading

External solids leave in the overflow and waste-sludge streams,

$$\dot{M}_X = Q_0(q_E X_E + w X_U). \quad (6.60)$$

Return sludge does not count as an external loss. With $X_i = t_X^\top c_i$, the whole-plant solids retention time is

$$\theta_X = \frac{\sum_{i=1}^{n_r} V_i X_i + M_{\text{cl}}}{\dot{M}_X}, \quad \dot{M}_X \geq \epsilon_X > 0. \quad (6.61)$$

Its numerator includes the clarifier inventory. Omitting that mass would underestimate solids age for the same external loss rate. A wasting ratio alone also does not determine solids age because $w X_U$ and the stored mass both matter (Alex et al., 2008).

The descriptive clarifier loading measures are

$$\ell_{\text{surface}} = \frac{Q_0 q_E}{A_{\text{cl}}}, \quad \ell_{\text{solids}} = \frac{10^{-3} Q_0 q_C X_N}{A_{\text{cl}}}. \quad (6.62)$$

The factor 10^{-3} converts grams to kilograms. The surface quantity is in m d^{-1} and the solids quantity is in $\text{kg m}^{-2} \text{d}^{-1}$. These quantities and θ_X are descriptive measures in the baseline comparison. They are not additional optimization limits.

The common engineering safeguards are

$$X_U \leq X_U^{\max}, \quad X_N \geq X_F^{\min} > 0, \quad \dot{M}_X \geq \epsilon_X > 0. \quad (6.63)$$

The positive feed-solids limit keeps particulate outlet reconstruction defined. The positive external-loss limit keeps solids retention time defined. The underflow bound represents the admitted thickening or solids-handling range. Their numerical values are plant-specific problem inputs selected before search. An underflow limit should be based on accepted sludge-handling conditions. The feed and loss floors should prevent undefined calculations without excluding physically relevant low-solids operation. These engineering inputs are distinct from the fixed $X_U^{\text{ref}} = 15,000 \text{ g m}^{-3}$ used only to normalize the wasted-solids objective.

6.6.3 Trust diagnostics without a mechanistic-accuracy guarantee

The feasible projection set is larger than the set of complete mechanistic steady states. An optimizer might exploit that gap. Four fixed diagnostics screen the selected responses,

$$\kappa_{\text{corr}} = \frac{\|D_X^{-1}(\hat{\chi} - \chi_{\text{raw}})\|_2}{\sqrt{n_X}}, \quad (6.64)$$

$$\kappa_{\text{lev}} = \phi^\top (\Phi^\top \Phi + n_{\text{dev}} \gamma R_\phi)^{-1} \phi, \quad (6.65)$$

$$\kappa_{\text{split}} = \left[\frac{1}{F_X} \sum_{f \in X} \left\{ \frac{X_N g_{U,f} - (t_X^\top g_U) c_{n_r,f}}{d_{\text{split},f}} \right\}^2 \right]^{1/2}, \quad (6.66)$$

$$\kappa_{\text{reac}} = \frac{\|D_{\text{bal}}^{-1} f_{\text{sm,reactor}}(\vartheta, \hat{\chi})\|_2}{\sqrt{n_r F}}. \quad (6.67)$$

The first measures standardized correction size. A large value identifies a raw response that requires a large movement to satisfy the imposed constraints. It does not establish that the projected response is inaccurate. The correction is constrained and reported separately from the engineering objective.

The second diagnostic is regularized feature leverage. It measures input support within the fitted quadratic feature map. Sahigara et al. (2012) compare applicability-domain assessments and motivate checking where a statistical model is being used.

Feature leverage does not detect every form of response error. A control vector inside its coordinate bounds may still have weak support in the joint feature space.

The third diagnostic checks the common particulate-recovery assumption in Equation (6.16). If every particulate component has the same recovery, its numerator vanishes. The projection enforces component conservation and densification but does not enforce that composition exactly. This diagnostic therefore checks a relation that can hold in the reduced mechanistic mapping while remaining outside the projection equalities.

The fourth diagnostic checks smooth reactor kinetics at the projected concentrations. It includes hydraulic transport, net biological conversion, and oxygen transfer on separate scales. It cannot supply a clarifier-layer residual because the surrogate has no internal layer concentrations. Its smoothing width is 10^{-8} under the definitions given below.

For a constraint row with p terms b_{ij} over n_{dev} development states, its fixed scale is

$$d_b = \max \left\{ 1, \left[\frac{1}{n_{\text{dev}} p} \sum_{i=1}^{n_{\text{dev}}} \sum_{j=1}^p b_{ij}^2 \right]^{1/2} \right\}. \quad (6.68)$$

The same convention defines inequality and split scales. The overflow row uses the mechanistic overflow flow and the out-of-fold fitted endpoint flow as its two terms. The floor of one is in the corresponding numerical row units. It is not a concentration floor.

Correction, split, and reactor limits use nearest-rank 95th percentiles of grouped out-of-fold development diagnostics. If sorted values are $v_{(1)}, \dots, v_{(n_{\text{dev}})}$, the limit is $v_{(\lceil 0.95 n_{\text{dev}} \rceil)}$. The leverage limit is the maximum development leverage under the final fitted feature map. These quantities are fixed before optimization. They are empirical guardrails rather than confidence bounds. A diagnostic with zero limit remains constrained to zero and uses one only as a numerical row scale.

The surrogate decision problem is consequently

$$\begin{aligned} \min_{\vartheta} \quad & J(\vartheta, \hat{\chi}(\vartheta, c_{\text{in}}), \varpi), \\ \text{subject to} \quad & \vartheta \in \mathcal{V}, \\ & g_{\text{eng}}(\vartheta, \hat{\chi}) \leq 0, \\ & g_{\text{trust}}(\vartheta, \chi_{\text{raw}}, \hat{\chi}) \leq 0, \\ & \hat{\chi} \text{ satisfies Equation (6.38)}. \end{aligned} \quad (6.69)$$

The lower problem is a convex quadratic projection. The upper problem is generally nonconvex. This distinction follows the nested-problem perspective described by Colson et al. (2007). A best retained feasible candidate is not a certified global optimum. The trust-region analysis of Alexandrov et al. (1998) explains why reliable approximation management needs more than a small average prediction error. The four fixed diagnostics here do not establish the fully linear approximation conditions of a convergent adaptive trust-region method.

6.7 A direct mechanistic comparator

6.7.1 The full-state constrained optimization problem

The direct route retains the controls and full reduced mechanistic state. It reconstructs $\chi(\vartheta, c_{\text{in}}, y)$ and evaluates the same engineering objective. Its finite-volume clarifier equations require a positive feed-solids denominator at solver iterates. An auxiliary variable $\widetilde{X}_F \geq X_F^{\min}$ supplies that denominator, with equality $\widetilde{X}_F = t_X^\top c_{n_r}$ imposed at feasibility.

Let f_{sm} collect all $n_r F$ reactor equations and all L smoothed layer equations. Let

$$g_{\text{env}}(y) = \text{col}(s_1 \mathbf{1}_{L-2} - s_{2:L-1}, s_{2:L-1} - s_L \mathbf{1}_{L-2}). \quad (6.70)$$

The direct nonlinear program is

$$\begin{aligned} \min_{\vartheta, y, \widetilde{X}_F} \quad & J(\vartheta, \chi(\vartheta, c_{\text{in}}, y, \widetilde{X}_F), \varpi), \\ \text{subject to} \quad & f_{\text{sm}}(\vartheta, c_{\text{in}}, y, \widetilde{X}_F) = 0, \\ & \widetilde{X}_F = t_X^\top c_{n_r}, \quad \widetilde{X}_F \geq X_F^{\min}, \\ & \vartheta \in \mathcal{V}, \quad y \geq 0, \quad g_{\text{env}}(y) \leq 0, \\ & g_{\text{eng,red}}(\vartheta, \chi) \leq 0. \end{aligned} \quad (6.71)$$

The reduced engineering vector removes the feed-solids row already imposed by the auxiliary equality and lower bound. The auxiliary does not alter a feasible physical state. It keeps the outlet mapping defined while a nonlinear solver moves through infeasible trial states.

6.7.2 Smooth switches and continuation

The exact kinetic and settling functions contain minima, maxima, positive parts, possible zero denominators, and a receiver switch. Derivative-based optimization benefits from smooth approximations to these operations. The chemical-process optimization treatment of Buzzi-Ferraris and Manenti (2013) provides the general derivative and conditioning context. For positive width $\delta = \epsilon S$, define

$$\max_{\delta}(a, b) = \frac{1}{2} \left[a + b + \sqrt{(a - b)^2 + \delta^2} \right], \quad (6.72)$$

$$\min_{\delta}(a, b) = \frac{1}{2} \left[a + b - \sqrt{(a - b)^2 + \delta^2} \right], \quad (6.73)$$

$$[v]_{+, \delta} = \frac{1}{2} \left[v + \sqrt{v^2 + \delta^2} \right], \quad (6.74)$$

$$\left(\frac{a}{b} \right)_{\delta} = \frac{ab}{b^2 + \delta^2}. \quad (6.75)$$

For negative v , the positive-part expression is evaluated as $\delta^2/[2(\sqrt{v^2 + \delta^2} - v)]$ to avoid subtraction of nearly equal values. These approximations change the equations near a switch. They do not make the exact and smooth models equivalent.

For receiver concentration s , the smooth switch is zero below $X_t - h$, one above $X_t + h$, and

$$6\zeta^5 - 15\zeta^4 + 10\zeta^3, \quad \zeta = \frac{s - X_t + h}{2h} \quad (6.76)$$

within the transition band. This polynomial matches the values and first two derivatives at the band endpoints. It gives smooth interpolation between the two gravitational-flux rules.

Each argument scale S is its development-set root mean square with a floor of one. Reactor arguments are pooled across stages. They include total oxidized nitrogen, fermentable substrate plus acetate, the hydrolysis denominator, the polyphosphate and stored-carbon denominators, and remaining storage capacity. Clarifier arguments include $s - f_{\text{ns}}X_N$ and gravitational solids flux. The velocity scale is 250 m d^{-1} . These scales stay fixed throughout continuation.

The direct route reduces smoothing through three stages,

$$(\epsilon, h) = (10^{-6}, 10), (10^{-7}, 3), (10^{-8}, 1), \quad (6.77)$$

where h is in g TSS m^{-3} . Each stage initializes from the preceding stage. Continuation helps a solver approach sharper switches while preserving a usable starting state. The final smooth candidate still requires an exact-model evaluation.

6.7.3 Jacobian structure and globalization

The direct residual is sparse in its mechanistic state. A reactor balance depends on its own component vector and the preceding stage. Recycle reconstruction adds dependencies on final-reactor and underflow quantities. A layer balance depends on nearby layer concentrations, feed solids, and hydraulic controls. The inventory and reporting maps are linear aggregations. These dependencies provide a structured Jacobian rather than a dense arbitrary response map.

Scaling is applied before testing Jacobian conditioning or comparing residual sizes. Different component bases include oxygen, COD, nitrogen, phosphorus, solids, and charge equivalents. An unscaled norm could be dominated by a large solids balance and conceal a nutrient-balance defect. Analytic derivatives or differentiated smooth equations supply the optimization Jacobian. Independent finite differences are retained for derivative checks. Sparse linear algebra exploits the local reactor and layer connections without omitting recycle terms.

The direct search uses the interior-point filter method studied by Wächter and Biegler (2006). Its line search manages progress in objective and constraint violation. This numerical globalization controls acceptance of local steps. It does not make a nonconvex plant objective globally convex or establish a global optimum. The surrogate route similarly separates numerical search progress from the final feasibility and stationarity audits.

6.8 Case-study domain and fixed parameters

6.8.1 Topology and free controls

The case study has five equal-volume reactors and ten clarifier layers. Reactors 1 and 2 are unaerated. Reactors 3–5 have independent aeration controls. The free-control vector is

$$\vartheta = (H, a_3, a_4, a_5, r_I, r_R, w)^\top \in \mathbb{R}^7. \quad (6.78)$$

For every stage, $h_i = H/5$. The unaerated settings are $a_1 = a_2 = 0$, the aerated transfer coefficients are $k_L a_i = 47a_i \text{ d}^{-1}$, and $\xi_i = 0$. These are case-study conditions rather than structural restrictions on the connected formulation.

Table 6.1: Plant geometry and operating domain.

Quantity	Value or interval
Fresh-influent flow Q_0	10,000 $\text{m}^3 \text{ d}^{-1}$
Reactors n_r , layers L , feed layer ℓ_F	5, 10, 5
Reactor volume fraction f_i^V	1/5 for every stage
Clarifier area and total depth	1,500 m^2 and 4.0 m
Clarifier total and individual layer volumes	6,000 m^3 and 600 m^3
Nominal total HRT H	6–36 h
Each of a_3, a_4, a_5	0–1 independently
Mixed-liquor recycle r_I	0–4
Return-sludge ratio r_R	0.25–1.25
Waste-sludge ratio w	0.001–0.05
Aeration reference E_A^{ref}	423,000 $\text{m}^3 \text{ d}^{-1}$
Wasted-solids reference X_U^{ref}	15,000 g TSS m^{-3}

The equal volumes provide a compact unaerated-to-aerated train. The layer depth is 0.4 m. The operating intervals preserve both positive clarifier outlet flow ratios. Together with 20 influent coordinates, the seven controls give 27 statistical inputs, 406 features, and 161 reduced-response coordinates. The mechanistic state has 110 coordinates. The projection has 76 equality rows and 12 physical inequality rows. Adding response nonnegativity gives 173 inequality rows.

Changing the number of reactors or free controls changes the statistical dimensions. Changing the layer count changes the mechanistic dimension and inventory mapping but does not add coordinates to the aggregate surrogate response. In either case, physical and predictive validation must be repeated. A small aggregate response does not establish insensitivity to the clarifier discretization.

6.8.2 Fresh-influent intervals

Table 6.2 defines independent intervals for the plant experiment. The nominal influent is their componentwise midpoint. The concentration bases and quality-map weights are those in Table 3.1. The plant alkalinity interval is 1.6–5.2 mol charge equivalents m^{-3} . It belongs to this connected-plant design and is not interchangeable with an interval specified for an isolated-reactor experiment.

Table 6.2: Independent fresh-influent intervals for the connected plant.

Component	Lower	Upper	Constituent basis
S_O	0	0.5	oxygen
S_F	20	180	COD
S_A	5	80	COD
S_{NH4}	12	55	nitrogen
S_{NO2}	0	3	nitrogen
S_{NO3}	0	8	nitrogen
S_{N2}	0	2	nitrogen
S_{PO4}	2	18	phosphorus
S_I	10	90	COD
S_{ALK}	1.6	5.2	charge equivalents
X_I	20	120	COD
X_S	60	280	COD
X_H	15	100	COD
X_{PAO}	5	60	COD
X_{PP}	2	20	phosphorus
X_{PHA}	1	30	COD
X_{AOB}	0.5	8	COD
X_{NOB}	0.5	8	COD
X_{MeP}	0	12	solids
X_{MeOH}	0	12	solids

All concentration values are in grams of the stated constituent per m^3 , apart from alkalinity. Numerically, g m^{-3} equals mg L^{-1} . Independent interval sampling is an experimental design. It does not assert that all sampled influent combinations occur with equal frequency at a real plant.

6.8.3 Stoichiometric, kinetic, and settling parameters

The constants in Tables 6.3–6.6 define the plant case. They are fixed parameters rather than fitted kinetic estimates. A parameter's concentration basis follows the associated component. Symbols with several process subscripts on the same row have

equal values. The two nitrifier populations remain separate, with different growth rates and oxygen half-saturation constants.

Table 6.3: Fixed yields, constituent contents, and conversion factors.

Parameter	Value	Unit or interpretation
Y_H, Y_{PAO}	0.625	g biomass COD per g substrate COD
Y_{PHA}	0.20	g stored COD per g phosphorus
Y_{PO4}	0.40	g phosphorus per g stored COD
Y_{AOB}	0.18	g biomass COD per g nitrogen
Y_{NOB}	0.08	g biomass COD per g nitrogen
f_{SI}	0	soluble inert fraction
f_{XI}	0.10	particulate inert fraction
$i_{N,SI}$	0.01	g nitrogen per g COD
$i_{N,SF}, i_{N,XS}$	0	g nitrogen per g COD
$i_{N,XI}$	0.02	g nitrogen per g COD
$i_{N,BM}$	0.07	g nitrogen per g biomass COD
$i_{P,SI}, i_{P,SF}, i_{P,XS}$	0	g phosphorus per g COD
$i_{P,XI}$	0.01	g phosphorus per g COD
$i_{P,BM}$	0.02	g phosphorus per g biomass COD
$i_{P,MeP}$	1/4.87	g phosphorus per g metal-phosphate solids
γ_{OH}	3.45	g metal-hydroxide solids per g phosphorus

6.9 Mechanistic sampling, fitting, and prediction assessment

6.9.1 Separate simulation streams and acceptance-conditioned data

The joint domain has seven control coordinates and twenty influent coordinates. Strength-one Latin-hypercube sampling places one point in each stratum of every coordinate. Independent random stratum orderings prevent a shared ordering from forcing artificial coordinate correlations. Open-unit jitter places a point inside each selected stratum. The comparison of sampling designs by McKay et al. (1979) provides the basis for this coverage-oriented design.

Table 6.4: Fixed hydrolysis and heterotrophic rate and saturation constants.

Parameter	Value	Unit
K_H	3.0	d^{-1}
$\eta_{\text{hyd},\text{NO}_3}, \eta_{\text{hyd},\text{NO}_2}$	0.6	dimensionless
$\eta_{\text{hyd},\text{fe}}$	0.4	dimensionless
$K_{\text{O},\text{hyd}}, K_{\text{O},\text{H}}, K_{\text{O},\text{PAO}}$	0.20	g oxygen m^{-3}
$K_{\text{NO}_3,\text{hyd}}, K_{\text{NO}_2,\text{hyd}}, K_{\text{NO}_x,\text{hyd}}$	0.5	g nitrogen m^{-3}
K_X	0.10	g substrate COD per g biomass COD
μ_H	6.0	d^{-1}
q_{fe}	3.0	d^{-1}
b_H	0.40	d^{-1}
$\eta_{\text{H},\text{NO}_3}, \eta_{\text{H},\text{NO}_2}$	0.9	dimensionless
K_F, K_{fe}, K_A	4.0	g COD m^{-3}
$K_{\text{NO}_3,\text{H}}, K_{\text{NO}_2,\text{H}}, K_{\text{NO}_x,\text{H}}$	0.5	g nitrogen m^{-3}
$K_{\text{NH}_4,\text{H}}, K_{\text{NH}_4,\text{PAO}}, K_{\text{NH}_4,\text{NOB}}$	0.05	g nitrogen m^{-3}
$K_{\text{PO}_4,\text{H}}, K_{\text{PO}_4,\text{PAO}}, K_{\text{PO}_4,\text{nit}}$	0.01	g phosphorus m^{-3}
$K_{\text{ALK},\text{H}}, K_{\text{ALK},\text{PAO}}$	0.10	mol equivalents m^{-3}

The study design attempts 8,000 development candidates and 2,000 holdout candidates from separate seeded streams. Rejected candidates are retained in the accounting and are not replaced. The two-start physical acceptance procedure determines which candidates enter a statistical partition. These attempted design sizes do not imply any accepted sample total.

For an exact numerical randomization convention, each coordinate uses a descending Fisher–Yates permutation of its strata. The jitter from an unsigned 64-bit random integer U is

$$\zeta_U = \frac{\lfloor U/2^{11} \rfloor + 0.5}{2^{53}} \in (0, 1). \quad (6.79)$$

This definition excludes exact stratum boundaries. Accepted states retain their candidate ordering before the grouped-fold permutation is applied.

Conditioning on mechanistic acceptance changes the population represented by the data. Regions with frequent integration, balance, branch, or stability failures may contribute few accepted states. The fitted model therefore represents the accepted portion of the design. A prediction score on that portion is not a reliability estimate for the complete rectangular operating domain.

6.9.2 Two initial states and the nonsmoothed steady solve

The first initial state assigns c_{in} to every reactor and fresh-influent TSS to every clarifier layer. The second retains the influent soluble concentrations and multiplies the

Table 6.5: Fixed storage, nitrifier, and chemical rate and saturation constants.

Parameter	Value	Unit
q_{PHA}	5.0	g stored COD per g biomass COD per d
q_{PP}	0.60	g phosphorus per g biomass COD per d
μ_{PAO}	0.56	d ⁻¹
η_{PAO,NO_3}	0.07	dimensionless
η_{PAO,NO_2}	0.90	dimensionless
$b_{PAO}, b_{PP}, b_{PHA}, b_{AOB}$	0.20	d ⁻¹
$K_{NO_3,PAO}, K_{NO_2,PAO}, K_{NO_x,PAO}$	0.5	g nitrogen m ⁻³
K_{PS}	0.20	g phosphorus m ⁻³
K_{PP}	0.01	g phosphorus per g biomass COD
K_{max}	0.34	g phosphorus per g biomass COD
K_{IPP}	0.02	g phosphorus per g biomass COD
K_{PHA}	0.01	g stored COD per g biomass COD
μ_{AOB}	1.81	d ⁻¹
μ_{NOB}	1.52	d ⁻¹
b_{NOB}	0.17	d ⁻¹
$K_{O,AOB}$	0.74	g oxygen m ⁻³
$K_{O,NOB}$	1.75	g oxygen m ⁻³
$K_{NH_4,AOB}, K_{NO_2,NOB}$	0.5	g nitrogen m ⁻³
$K_{ALK,nit}, K_{ALK,PRE}, K_{ALK,chem}$	0.50	mol equivalents m ⁻³
k_{PRE}	1.0	m ³ per g solids per d
k_{RED}	0.60	d ⁻¹

Table 6.6: Fixed settling and oxygen-transfer parameters.

Quantity	Value
Maximum settling velocity v_{max}	250 m d ⁻¹
Theoretical settling velocity v_0	474 m d ⁻¹
Hindered-settling coefficient r_h	0.000576 m ³ per g TSS
Low-concentration coefficient r_p	0.00286 m ³ per g TSS
Nonsettleable fraction f_{ns}	0.00228
Receiver threshold X_t	3,000 g TSS m ⁻³
Oxygen saturation S_O^*	8.5 g oxygen m ⁻³
Maximum aerated-stage k_La	47 d ⁻¹

Table 6.7: Randomization and fitting settings specified before assessment.

Setting	Value
Development Latin-hypercube seed	100042
Holdout Latin-hypercube seed	100043
Influent-scenario Latin-hypercube seed	314159
Grouped-fold permutation seed	271828
Number of fitting folds	5
Raw-response and logarithmic ridge grids	$\{10^{-8}, 10^{-7}, \dots, 10^2\}$
Standard-deviation divisor	fitting-set size
Near-constant coordinate cutoff	$10^{-12} \max(1, \max_i v_i)$
Low-overflow group	at or below development 25th percentile
High-overflow group	at or above development 90th percentile

particulate concentrations in reactors 1–5 by (1.5, 2, 2.5, 3, 3.5). Its final-reactor TSS sets a layer profile with multipliers

$$(0.002, 0.005, 0.01, 0.03, 0.10, 0.50, 1.25, 2.25, 3.25, 4.00) \quad (6.80)$$

from surface to hopper. The different solids distributions probe sensitivity to the initial biomass and storage state. Agreement from two starts is a finite numerical check. It is not a proof of a unique steady state for all possible initial conditions.

Time integration uses the original nonsmoothed equations. A logarithmic state transformation keeps integrated coordinates positive. Before this transformation alone, an initial coordinate smaller than 10^{-8} is raised to 10^{-8} in its own physical unit. For $y_f = \exp(\eta_f)$, the transformed balance is $\dot{\eta}_f = f_f(y)/y_f$. This initialization safeguard is distinct from clipping a final predicted concentration.

The backward differentiation method provides an integration route for stiff equations (Curtiss & Hirschfelder, 1952). The horizon is

$$T = \max\{400, 50/\max(w, 0.001)\} \text{ d.} \quad (6.81)$$

The calculation can stop early when its scaled residual falls below 10^{-9} . Dynamic relaxation supplies an initial state for algebraic steady calculation, consistent with the continuation rationale of Kelley and Keyes (1998) and its activated sludge use in Selerio (2026). A bounded least-squares polishing calculation is applied only if the integrated endpoint fails the physical audit. Solver convergence and physical acceptance use separate tolerances.

Table 6.8: Steady-state generation and physical acceptance settings.

Setting	Value
Integration relative tolerance	10^{-7}
Transformed integration absolute tolerance	10^{-9}
Maximum integration step	$T/100$
Polishing evaluation limit	5,000
Polishing step, function, and gradient tolerances	10^{-9} each
Term-scaled balance tolerance	10^{-6}
State nonnegativity tolerance	10^{-10}
Reaction-rate nonnegativity tolerance	10^{-12}
Two-start scaled state agreement	10^{-6}
Rightmost reduced-state Jacobian eigenvalue	at most -10^{-8} d^{-1}
Eigenvalue agreement between two step sizes	10^{-6} d^{-1}
Matrix conservation and rank audit tolerance	10^{-10}

For a balance with separately evaluated terms b_j , the acceptance residual is

$$r_{\text{bal}} = \frac{|\sum_j b_j|}{\max(1, \max_j |b_j|)}. \quad (6.82)$$

Reactor generation scales repeat $\max(1, c_{\text{in},f}^U)$ for each component. Layer scales use $\max(1, X_N)$. State agreement is tested in these scaled coordinates.

Reduced-state local stability is assessed by the rightmost eigenvalue of the dynamic Jacobian. Finite differences use normalized steps $\epsilon_{\text{mach}}^{1/3}$ and $2\epsilon_{\text{mach}}^{1/3}$. Central differences are used away from the nonnegative boundary. Second-order forward differences are used near it. Both eigenvalue estimates must pass the upper-bound and agreement checks. This is stability evidence for the reduced reactor-and-aggregate-layer model. It does not establish stability of unrepresented species profiles or of a real plant.

Both starts must provide finite states, satisfy nonnegativity and rate checks, pass component and invariant conservation checks, obey the operating domain and layer envelope, and meet the layer-flux residual tolerance. Generation also requires identical branch-classification tuples between the two starts. These tuples record the active kinetic and settling branches. A state that meets a residual test but fails another physical check is not admitted as a training target.

6.9.3 Fold-local fitting and two separately selected penalties

Every component and location from one accepted plant state stays in the same fold. This grouping prevents the same simulated plant from contributing training information and an assessment target within a fold. Fold sizes differ by at most one. Centers, input scales, quadratic-feature scales, and response scales are recomputed inside each

fitting fold. The holdout contributes none of these quantities. The information-leakage analysis of Kaufman et al. (2012) explains why assessment information must be kept outside fitting choices.

Each candidate penalty is assessed by the standardized root mean square error of the complete raw response. The largest penalty within one standard error of the smallest mean score is selected. The rule favors stronger regularization when the score does not clearly distinguish the smaller penalty (Hastie et al., 2009). Projected errors do not choose the raw-response penalty. The logarithmic model uses the same folds, grid, and one-standard-error rule but is scored on its logarithmic target. Its penalty is selected separately.

The augmented ridge equations are solved with column-pivoted orthogonal-triangular factorization and audited using singular-value decomposition. A coordinate whose standard deviation meets the near-constant cutoff in Table 6.7 is rejected rather than assigned an arbitrary scale. Positive ridge penalties and the unpenalized intercept give a unique coefficient solution under the stated fitting design. The fitted penalties and coefficient matrices are retained as estimation quantities. They cannot be inferred from the candidate grid alone.

After selection, both models are refitted once on all accepted development states. Grouped out-of-fold predictions supply trust calibration and projection assessment. The overflow equality for an out-of-fold state uses the corresponding out-of-fold logarithmic model, not the final fit that has seen that state. The final models supply holdout predictions and optimization evaluations.

The overflow target must be strictly positive before logarithmic fitting. Predicted logarithms and their back-transformations must be finite. The development-set low-overflow and high-overflow strata are fixed by the percentiles in Table 6.7. Separate stratum assessment can reveal an error pattern concealed by a central concentration range. The imbalanced-regression analysis of Shahbazi et al. (2026) motivates attention to sparse or extreme response ranges. It does not establish imbalance as the cause of an error in this plant experiment.

6.9.4 Scientific admission and projection audits

Admission of the connected surrogate requires finite and independently audited out-of-fold projections for all accepted development states. The complete raw-response normalized root mean square error and the raw clarifier-inventory normalized error must each be below one. Because errors use development response scales, the threshold requires aggregate error below one scale unit. It is a case-specific admission criterion. It is not a universal standard of surrogate quality.

Every projection is assessed for equality feasibility, physical inequalities, nonnegativity, dual feasibility, stationarity, and complementarity. The overflow-closure residual is also reported separately. A failed admission criterion remains a failure rather than

triggering an unreported refit or changed penalty selection rule.

Table 6.9: Projection solve and independent audit settings.

Setting	Value
Solver absolute and relative tolerances	10^{-12} each
Initial iteration limit	100,000
Polishing	enabled
Retry values of algorithm parameter ρ_{alg}	0.01 and 10
Retry iteration limit	200,000 each
Adaptive ρ_{alg} during retries	disabled
Scaled equality and inequality tolerances	10^{-8} each
Dual, stationarity, and complementarity tolerances	10^{-8} each
Physical nonnegativity tolerance	10^{-10}
Active-set identification tolerance	10^{-7}

6.9.5 Prediction metrics and their denominators

For a response coordinate f and assessment state i , define the signed error $a_{if} = p_{if} - t_{if}$, where p is predicted and t is mechanistic. Physical-unit root mean square error, mean absolute error, and bias are

$$\left(\text{mean}_i a_{if}^2\right)^{1/2}, \quad \text{mean}_i |a_{if}|, \quad \text{mean}_i a_{if}. \quad (6.83)$$

Normalized joint-response metrics replace a_{if} by $a_{if}/(D_\chi)_{ff}$ and state whether they pool errors across coordinates. Coordinatewise coefficients of determination use

$$R_f^2 = 1 - \frac{\sum_i (p_{if} - t_{if})^2}{\sum_i (t_{if} - \bar{t}_f)^2}. \quad (6.84)$$

A constant assessment target has an undefined denominator and is recorded accordingly. An arithmetic mean of coordinate scores is distinct from a score calculated on pooled observations.

Location-quality summaries cover the mixer, five reactors, overflow, and underflow. Outlet flows are divided by their own positive flow ratios before applying the quality map. For location ℓ and composite j , define the holdout reporting range and normalized error as

$$r_{\ell j} = \max_i t_{i\ell j} - \min_i t_{i\ell j}, \quad e_{i\ell j} = \frac{p_{i\ell j} - t_{i\ell j}}{r_{\ell j}}. \quad (6.85)$$

For positive ranges, an aggregate score pools the 8×4 location-composite coordinates. A location score pools four composites and a composite score pools eight locations. The displayed R^2 summaries average the corresponding coordinatewise scores. The holdout ranges serve only reporting. They do not enter fitting, admission, objective normalization, or trust calibration.

The overflow model receives concentration-scale and logarithmic assessment separately. The concentration report includes root mean square error, mean absolute error, bias, normalized error, and R^2 . The logarithmic report includes root mean square error and bias. Positivity of Equation (6.33) does not replace these assessments.

The experimental design includes knowledge from exploratory simulation when specifying its response, closure, and acceptance rules. The separate holdout stream therefore supports descriptive post-selection assessment. It is not independent external confirmation. Holdout and optimization responses do not change the fitted coefficients, preprocessing, priorities, trust limits, engineering safeguards, or numerical settings. This separation controls later information leakage while preserving the appropriate limits on interpretation (Kaufman et al., 2012).

6.10 Operating searches and convergence evidence

6.10.1 A common baseline and an explicit basin-sensitivity experiment

The baseline search for each route starts at the center of the normalized operating box. For each coordinate,

$$\eta_j = \frac{\vartheta_j - \vartheta_j^L}{\vartheta_j^U - \vartheta_j^L} \in [0, 1]. \quad (6.86)$$

The nominal influent is the midpoint of the fresh-influent intervals. The baseline explores one initial basin. Its interpretation remains local even if the numerical search terminates successfully.

A planned multistart sensitivity experiment evaluates basin dependence separately from this baseline. It uses the same 15 starting control vectors for both routes,

$$\eta^{(0)} = \frac{1}{2}\mathbf{1}_7, \quad \eta^{(j,\pm)} = \frac{1}{2}\mathbf{1}_7 \pm \frac{1}{4}e_j, \quad j = 1, \dots, 7. \quad (6.87)$$

Each start receives the same route-specific budget and audit rules. A direct start obtains its mechanistic initial state from an independently accepted solve at those fixed controls. Failed initial states remain failures. The best independently validated feasible candidate is retained, with all starting-point outcomes and total search work reported. This planned sensitivity analysis does not replace the single-start baseline or claim exhaustive basin coverage.

The distinction matters when interpreting a faster search. An algorithm that visits fewer basins can finish quickly while missing a better candidate. The expensive-function benchmarking study of Bliet et al. (2023) motivates keeping objective quality, evaluation work, and computational time as separate comparison dimensions.

6.10.2 Surrogate search and derivative fallback

The initial surrogate outer method is sequential least squares programming, based on the constrained optimization approach of Kraft (1988). Each trial obtains a fully audited joint projection. The active-set derivative chain is used only after the rank, conditioning, multiplier, and independent derivative checks pass. An active-set change requires reconstruction of the projection and its derivatives.

When that chain cannot be validated, a deterministic value-only quadratic-approximation method continues within the normalized bounds (Ragonneau, 2022). It uses the same audited projection and does not substitute an approximate lower solve. Finite differences are used for independent derivative audits rather than as an unqualified derivative replacement. The best feasible visited point is solved again from an independent projection initialization before retention.

Table 6.10: Surrogate search, sensitivity, and finite-poll settings.

Setting	Value
Initial outer iteration limit	250
Initial outer function tolerance	10^{-10}
Active slack tolerance	10^{-7}
Required active inequality multiplier	greater than 10^{-8}
Domain-aware active-set perturbation	10^{-6}
Stationary-system condition number times machine precision	at most 10^{-8}
Sensitivity residual and scaled state reproduction	10^{-8} each
Independent derivative steps	10^{-6} and 5×10^{-7}
Derivative relative-error tolerance	10^{-5}
Upper feasibility and stationarity tolerance	10^{-6} each
Value-only fallback iteration and evaluation limits	250 each
Initial and final fallback trust radii	0.25 and 10^{-6}
Fallback feasibility tolerance	10^{-6}
Coarse and fine poll radii	10^{-3} and 10^{-4}
Coarse and fine direction counts	14 and 106
Absolute and relative sufficient-decrease tolerances	10^{-8} each
Direction-rank tolerance	10^{-10}
Certification evaluation limit	10,000
Ray growth factor and maximum probes	2 and 16
Objective relative tie tolerance	10^{-10}

6.10.3 First-order evidence and finite-resolution polling

A candidate receives first-order stationarity evidence only when the derivative chain, lower projection certificate, and upper constrained-stationarity audit pass. This is a statement about the declared surrogate optimization problem at that candidate. It does not establish a local minimum, a global optimum, or stationarity of the original mechanistic objective.

If the derivative evidence is unavailable, the convergence procedure performs complete feasible-direction polls at normalized radii 10^{-3} and 10^{-4} . The coarse poll contains the 14 signed coordinate directions. The fine poll adds 84 signed pairwise diagonals and eight regular-simplex directions from a Helmert construction. The pairwise count is $4\binom{7}{2}$. Coupled directions can reveal feasible improvements that an axis-only test misses near a constraint.

Each complete poll must contain a feasible nonzero displacement. A rank calculation is diagnostic because active constraints can reduce the dimension of the feasible cone. An accepted improvement triggers a new complete poll. A fine-radius move also triggers revalidation at the coarse radius. After an improving step, up to 16 search-only probes grow along the same ray by successive factors of two. These probes may accelerate search but supply no convergence certificate.

Absence of sufficient decrease in both complete revalidated polls gives evidence only for the tested directions and radii. An incomplete poll, a failed calculation, or an exhausted certification budget leaves stationarity unresolved. The limiting direct-search analysis of Audet and Dennis (2006) requires suitable refining directions and an asymptotic argument. Two finite polls do not provide that argument. Neither convergence tier establishes global optimality.

6.10.4 Direct continuation, recovery, and endpoint checks

The direct route uses the interior-point filter algorithm of Wächter and Biegler (2006) through the three smoothing stages. Its endpoint audit independently checks all smooth reactor and layer equations, inequalities, nonnegativity, multipliers, and stationarity. The full-state audit is separate from the final solver status.

If the initial direct search fails, one recovery may start from the same influent's certified surrogate controls and repeat the same smoothing sequence. Recovery is not attempted after an initially successful search or from an uncertified surrogate candidate. Because this recovery uses a surrogate decision, it is a dependent rescue calculation rather than independent basin exploration. Its status and computational cost are reported separately.

Table 6.11: Direct continuation and exact-evaluation qualification settings.

Setting	Value
Interior-point iteration limit per smoothing stage	2,500
Convergence and constraint tolerances	10^{-8} each
Dual and complementarity tolerances	10^{-6} each
Endpoint equality acceptance	10^{-8}
Endpoint inequality and stationarity acceptance	10^{-6} each
Endpoint active-set tolerance	10^{-7}
Receiver-switch ambiguity distance from X_t	1 g TSS m ⁻³
Other branch ambiguity distances	$10^{-6}S$
Two-start agreement under both prescribed state scales	10^{-6}
Engineering feasibility tolerance after exact evaluation	10^{-6}

A validated feasible candidate can be retained when its search budget is exhausted or stationarity remains unresolved. Its convergence qualification remains visible. A case with no validated feasible candidate remains a search failure. A favorable success flag alone is insufficient for either route.

6.11 Independent mechanistic verification and fair comparison

6.11.1 Evaluation at fixed selected controls

Every available selected decision is evaluated with its controls and influent held fixed. Both initial states are solved on the original nonsmoothed reactor and ten-layer clarifier equations. This calculation does not use the statistical prediction as its treatment outcome. It also does not use the smooth direct state as proof of an exact steady state. The role of an original-model check follows the feasibility and validation requirements discussed by Bhosekar and Ierapetritou (2018).

Exact evaluation uses the physical checks in Table 6.8. Two-start state agreement is tested under both generation scales and fixed state scales. The fixed scales are development-set coordinatewise median absolute deviations with a floor of one. A valid evaluation must pass mass conservation, nonnegativity, layer residuals, the domain, the endpoint envelope, and the reduced-state stability test.

Branch tuples must agree unless an evaluation start is boundary-ambiguous. Ambiguity bands cover the receiver threshold, zero storage capacity, zero settling offset, zero or maximum settling velocity, and equal adjacent gravitational fluxes. An ambiguity flag qualifies the state-level branch comparison. It does not prove that every mismatching branch falls within its own ambiguity band. Agreement of the physical states and all

other audits is still required. Smooth stationarity or smooth-reference equivalence near such a boundary requires that qualification.

This verification is especially relevant near a prescribed nutrient limit. The aeration-scheduling framework of Ren et al. (2026) explicitly considers influent disturbances and effluent ammonium safeguards. Its safety setting is not identical to the present TN or TP objective. It reinforces the need to evaluate the particular effluent quantity and limit that governs a decision. A nonnegative or conservation-corrected concentration can still misclassify compliance.

6.11.2 Native and common-reference objectives

Let route S denote the connected surrogate and route M the direct smooth mechanistic search. Their native responses are $\chi_{\text{nat},S} = \hat{\chi}$ and $\chi_{\text{nat},M} = \chi_{\text{sm},M}$. The independently solved responses are $\chi_{\text{ref},S}$ and $\chi_{\text{ref},M}$. Raw and projected surrogate predictions are also evaluated, where defined, at both retained control vectors.

For positive development-set effluent standard deviations σ_j , native quality normalization is

$$D_{T,S} = \text{diag}(\max(1, \sigma_j)), \quad D_{T,M} = \text{diag}(\sigma_j). \quad (6.88)$$

The unit floor is in the physical unit of each quality coordinate. The common reference objective uses $D_{T,M}$ for both decisions. Actual fitted standard deviations are estimation quantities and are not specified as theoretical constants.

The native-to-reference and route-to-route differences are

$$\Delta J_{\text{nat-ref},k} = J_{\text{nat},k}(\vartheta_k, \chi_{\text{nat},k}) - J_{\text{ref}}(\vartheta_k, \chi_{\text{ref},k}), \quad k \in \{S, M\}, \quad (6.89)$$

$$\Delta J_{S-M} = J_{\text{ref}}(\vartheta_S, \chi_{\text{ref},S}) - J_{\text{ref}}(\vartheta_M, \chi_{\text{ref},M}). \quad (6.90)$$

The first can reflect both response approximation and route normalization. The second compares independently evaluated decisions using a common objective basis. Neither is called regret because a global optimal objective is unavailable.

Standardized response errors at each decision retain the raw, projected, and native quantities separately. A symmetric relative objective difference divides ΔJ_{S-M} by $\max(1, |J_S|, |J_M|)$. A direct-relative percentage divides $100\Delta J_{S-M}$ by J_M only when that denominator is nonzero. Normalized control differences are

$$d_j = \frac{\vartheta_{S,j} - \vartheta_{M,j}}{\vartheta_j^U - \vartheta_j^L}, \quad d_{\text{rms}} = \left(\frac{1}{7} \sum_{j=1}^7 d_j^2 \right)^{1/2}, \quad d_{\text{max}} = \max_j |d_j|. \quad (6.91)$$

These summaries distinguish similar objective values from similar controls.

Selected-decision removal for composite j is

$$\mathcal{R}_j = 100 \left(1 - \frac{z_{E,j}}{z_{\text{in},j}} \right), \quad (6.92)$$

when the fresh-influent denominator is positive. Predicted-to-reference removal differences use percentage points. Concentration percentage errors instead divide by the mechanistic effluent concentration. For a hypothetical influent of 200 mg L^{-1} and effluent values of 2 and 4 mg L^{-1} , the effluent prediction error is 100% relative to 2. The removal difference is only one percentage point. Reporting both makes their different denominators clear.

6.11.3 Validity, engineering eligibility, and failures

A paired objective comparison requires both routes to supply natively feasible candidates, valid exact evaluations, and satisfaction of the common retained engineering safeguards after evaluation. A valid physical evaluation that violates an underflow, feed-solids, or external-loss safeguard still retains its diagnostic objective and treatment response. It does not enter the eligible paired comparison. Solids retention time and loading measures are reported for every finite evaluation and do not add eligibility thresholds.

Table 6.12: Distinct questions retained in failure and qualification accounting.

Assessment	Question answered
Generation acceptance	Did both original-model starts pass the required physical and numerical checks?
Projection feasibility	Did the fitted closure and physical constraints admit an audited joint response?
Native engineering feasibility	Did the search candidate satisfy its declared optimization constraints?
Convergence qualification	Did derivative audits or complete finite polls supply their stated evidence?
Exact-model validity	Did independent original-model evaluation converge to an accepted state?
Retained engineering feasibility	Did the exact response pass the common safeguards?
Paired eligibility	Were both decisions available and admissible on the common reference basis?

These status fields are not collapsed into one success label. Missing or nonfinite quantities, an undefined feature scale, a failed factorization, or an inconsistent numerical audit remain computational failures. A finite state that fails a scientific threshold remains a diagnostic state with the failed criterion identified. Generation rejection, optimization failure, projection infeasibility, and branch qualification have different causes and denominators.

6.11.4 Influent scenarios and time accounting

Ten influent vectors from a separate twenty-dimensional Latin hypercube define the scenario experiment. They use the seed in Table 6.7 and do not enter fitting, calibration, or tuning. Both routes are attempted for the nominal influent and every scenario. Available selected decisions receive exact evaluation even when the other route fails. These cases examine sensitivity within a specified set. They do not estimate population-level reliability under arbitrary disturbances.

Optimization time is the duration of the route’s initial search. For the surrogate, it includes regression, projection, and any value-only fallback inside that search. For the direct route, it includes the initial three-stage smoothing sequence. Development sampling, fitting, convergence certification, conditional recovery, and exact-model evaluation are outside this interval. Failed initial attempts remain in the time accounting. Scenario aggregates use the ten influent cases and keep the nominal case separate. Both routes use the same processor allocation and thread count.

Search time is accompanied by evaluated points, iterations, projection retries, fallback use, completed smoothing stages, recovery status, and convergence qualification. The benchmark concerns identified by Blik et al. (2023) make these workload quantities useful when comparing expensive-function methods. Time alone cannot establish equal decision quality or equal completeness of a search.

A complete workflow account additionally includes development, certification, recovery, and verification. These costs are reported separately rather than silently absorbed into a search-speed claim. Repeated use can amortize development effort, but the relevant number of later decisions depends on the full per-decision cost and on acceptable mechanistic decision quality. A reduction in the selected search interval therefore does not by itself establish a reduction in the complete computational task.

The resulting evaluation keeps four types of evidence separate. Statistical assessment concerns approximation of accepted mechanistic states. Projection audits concern the imposed network and inventory constraints. Search audits concern local numerical evidence for the declared optimization problem. Original-model verification concerns treatment and feasibility at the selected controls. This separation allows a connected surrogate to be studied as a decision aid while preserving the limits of its empirical closure and aggregate clarifier representation.

Chapter 7

Integrated Assessment and Research Boundaries

7.1 A common basis for evaluating the prediction chain

The research links a mechanistic component representation, statistical approximation, physical enforcement, interpretation, and operating choice. Each part introduces a specific question about credibility. The mechanistic representation defines the reference system. The statistical predictor approximates its response. The correction restricts the returned state to the imposed feasible set. Interpretation describes selected parts of the prediction map. Optimization uses the returned response to select controls. Evidence is needed at each point where the intended claim changes.

The raw benchmark and the correction benchmark share a particular prediction. This permits a paired assessment of what changes when enforcement is applied. A comparison between two independently trained models answers a different question because it also contains differences in fitting and hyperparameter selection. The structured regression adds another distinction. Its raw head, affine correction, and final constrained output are separately assessable objects. Reporting only the final output would leave the contribution of the learned structure unclear.

The connected plant adds several linked outputs. Its correction must be assessed across the mixer, reactor train, clarifier outlets, and solids inventory. A change that improves one location can worsen another while preserving joint constraints. Location-specific errors therefore remain necessary alongside a plant aggregate. This follows the general principle that approximation quality must be assessed in the quantities used by the intended decision (Bhosekar & Ierapetritou, 2018; McBride & Sundmacher, 2019).

Table 7.1: Claims and the evidence required to assess them.

Claim	Assessment evidence	Limit of the inference
Accurate approximation	Held-out component and composite errors with defined scales	Agreement concerns the selected reference model and tested domain
Physical admissibility	Original equality residuals, non-negativity checks, and inventory bounds	Unimposed kinetic and dynamic relations remain unverified
Sample efficiency	Common learning curves and variation across partitions	The conclusion depends on the sampling and fitting budget
Transparent structure	Named blocks, physical-unit coefficients, and sensitivity calculations	Coefficients are predictive associations rather than identified causal laws
Reliable operating choice	Independent mechanistic solve at selected controls	A local search does not establish a global optimum
Computational benefit	Separate development, search, prediction, correction, and verification costs	Search-time savings alone do not establish a lower total cost

7.2 Examples that separate the assessment questions

Consider a hypothetical two-component prediction problem in which the total inventory is ten concentration units. The raw prediction is $(8, 3)$, while the reference state is $(6, 4)$. The raw prediction is nonnegative but violates the total inventory. A conservative correction might return $(7, 3)$. It satisfies the total and has a different prediction error. Conservation has been established by the equality check. Accuracy still requires comparison with the reference state. This example illustrates two assessment questions rather than a measured treatment response.

Now consider a hypothetical conservative raw state $(11, -1)$ under the same inventory. Clipping the negative value gives $(11, 0)$ and destroys conservation. An equality correction alone can also retain a negative component. The final assessment must therefore check both properties on the same returned state. Separate marginal summaries can be misleading if one prediction satisfies only conservation and another satisfies only nonnegativity (Haddad et al., 2010; Kircher & Votsmeier, 2025).

A decision example makes a further distinction. Suppose two operating choices have surrogate-predicted dimensionless objectives of 0.90 and 0.95, with smaller values preferred. Independent mechanistic evaluation could instead give 1.10 and 0.98. Both projected surrogate states could satisfy every imposed balance, while the preferred

decision changes under the reference evaluation. The example contains no optimization finding. It explains why physical consistency and average error cannot establish the ordering of selected decisions (Alexandrov et al., 1998; Pedrozo et al., 2025).

Interpretation also depends on the complete map. A positive quadratic aeration coefficient in a raw driver does not imply that every final effluent quantity increases with aeration. Coupling transforms the driver into component predictions. The reporting map combines components. The active output constraints can alter local derivatives again. Physical-unit interpretation must therefore identify whether it concerns a driver coefficient, a raw prediction derivative, or a derivative of the final projected response (Brunton et al., 2016; Shen et al., 2023).

7.3 Data separation and the status of operating evidence

All quantities learned from data require a declared fitting reference. Input transformations, output scales, hyperparameters, regularization, overflow solids fitting, and trust calibration belong to the development information. The final assessment information is reserved for evaluating the specified procedure. This separation must remain consistent when the procedure contains a learned model followed by a constrained calculation (Kaufman et al., 2012).

The standalone reactor benchmark uses nested cross-validation to separate model selection from outer assessment. Its beyond-domain samples are simulated separately and are not used to tune the fitted predictor. The structured-regression protocol uses its declared training and holdout partitions and repeated learning-curve assessment. These designs answer related questions with different units of evidence. Their scores should not be pooled into one model ranking as if they arose from a common assessment sample.

The connected plant design distinguishes development, holdout prediction assessment, and operating verification. It also recognizes that engineering choices can be informed by earlier inspection of a simulated response. Where an analysis setting follows such inspection, mechanistic evaluation of the selected controls provides conditional evidence for that declared setting. It does not become a fully prospective assessment merely because another solve is performed. A later independent evaluation with frozen choices is needed for a stronger prospective claim (Kaufman et al., 2012; Zhang et al., 2026).

Failure accounting is part of this evidence. A failed mechanistic solve, an infeasible projection, an operating search that fails a stopping criterion, and a control that violates the original engineering constraints are different events. Omitting them changes the apparent domain and the apparent computational cost. Acceptance rules and failure counts must therefore accompany empirical assessment when it is undertaken.

7.4 Limits of physical guarantees

The reactor invariant constraints follow from the stoichiometric matrix and the compatibility of the external forcing with those invariants. They apply to the declared component representation and reactor boundary. They do not establish that every feasible concentration vector is a reachable reactor steady state. The feasible set generally contains many states that satisfy inventories and nonnegativity without satisfying the nonlinear rate equations.

The plant constraints extend this logic to connected unit outputs. Mixing, flow-weighted clarifier balances, soluble-component continuity, and solids inventory bounds address specified physical relationships. Their satisfaction does not reconstruct the full layered clarifier or prove dynamic stability. The direct reference equations remain necessary for judging the selected operating controls.

Numerical feasibility also needs a declared tolerance. Exact equalities describe the mathematical formulation. Finite arithmetic and iterative solvers return approximate numerical states. A solver termination message is insufficient without residual checks in the original units or in a clearly stated scaled form. The same principle applies to the direct mechanistic comparator (Boyd & Vandenberghe, 2004; Buzzi-Ferraris & Manenti, 2013).

7.5 Limits of generalization and interpretation

Simulated training data reflect the selected process model, parameter values, and input design. A wide numerical range does not guarantee coverage of every important joint condition. A steady-state input description does not encode all dynamic or historical influences. The distinct standalone and plant alkalinity ranges also prevent an automatic transfer claim between those domains. Domain checks need to refer to the actual training representation.

Interpretation is conditional on the fitted predictor. Second-order blocks support transparent inspection, but correlated inputs and the coupling variables can produce nonunique coefficient combinations. Resampling can assess stability of structural patterns. It cannot turn a predictive coefficient into a causal biological constant. Sensitivity analysis should identify the input unit, the operating point, and the correction stage to which it applies (Brunton et al., 2016; Machado et al., 2014).

A corrected point prediction also lacks an automatic uncertainty guarantee. Model ensembles, probabilistic reconciliation, and hard-constrained probabilistic learning provide other approaches to uncertainty (Hansen et al., 2024; Lakshminarayanan et al., 2017; Utkarsh et al., 2025). Their existence does not make a deterministic correction a calibrated distribution. Uncertainty remains a separate extension requiring its own assumptions and validation.

7.6 Engineering interpretation of the planned assessment

The intended assessment can identify which prediction procedures are suitable for particular uses. A rapid unconstrained approximation may be useful for exploratory analysis if its limitations are visible. A corrected component state can support screening that requires declared balances and nonnegativity. An interpretable constrained procedure can support inspection of named response terms. A connected plant surrogate can support an operating search when selected controls are independently checked.

The assessment must retain these different uses when interpreting empirical evidence. A smaller error does not settle every engineering requirement. A stronger physical guarantee does not settle every accuracy question. A faster search does not settle every cost question. The methodological contribution lies in making each requirement explicit and connecting it to evidence that can assess it. Treatment outcomes and empirical performance remain questions for the subsequent evaluation.

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